



Central Coast Regional Water Quality Control Board

December 1, 2023

Sent Via Electronic Mail

William Via
Developmental Services Director
375 Fifth Street
Hollister, CA 95023
Email: William.Via@hollister.ca.gov

Dear William Via:

**THE CITY OF HOLLISTER RECYCLED WATER USE AREA, SAN BENITO COUNTY
– NOTICE OF APPLICABILITY, ENROLLMENT IN ORDER WQ 2016-0068-DDW,
WATER RECLAMATION REQUIREMENTS FOR RECYCLED WATER USE AND
TRANSMITTAL OF MONITORING AND REPORTING PROGRAM ORDER NO. R3-
2023-0027**

Central Coast Regional Water Quality Control Board (Central Coast Water Board) staff reviewed the City of Hollister's October 2022 notice of intent to obtain coverage under Order WQ 2016-0068-DDW, Water Reclamation Requirements for Recycled Water Use (General Permit). According to the information provided, the proposed recycled water use project meets the conditions of the General Permit. This letter serves as a notice of applicability for enrollment in the General Permit. Attachment 1 includes ownership, facility, and facility operation summary information, Attachment 2 includes maps of the site location and areas to receive recycled water, Attachment 3 includes your Monitoring and Reporting Program, and Attachment 4 contains your title 22 engineering report.

The City of Hollister must comply with the following:

1. **General Permit** – The City of Hollister must comply with all conditions and requirements of the General Permit. As described in the General Permit, ongoing operation, maintenance, monitoring, and reporting are required. A copy of the General Permit is available electronically at the following link:

https://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2016/wqo2016_0068_ddw.pdf

JANE GRAY, CHAIR | RYAN E. LODGE, EXECUTIVE OFFICER

895 Aerovista Place, Suite 101, San Luis Obispo, CA 93401 | www.waterboards.ca.gov/centralcoast

2. **Monitoring and Reporting Program** –The City of Hollister must comply with the requirements of Monitoring and Reporting Program No. R3-2023-0027 enclosed with this letter (Attachment 3).

Per the Monitoring and Reporting Program, you are required to submit an annual report by **April 1st following the monitoring year**. Your first annual report is due on **April 1, 2024**.

Monitoring reports must be provided electronically in searchable PDF with the Central Coast Water Board's current transmittal sheet found at the link below as the cover page. The transmittal sheet must be signed.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/docs/transmittal_sheet.pdf

The City of Hollister must submit all reports/documents and laboratory data (using the transmittal sheet as the cover page) to the publicly accessible State Water Board's GeoTracker^{1,2} database with applicable Electronic Submittal of Information (ESI) requirements under the wastewater system-specific global identification number WDR100031912 at:

https://www.waterboards.ca.gov/ust/electronic_submittal/index.html

3. **Fees** – The Central Coast Water Board has elected to not charge a fee for enrollment in this permit at this time. In the future, the Central Coast Water Board may choose to charge an annual fee for continued enrollment into the General Permit at a rate to coincide with a Threat and Complexity rating of 3C. A copy of the current state fee schedule is available electronically at the following link:

https://www.waterboards.ca.gov/resources/fees/water_quality/

4. **Notification** – The Central Coast Water Board will be notified of your enrollment at a regularly scheduled public meeting on February 15-16, 2024. Details about that meeting are available on our website at:

http://www.waterboards.ca.gov/centralcoast/board_info/agendas/

5. **Future Discharge Modification** – Pursuant to California Water Code section 13260, you must inform the Central Coast Water Board at least 120 days prior

¹ Information for first-time GeoTracker users is available at:

https://www.waterboards.ca.gov/ust/electronic_submittal/docs/beginnerguid2.pdf

² Additional information available at: <http://geotracker.waterboards.ca.gov/>

to modifying your discharge. If there are any significant changes in either treatment or disposal methodologies, or the volume or character of the treated wastewater, you must notify the Central Coast Water Board immediately of such changes.

6. **Responsible Party** – The City of Hollister is responsible for the management and use of the recycled water in compliance with the conditions of the General Permit. Any noncompliance with this General Permit constitutes a violation of the California Water Code and subjects The City of Hollister to enforcement action and/or termination of enrollment under this General Permit.
7. **Change in Ownership** – In the event of any change in control or ownership of the property, The City of Hollister must notify the succeeding owner or operator of the existence of this General Permit by letter, a copy of which must be immediately forwarded to the Central Coast Water Board. In addition, the succeeding owner must submit a notice of intent for enrollment in the General Permit. Once both actions have taken place, the new owner or operator can be enrolled in the General Permit and your enrollment in the General Permit can be terminated.
8. **Division of Drinking Water Conditions** – The City of Hollister must comply with any conditions contained in the current Division of Drinking Water acceptance letters and use of recycled water in accordance with their latest accepted title 22 Engineering Report.

If you have any questions, please contact **Mark Lemus at (805) 549-3703 or by email at Mark.Lemus@waterboards.ca.gov**, or Jennifer Epp at (805) 594-6181 or by email at Jennifer.Epp@waterboards.ca.gov.

Sincerely,

Jennifer Epp
Digitally signed by Jennifer Epp
Date: 2023.11.30 15:18:51 -08'00'
Water Boards

for Ryan E. Lodge
Executive Officer

Attachments:

- Attachment 1: Ownership, Facility, and Facility Operation Summary Information
- Attachment 2: Figures – Site Location and Area to Receive Recycled Water by City of Hollister
- Attachment 3: Monitoring and Reporting Program No. R3-2023-0027
- Attachment 4: City of Hollister Water Recycling Project Engineers Report for Production, Distribution and Use of Recycled Water, April 2008

cc:

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Jesse Woodard, Central Coast Water Board, Jesse.Woodard@waterboards.ca.gov
Mark Lemus, Central Coast Water Board, Mark.Lemus@Waterboards.ca.gov
WDR Program, GeoTracker RB3-WDR@Waterboards.ca.gov

ECM/CIWQS = CW-230555

GeoTracker No. = GT-100031912

ECM Subject Name = City of Hollister NOA Order WQ 2016-0068-DDW
\\ca.epa.local\RB\RB3\Shared\WDR\WDR Facilities\San Benito Co\City of
Hollister\2023\2023 Enrollment\NOA\0068\Jennifer's Review\Hollister
0068_NOA&MRP_final.docx

ATTACHMENT 1

WASTEWATER SYSTEM OPERATION SUMMARY AND SITE-SPECIFIC REQUIREMENTS AND LIMITATIONS

1. OWNERSHIP, FACILITY, AND FACILITY OPERATION SUMMARY INFORMATION

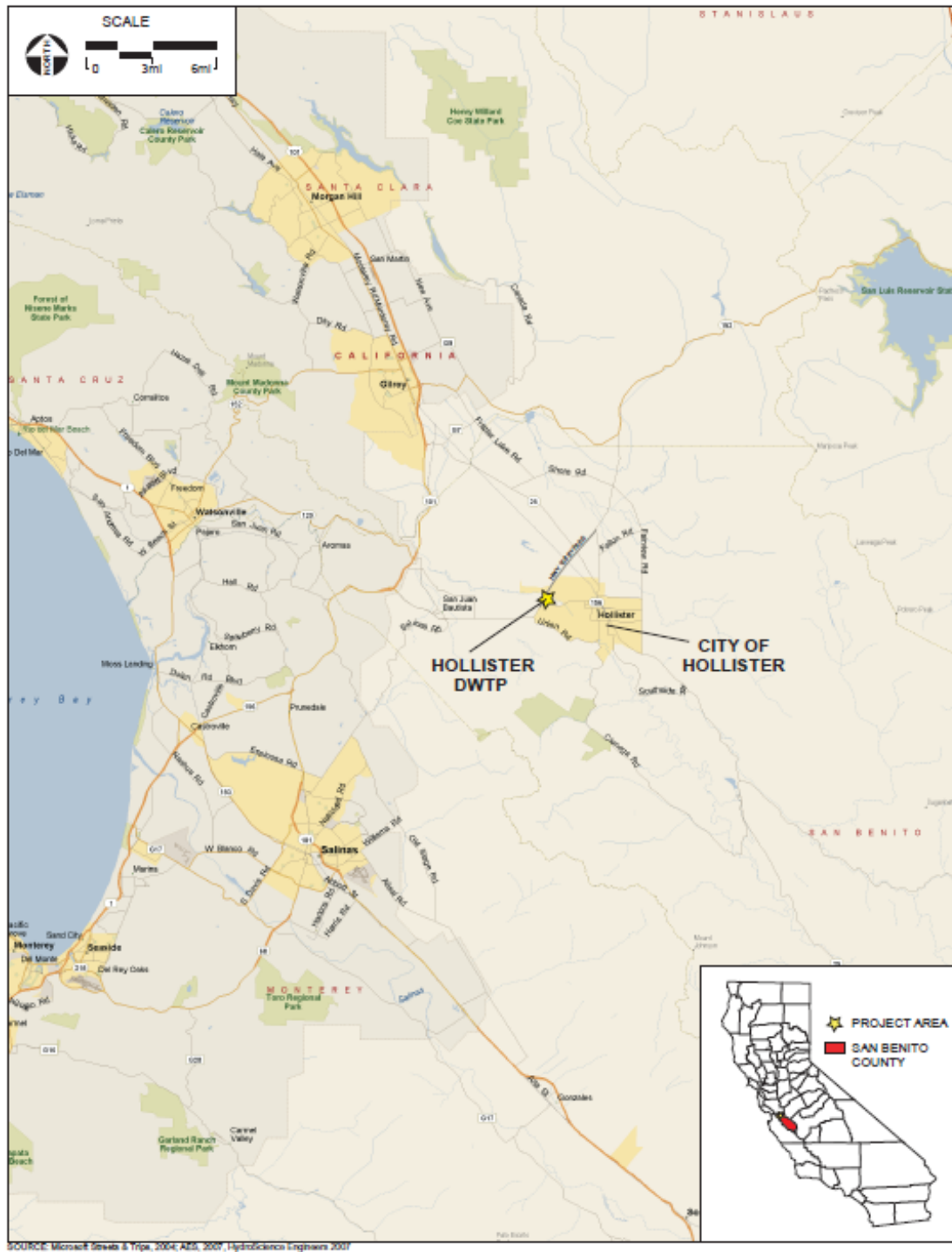
Ownership and facility information submitted by City of Hollister to the Central Coast Water Board is shown in Table 1.

Table 1: Ownership and Facility Information

Facility Name	The City of Hollister Recycled Water Use Area
Facility Physical Address:	2100 San Juan Road Hollister, CA, 95023
Latitude	36.848231
Longitude	-121.431094
Facility Mailing Address:	1321 South Street, Hollister, CA 95023
Owner and Operator of Recycled Water Use Site:	The City of Hollister Contact: William Via 1321 South Street, Hollister, CA 95023 (831) 636-4370 William.Via@hollister.ca.gov
Legally Responsible Official:	David Mirrionne phone: 831-636 - 4300 e-mail: David.Mirrione@hollister.ca.gov
Threat to Water Quality	3
Complexity	C
For Internal Use Only	
Fee Code	(20)
Primary Place Type	Recycled Water Use Area
Facility Type	Domestic
Facility Waste Type	Recycled Water
Regulatory Measure Type	Enrollee – WDR
Reclamation Included	Yes
EPA Approved Pretreatment Program	No

- The City of Hollister's tertiary wastewater system meets California Code of Regulations title 22, section 60301.230 criteria and is permitted for the production of recycled water through enrollment on October 24, 2023 in General Waste Discharge Requirements Order No. R3-2020-0020 for Discharges from Domestic Wastewater Systems With Flows Greater than 100,000 Gallons Per Day.
- The recycled water produced by the City of Hollister's wastewater system is used at Brigantino Park (covered by this General Permit enrollment) and at sites administered by San Benito County Water District (covered by a separate permit).
- The City of Hollister is responsible for the production of recycled water and the distribution and use of recycled water at Brigantino Park for landscape irrigation. This General Permit enrollment covers the distribution and use of recycled water at Brigantino Park for landscape irrigation.
- The San Benito County Water District will be permitted separately as an administrator and distributor of recycled water produced by the City of Hollister.
- In April 2008, the City of Hollister submitted a title 22 engineering report describing all specific uses and use area requirements to the California Department of Public Health, now State Water Resources Control Board Division of Drinking Water, for review and approval.
- In 2009, the California Department of Public Health provided final comments, recommendations, and conditional approval for the use of recycled water for landscape irrigation at Brigantino Park.
- The City of Hollister's Brigantino Park is the only user of tertiary recycled water in the City of Hollister's service area (Attachment 2).
- The City of Hollister must submit supplemental information for review and approval (including a revised title 22 engineering report) to the Central Coast Water Board and Division of Drinking Water prior to adding new recycled water users or locations.

ATTACHMENT 2 FIGURES



Engineering Report on the Production, Distribution, and Use of Recycled Water

Figure 1-1
Regional Location

FIGURE 1. SITE LOCATION MAP

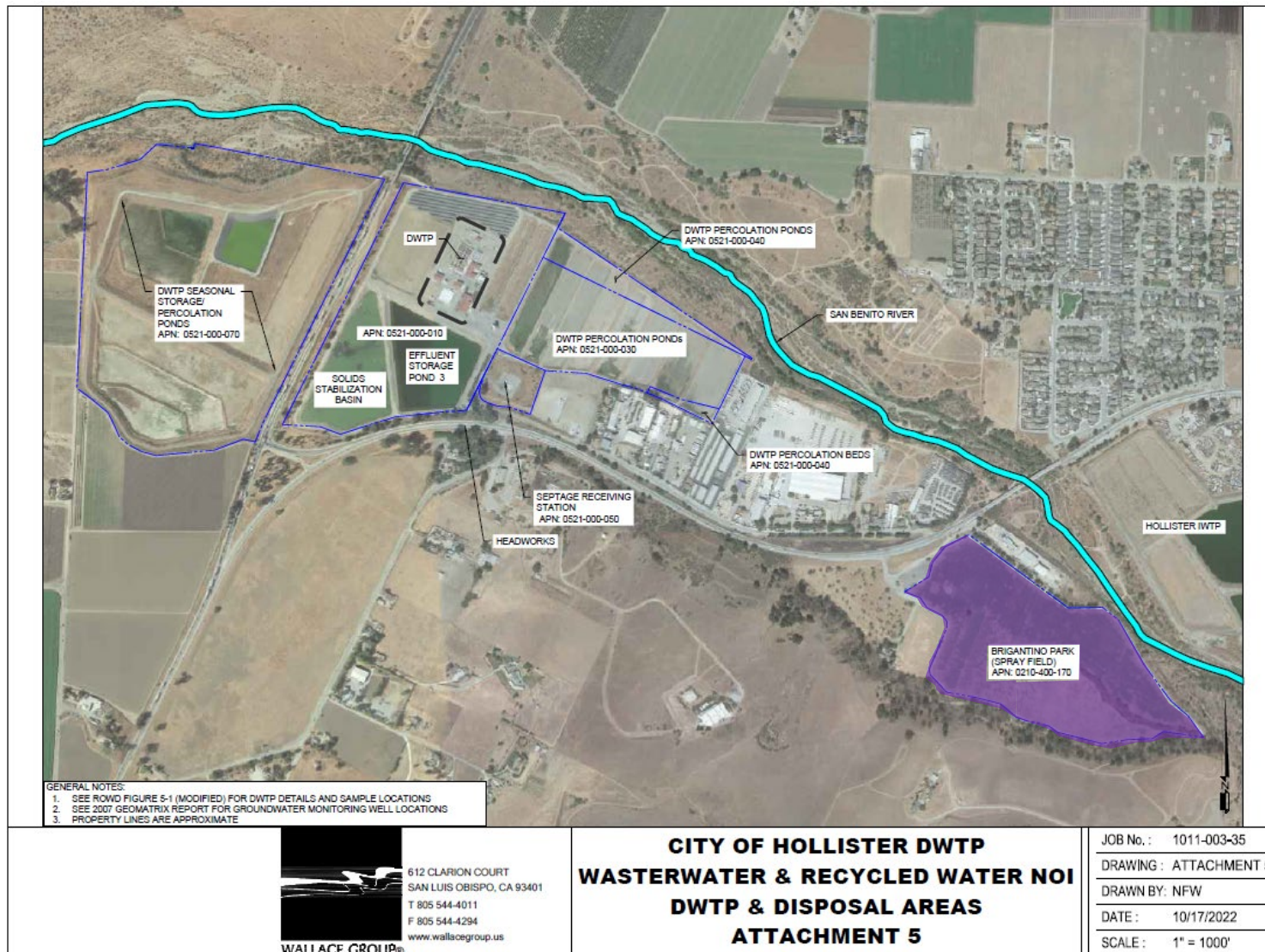


FIGURE 2. LOCATION OF BRIGANTINO PARK RECYCLED WATER IRRIGATION AREA

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

**MONITORING AND REPORTING PROGRAM NO. R3-2023-0027
For
THE CITY OF HOLLISTER, SAN BENITO COUNTY**

This monitoring and reporting program describes requirements for monitoring a recycled water system.

The City of Hollister has applied for and received coverage for the recycled water system that is subject to the notice of applicability dated December 1, 2023 for coverage in Order WQ 2016-0068-DDW, Water Reclamation Requirements for Recycled Water Use (General Permit).

If water quality samples are collected, the samples must be representative of the volume and nature of the discharge or matrix of material sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample must be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) must be approved by the Central Coast Water Board.

Field test instruments (such as those used to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a California Environmental Laboratory Program (ELAP) certified laboratory or:

1. The user is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
3. Instruments are serviced by the manufacturer or authorized representative at the recommended frequency; and
4. Field calibration reports are maintained and available for at least three years.

Monitoring requirements listed below may duplicate existing requirements under other orders. Duplication of sampling and monitoring activities are not required if the monitoring activity satisfies the requirements of this monitoring and reporting program. The facility may continue using existing sampling collection equipment that is consistent with the applicable facility order. However, due to short sample holding times, bacteriological samples collected to verify disinfection effectiveness must be grab samples. In addition to submitting the results under another order, the results must be submitted in the reports required by this monitoring and reporting program.

1. RECYCLED WATER MONITORING

Because recycled water is used for irrigation of landscape areas, priority pollutant monitoring is required at the production facility. Priority pollutants are listed in Appendix A of 40 Code of Federal Regulations (CFR) Part 423. Sampling and reporting frequency are shown in Table 1.

Table 1. Priority Pollutant Sampling and Reporting Frequency

Constituent	Sample frequency	Reporting Frequency
Priority Pollutants	Annually	Annually

2. DISINFECTION SYSTEM MONITORING

Samples shall be collected from downstream of the disinfection system and analyzed by an approved laboratory per title 22, section 60321(a). Disinfection monitoring requirements are described in Table 2.

Table 2. Disinfection System Monitoring Requirements

Constituent/ Parameter	Units	Sample Type	Sample Frequency	Reporting Frequency
Total Coliform Bacteria	MPN/100 mL	Grab	Daily	Annually
Turbidity	NTU	Meter	Continuous	Annually

MPN = most probable number

NTU = nephelometric turbidity units

3. USE AREA MONITORING

The City of Hollister must monitor use areas at a frequency appropriate to determine compliance with this General Permit and the City of Hollister's recycled water use program requirements. A recycled water user takes physical possession of the recycled water from the producer and/or distributor for an approved beneficial recycled water use that is consistent with the Uniform Statewide Recycling Criteria. The City of Hollister may assign monitoring responsibilities to the user as part of the Water Recycling Use Permit program; The City of Hollister retains responsibility to ensure the data is collected and to prepare and submit the annual report.

The following must be recorded for each user with additional reporting for use areas as appropriate. The frequency of use area inspections must be based on the complexity and risk of each use area. Use areas may be aggregated to combine acreage for calculation or observation purposes. Use area monitoring requirements are shown in Table 3.

Table 3. Use Area Monitoring Requirements

Parameter	Units	Monitoring Type	Frequency	Reporting Frequency
Recycled Water User	NA	NA	NA	Annually
Recycled Water Flow	gpd	Meter ^(a)	Monthly	Annually
Recycled Water Volume	acre-feet	Calculated	Monthly	Annually
Acreage Applied ^(b)	Acres	Calculated	NA	Annually
Application Rate	inches/acre/year	Calculated	NA	Annually
Soil Saturation/Ponding	NA	Observation	Quarterly	Annually
Nuisance Odors/Vectors	NA	Observation	Quarterly	Annually
Discharge Off-Site	NA	Observation	Quarterly	Annually
Notification Signs ^(c)	NA	Observation	Quarterly	Annually

gpd = gallons per day

NA = not applicable

^(a) Meter requires meter reading, a pump run time meter, or other approved method.

^(b) Acreage applied denotes the acreage to which recycled water is applied.

^(c) Notification signs must be consistent with the requirements of California Code of Regulations, title 22, section 60310 (g).

4. DUAL PLUMBED RECYCLED WATER SYSTEMS

If dual plumbed recycled water systems are proposed, consult with State Water Board for additional reporting, design, and operation requirements. The frequency of testing for cross connection and backflow prevention devices must be as listed below or more frequently if specified by State Water Board.

Table 4. Dual Plumbed Recycled Water Systems Monitoring

Requirement	Frequency	Reporting Frequency
Cross Connection Testing	Four Years ^(a)	30 days/Annually ^(b)
Backflow Incident	--	24 hours from discovery
Backflow Prevention Device Testing and Maintenance	Annually ^(c)	Annually

^(a) Testing must be performed at least every four years, or more frequently at the discretion of the State Water Board Division of Drinking Water.

^(b) Cross connection testing must be reported pursuant to California Code of Regulations, title 22, section 60314. The report must be submitted to State Water Board within 30 days and included in the annual report to the Central Coast Water Board.

^(c) Backflow prevention device maintenance must be tested by a qualified person as described in California Code of Regulations, title 17, section 7605.

5. REPORTING

In reporting monitoring data, the City of Hollister must arrange the data in tabular form so that the date, data type (e.g., flow rate, bacteriological, etc.), and reported analytical or visual inspection results are readily discernible. The data must be summarized to illustrate compliance with this General Permit and notice of applicability as applicable. The results of any monitoring done more frequently than required at the locations specified in the monitoring and reporting program must be reported in the next regularly scheduled monitoring report and must be included in calculations as appropriate.

A. Annual Report

An annual report must be submitted to the Central Coast Water Board by **April 1 following each monitoring year**. The annual report must include the following:

1. A summary table of all recycled water users and use areas. Maps must be included to identify use areas. Newly permitted recycled water users and use areas must be identified. When applicable, if there was a supplement to the Title 22 engineering report and the State Water Board approval letter supporting those additions, reference the date and attach a copy to the annual report.
2. A summary table of all inspections and enforcement activities initiated by the City of Hollister. Include a summary of discovered non-compliance events, a discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into compliance with the notice of applicability and/or General Permit. Copies of documentation of any enforcement actions taken by the City of Hollister must be provided.
3. An evaluation of the performance of the recycled water treatment facility, including discussion of capacity issues, system problems, and a forecast of the flows anticipated in the next year.
4. Tabular and graphical summaries of all monitoring data collected during the year.
5. The name and contact information for the recycled water operator responsible for operation, maintenance, and system monitoring.
6. Modifications to the distribution system.

B. Electronic Submittal

All monitoring reports must be provided electronically in a searchable PDF format, with the Central Coast Water Board's current transmittal sheet found at the link below as the cover page.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permiiting/docs/transmittal_sheet.pdf

The City of Hollister must submit all reports/documents and laboratory data to the State Water Resources Control Board's GeoTracker³ database for the City of Hollister Recycled Water Program (GeoTracker No. WDR100031912) over the internet at:

http://www.waterboards.ca.gov/ust/electronic_submittal/index.shtml

Table 5 below summarizes all the electronic reporting requirements. The Central Coast Water Board may request submittal of some documents on paper, particularly drawings or maps that require a large size to be readable, or in other electronic formats where evaluation of data is required.

Table 5. GeoTracker Electronic Submittal Information (ESI) Data Requirements

Electronic Submittal	Description of Action	Action	Frequency
Reports and documents	Complete copy of all documents including monitoring reports (in searchable PDF format) and any other associated documents related to the facility.	Upload directly to GeoTracker all monitoring reports (in searchable PDF format) and any other associated documents.	On or before the due dates required by this monitoring and reporting program and for other documents when requested by Central Coast Water Board staff.
Laboratory Data	All analytical data (including geochemical data) in electronic format (EDF). This includes all water, soil, and vapor samples collected when monitoring a discharge.	Direct your State Certified Laboratory staff to upload all laboratory data directly to GeoTracker.	On or before the due date of the required monitoring report.
Sampling Point Location Data (Geo XY)	Name, classify, and identify the location (latitude and longitude) of all sampling points (monitoring wells, drinking water wells, effluent sample location). Monitoring wells must be surveyed, influent and effluent sample locations must be identified on the GeoTracker mapping tool under "non-surveyed data." These data points are required prior to laboratory data uploads.	Upload the location data (surveyed and non-surveyed) to the GeoTracker GEO_XY file.	Every time a permanent monitoring point is established.

³ Information for first-time GeoTracker users is available here:
https://www.waterboards.ca.gov/ust/electronic_submittal/docs/beginnerguide2.pdf

Electronic Submittal	Description of Action	Action	Frequency
Geo Map	Site layout, map of facilities, wastewater treatment system, and disposal area(s).	Upload the site layout PDF to the GeoTracker site plan file.	Year one and every five years thereafter and when the facilities are modified.

6. LEGAL REQUIREMENTS

The Central Coast Water Board's requirements that the City of Hollister submit the technical and monitoring reports described in this monitoring and reporting program are made pursuant to [section 13267 of the California Water Code](#). Failure to submit reports in accordance with schedules established by this General Permit and notice of applicability with attachments or failure to submit a report of sufficient technical quality to be acceptable to the Central Coast Water Board may subject the City of Hollister to enforcement action pursuant to [section 13268 of the Water Code](#).

The Central Coast Water Board needs the required information to ensure compliance with the notice of applicability and the General Permit. The City of Hollister is required to submit this information because it is subject to the General Permit and is responsible for the discharge.

The burden, including costs, of the reports bears a reasonable relationship to their need and the benefits to be obtained. The City of Hollister must implement the above monitoring program as of the date of this monitoring and reporting program. The Central Coast Water Board may rescind or modify the monitoring and reporting program at any time. The City of Hollister must not implement any changes to this monitoring and reporting program unless and until a revised monitoring and reporting program is issued by the Central Coast Water Board

Ordered by:

Jennifer Epp  Digitally signed by Jennifer Epp
Date: 2023.11.30 15:18:15 -08'00'
Water Boards

for Ryan E. Lodge
Executive Officer

ATTACHMENT 4

ATTACHMENT 4



City of Hollister Water Recycling Project

Engineers Report for Production, Distribution and Use of Recycled Water

April 2008



Prepared by

RMC Water and Environment and HydroScience Engineers



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Appendices

Appendix A -	Draft Recycled Water Use Manual and Rule of Service Handbook
Appendix B -	Recycled Water Program Material
Appendix C -	Recycled Water Use Area Applications and Site Plans
Appendix D -	Transmission Line Plans

List of Abbreviations

ADF	average design flow
ADWF	average dry weather flow
AWWA	American Water Works Association
CCB	chlorine contact basin
CCR	California Code of Regulations
CEQA	California Environmental Quality Act
City	City of Hollister
DHS	Department of Health Services [Note: Now referred to as Department of Public Health]
DI	ductile iron
DPH	California Department of Public Health [Note: Formerly the Department of Health Services]
DWTP	Domestic Wastewater Treatment Plant
EIR	Environmental Impact Report
gpm	gallons per minute
HDPE	high density polyethylene
HDLPE	high density linear polyethylene
HRT	hydraulic residence time
IWTP	Industrial Wastewater Treatment Plant
mgd	million gallons per day
MLSS	mixed liquor suspended solids
MRP	Master Reclamation Permit
ORP	oxidation-reduction potential
PLC	Programmable logic controllers
Project	City of Hollister's Recycled Water Project
Program	City of Hollister Recycled Water Use Program
PVC	polyvinyl chloride
Report	Engineering Report on the Production, Distribution, and Use of Recycled Water
RW Use Handbook	Recycled Water Use Manual and Rules of Service Handbook
RWQCB	Central Coast Regional Water Quality Control Board
SCADA	Supervisory Control and Data Acquisition
SEIR	Supplemental Environmental Impact Report
SSB	sludge storage basin
VFD	variable frequency drive
WDR	Waste Discharge Requirement

Chapter 1 Introduction

This Engineering Report on the Production, Distribution, and Use of Recycled Water (Report) is submitted to support approval of disinfected tertiary recycled water (RW) use in the City of Hollister (City) and San Benito County. The City is seeking a Master Reclamation Permit (MRP) from the Regional Water Quality Control Board. After approval by the California Department of Public Health (DPH), this Report will be submitted, along with DPH comments, to the Central Coast Regional Water Quality Control Board (RWQCB) to support adoption of a MRP for the City.

1.1 Basis

This Report fulfills the requirements of Title 22, Division 4, Chapter 3, Article 7, Section 60323 of the California Code of Regulations (CCR). It describes the manner in which the City proposes to comply with the water recycling criteria set forth in Sections 60301 through 60355 of Title 22. This Report is organized in the format presented in the *Guidelines for the Preparation of an Engineering Report for the Production, Distribution, and Use of Recycled Water*, dated March 2001 and prepared by the State Department of Health Services (DHS).

1.2 Recycled Water Source and Treatment

The origin of the wastewater to be recycled will be domestic wastewater generated in the City of Hollister service area. The wastewater will be treated, and recycled water produced, at the City's newly constructed 4.0 million gallons per day (mgd) capacity Domestic Wastewater Treatment Plant (DWTP). The tertiary DWTP will utilize ultrafiltration technology (immersed membrane bioreactor, MBR) and chlorination disinfection to achieve turbidity and bacteriological standards for Disinfected Tertiary Recycled Water, and meet all requirements of 22CCR§60301.23. The use of recycled water will generally be for irrigation and other approved uses. Disinfected tertiary recycled water will also be stored in ponds located at the DWTP.

Figure 1-1 shows the regional location of the City of Hollister's Recycled Water Project (Project). Figure 1-2 show the greater Hollister and San Benito County area, the location of the DWTP, and the proposed initial recycled water use sites. Additional recycled water customers are expected to be added in the future as there is significant potential expansion for agricultural and other uses in the region.

1.3 Producer, Distributor, and User

The City will be the Producer and Distributor of the recycled water. As Producer/Distributor under the MRP, the City will be responsible for establishing and enforcing rules governing the use of recycled water by Users. The City has developed a draft Recycled Water Program for management of the recycled water facilities, distribution, and use. Draft user guidelines, rules and regulation, and other program material are included in Appendices A and B for review.

1.4 Distribution System

The recycled water will be distributed to use sites by a system of 12-inch to 20-inch diameter transmission and distribution pipelines which are, or will be, constructed, owned, and operated by the City.

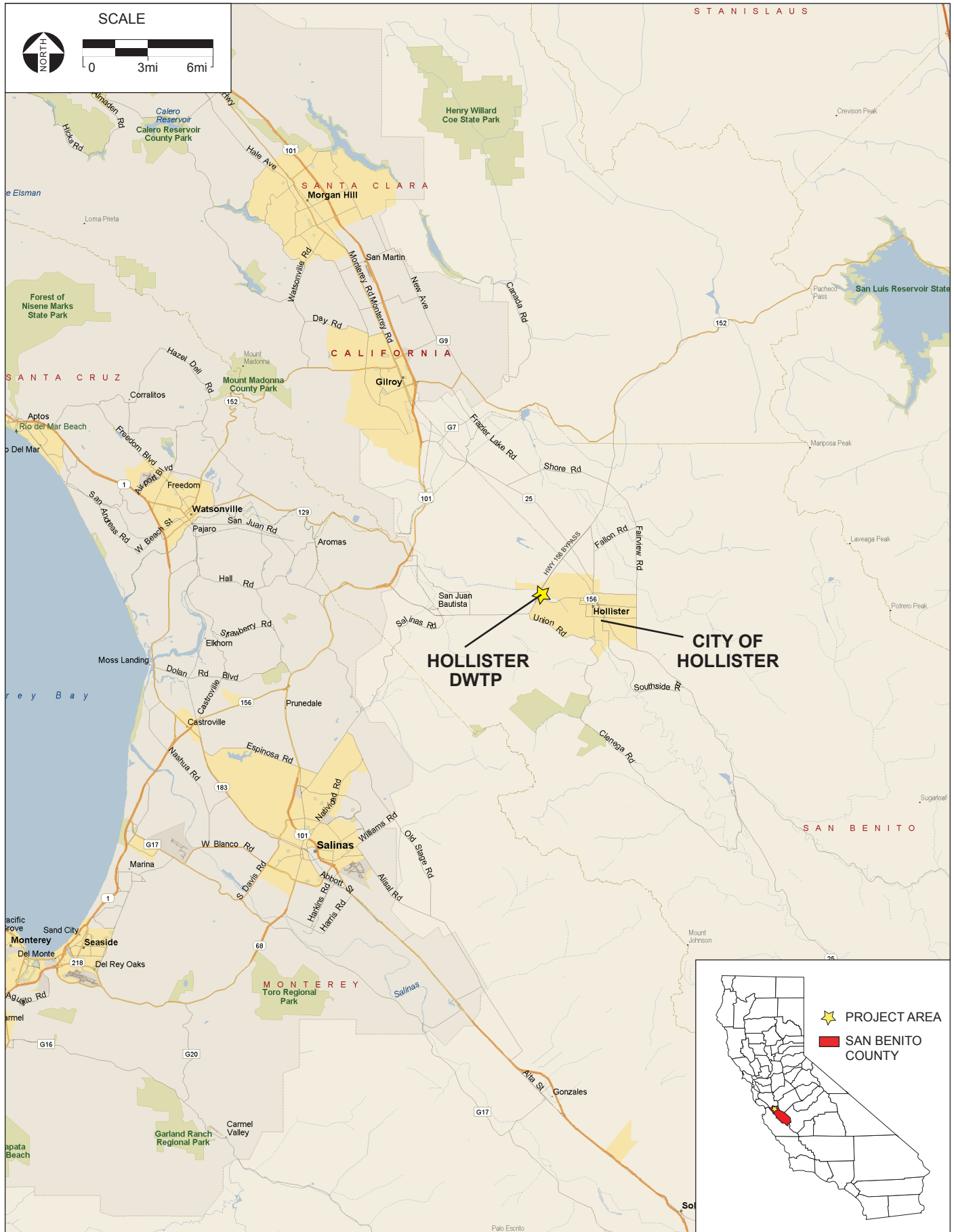
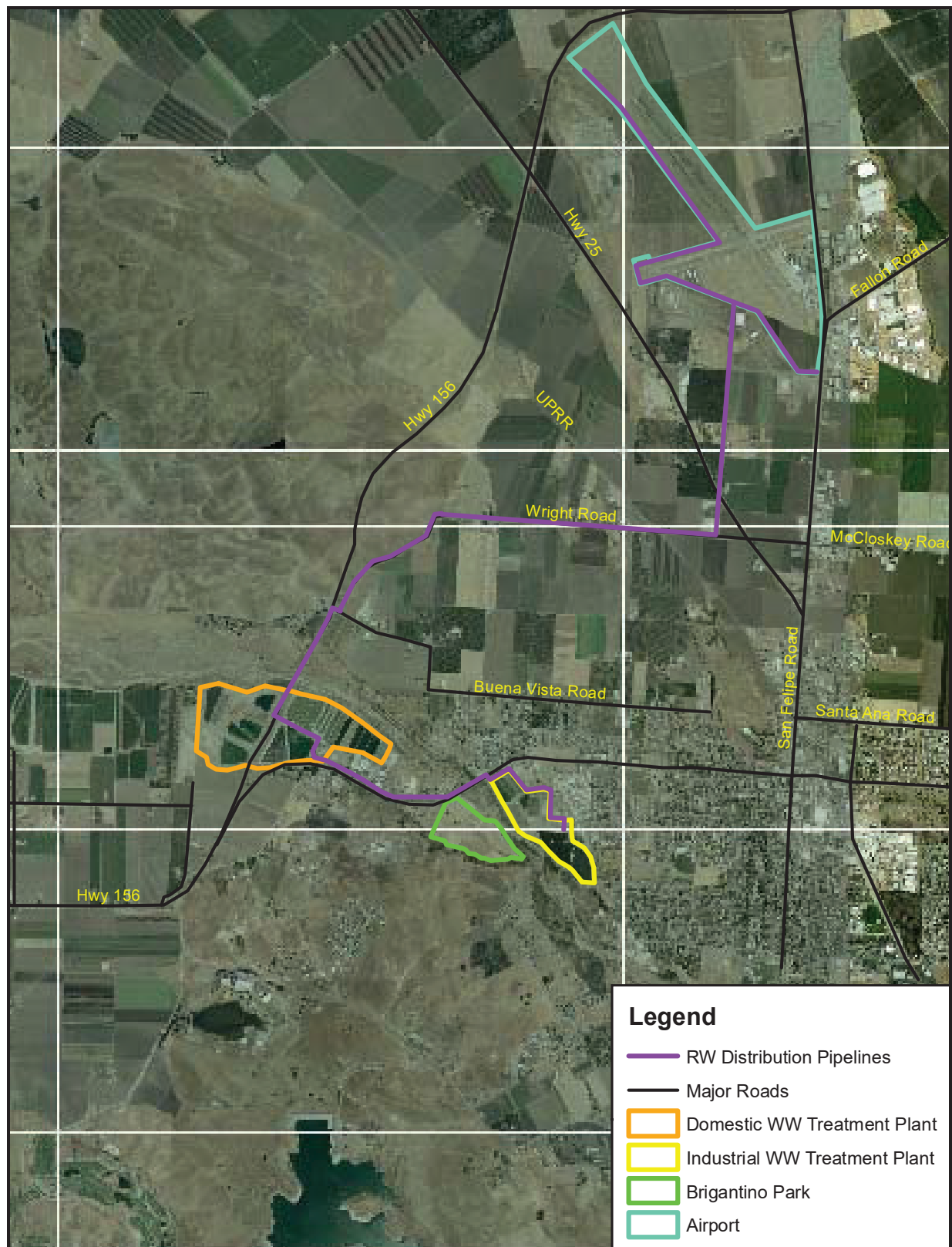


Figure 1-1

Figure 1-2: Project Area Map



1.5 Environmental Review

The initial phase of the proposed recycling project has been evaluated for compliance with the provisions of the California Environmental Quality Act (CEQA). An Environmental Impact Report (EIR) on the Project was certified by the lead agency (the City) on 31 October 2006. The SCH number for the initial EIR is SCH2006012149. The initial EIR addressed the construction of the MBR facilities at the DWTP, a septage receiving station, and the seasonal storage ponds. The initial EIR also provided a project-level assessment of recycled water irrigation at the Hollister Municipal Airport and the San Juan Oaks Golf Club.

The initial EIR also provided a program-level assessment of recycled water irrigation at the Pacific Sod Farm and on ranch lands in the eastern Flint Hills. A Supplemental EIR (SEIR) addressed CEQA analysis of additional candidate disposal/re-use sites at the project level of analysis. The SEIR is expected to be certified in March 2008. The SEIR also addressed future agricultural re-use.

1.6 Construction Schedule

Construction of the wastewater treatment plant is under way and is expected to be completed in Fall 2008. Construction of additional seasonal storage for recycled water began in July 2007 and is expected to be complete by December 2008. Construction of the recycled water conveyance and application facilities is scheduled to begin in March 2008, with recycled water use to begin at the first irrigation sites at the end of 2008 as appropriate. Details for the Airport and Brigantino Park, the first recycled water use sites, are provided in Appendix C and have a target constructed completion by December 2008.

Chapter 2 Recycled Water Project

This chapter is intended to provide the information required by Section 2.0 in the *Guidelines for the Preparation of an Engineering Report* (DHS, March 2001). This chapter describes how the City proposes to produce recycled water meeting Title 22 requirements for disinfected tertiary recycled water. Topics addressed in this chapter include: identification of agencies and entities involved in the Project (Section 2.1); how the City proposes to administer the use of recycled water (Section 2.2); identification of Producer/Distributor/User (Section 2.3); characterization of the raw wastewater (Section 2.4); description of the equipment and treatment process to be employed to produce recycled water (Section 2.5); plant reliability features (Section 2.6); identification of backup water supply (Section 2.7); proposed monitoring and reporting procedures (Section 2.8); and contingency plans (Section 2.9).

2.1 General

2.1.1 Responsible Entities

Initially, the operation of all aspects of the Project will be entirely under the authority of the City. Eventually, it is expected that the City will distribute recycled water produced at the City's DWTP to users other than the City. New users will be required to meet the requirements of the City's recycled water program that is further described in Chapter 4. Table 2-1 lists the responsible parties for various elements of the Project, or who have performed professional services in connection with Project.

Table 2-1: Summary of Responsible Entities

Project Element	Entity
Environmental Analysis	Analytical Environmental Services
Engineering and Design (DWTP)	HydroScience Engineers, Inc.
Construction (DWTP)	C. Overaa & Co.
Engineering and Design (Recycled Water Distribution and Use Sites)	RMC Water and Environment
Construction (Recycled Water Airport Pipeline)	To be determined
Construction Management	Harris and Associates
Treatment Plant and Recycled Water Production - Operation and Maintenance	City of Hollister, Dennis Rose
Recycled Water Distribution	City of Hollister, Dennis Rose

2.1.2 WWTP Staff Training

The DWTP is a Class IV wastewater treatment plant as defined in CCR Title 23. Section 3680 of Title 23 requires that supervisors and shift supervisors at a Class IV WWTP possess a minimum Grade III State Operator Certification. Additionally, a minimum 50 percent of the plant operators must possess a minimum Grade II Operator certification, or valid operator-in-training certificate at the Grade II or higher level.

2.2 Rules and Regulations

The City will function as both Producer and Distributor of recycled water to users. Users will include the City as well as private entities. The allowable uses of recycled water under the City of Hollister program are described in Chapter 4. The uses in Chapter 4 are all allowable uses under Title 22. No dual plumbed uses will be allowed under the City's proposed program. If dual plumbed uses are considered in the future, the City will further develop the recycled water program to adequately address dual plumbed system regulations and resubmit the Title 22 Engineers Report to DPH for review.

The City has developed a draft Recycled Water Use Manual and Rules of Service Handbook defining the regulations and rules governing the use of the recycled water. This draft document, herein referred to as the RW Use Handbook is included as Appendix A. Additional recycled water program materials such as standard details, annual self monitoring report, construction inspection, and other forms are included in Appendix B. Additional recycled water program information and details on the design are included in Chapter 4 of this Report.

2.3 Producer, Distributor and User Roles

This section describes the responsibilities that the City and recycled water users will have in the development, implementation, and operation of the recycled water system. The City will identify a Recycled Water Supervisor who will oversee all programs and facilities related to the use of recycled water distribution by the City. Each recycled water use site will have a designated User Supervisor in charge of the on-site use of recycled water. Additional details on the responsibilities of each party are described below.

2.3.1 City of Hollister Producer Responsibilities

As the owner of the treatment plant, storage and conveyance facilities, and all land to be used for land application, the City is the agency responsible for the production of recycled water. The City will be responsible for operations and maintenance of the DWTP as well as monitoring and reporting associated with the treatment process.

2.3.2 City of Hollister Distributor Responsibilities

The City of Hollister will be the Distributor of recycled water as defined in Title 22. As the Distributor, the City will be responsible for:

- Establishing and enforcing operating rules and regulations for use of recycled water;
- Overseeing the implementation of programs for use of recycled water;
- Developing a Public Notification Program (signage, worker training program); and
- Conducting the Monitoring and Reporting Program.

The Recycled Water Supervisor will:

- Be knowledgeable of the entire recycled water distribution system and of all water reuse criteria outlined in the MRP;
- Be responsible for implementing and overseeing programs and conditions under the MRP;
- Be the City's primary contact person in all matters relating to recycled water;
- Ensure that all User Supervisors (described in Section 2.3.3) are properly trained regarding the use of recycled water; and
- Be responsible for monitoring and reporting subject to permit requirements.

The City is developing a RW Use Handbook, to aid the Recycled Water Supervisor in conveying the City's programs and requirements to the Users. A draft version of the RW Use Handbook is attached to this report as Appendix A.

2.3.3 User Responsibilities

Each recycled water use site owner/operator will appoint a User Supervisor to be the main point of contact with the City regarding recycled water use. The User Supervisors will be responsible for ensuring compliance with the programs and procedures for the use of recycled water, as outlined in the Reclamation Permit and the City's RW Use Handbook. The User Supervisor will be responsible for:

- Ensuring that all use site facilities that serve recycled water are maintained and operated in accordance with the Reclamation Permit and RW Use Handbook;
- Operating and controlling the irrigation systems in a manner to prevent human consumption of recycled water, to control and limit runoff, and to prevent contamination of nearby domestic supply wells, as applicable;
- Ensuring that all personnel are educated in practices and procedures for working with recycled water;
- Installing and maintaining signs around the use site;
- Preventing cross-connection between recycled water facilities and domestic water facilities;
- Notifying, preparing and submitting reports to the City when there are system failures that cause unauthorized discharges;
- Notifying the City for its approval of any proposed modification or additions to facilities.

In addition to a site specific Recycled Water Use Permit, User Supervisors will be given a copy of the Reclamation Permit and the City's RW Use Handbook for guidance on the use of recycled water. The User Supervisor will be required to have these documents available at all times for inspection by the City, Regional Water Quality Control Board staff, or State/County Health Officers.

2.4 Raw Wastewater

Operator records for the period 01 January 2003 through 30 January 2008 were used to compile the summary of DWTP influent chemical analyses for general mineral parameters in Table 2-2. There is one industrial discharger (a Cannery) that discharges to the Industrial Wastewater Treatment Plant (IWTP). This industrial discharge does not have any impact on the recycled water supply from the DWTP. The cannery wastewater is treated and disposed at the IWTP, and is segregated from all treatment and disposal operations at the DWTP. The operation of the IWTP may be considered a recycled water source control program for constituents known to be elevated in cannery wastewater, such as TDS and nutrients.

Table 2-2: DWTP Influent Characterization – General Minerals

Parameter	Concentration (mg/l)				
	Min	Average	Max	Median	95 th Percentile
Total suspended solids	115	369	1,350	356	560
Total dissolved solids	824	1,178	2,224	1,202	1,462
Biochemical oxygen demand	162	341	589	343	424
Nitrate, as N	0.5	14.1	43.2	11.9	34.7
Nitrite, as N	0.01	0.12	0.87	0.10	0.20
Ammonia as N	10.2	46.1	78.3	46.4	67.1
Total organic N	0.6	18.5	107.1	9.8	59.1
Sodium	165	248	612	244	335
Chloride	156	295	914	284	432
Sulfate	13	209	355	210	286

Notes:

1. Source: DWTP Operator self-monitoring records, 2003-2007.

2.5 Treatment Processes

Figure 2-1 presents a schematic of the wastewater plant treatment process. The site layout of the new DWTP wastewater treatment facilities is shown in Figure 2-2. Figure 2-2 through 2-6 illustrate the design of the new MBR facility. A summary of the relevant process design criteria for recycled water is presented in Table 2-3 below. The sections which follow describe the design and operation of the wastewater plant processes.

Table 2-3: DWTP Design Criteria and Process Capacities

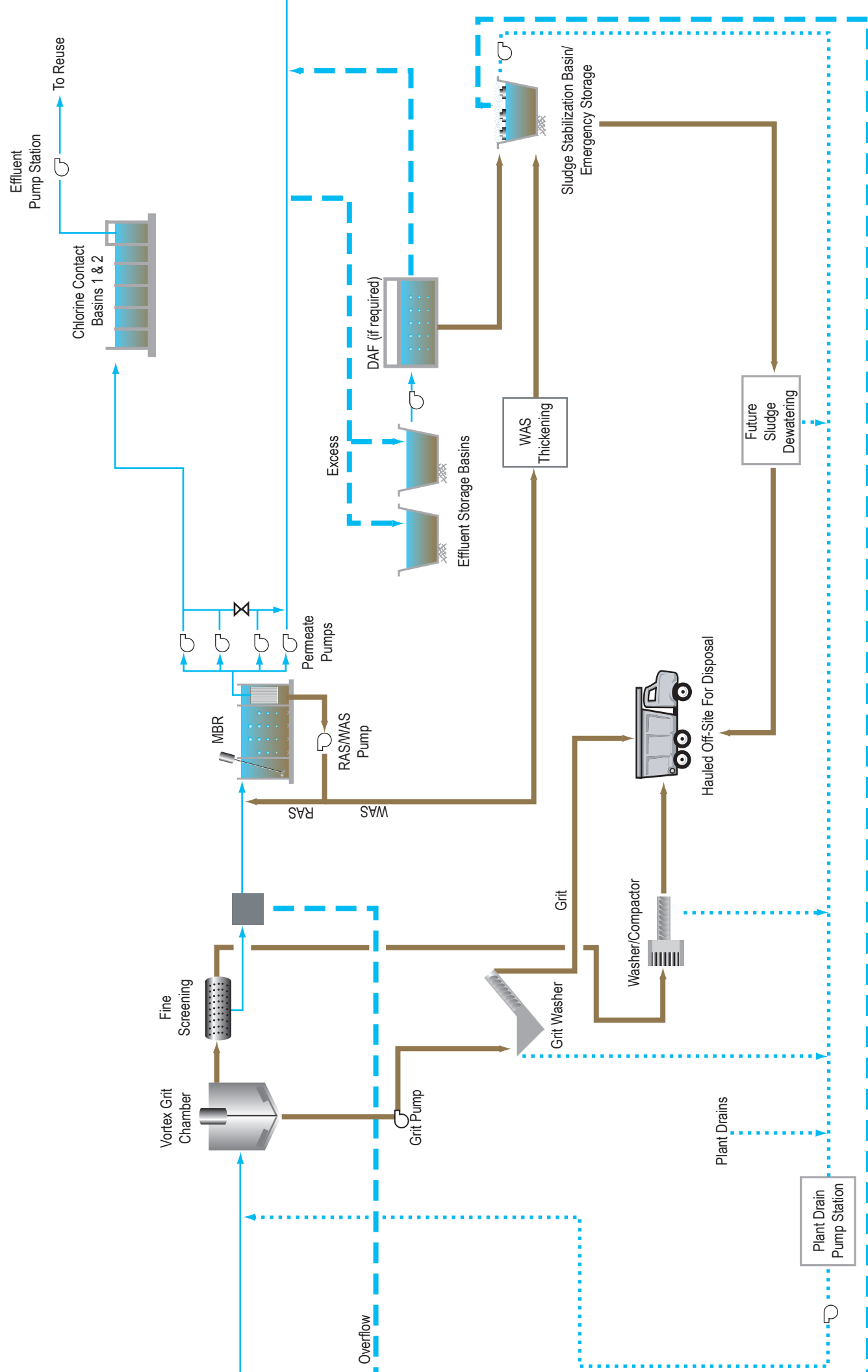
Parameter	Value
Design Flows	
ADWF (Average Dry Weather Flow)	4.5 mgd (3125 gpm)
PWWF (Peak Wet Weather Flow)	5.0 mgd (3470 gpm)
ADF (Average Design Flow)	5.0 mgd (3470 gpm) (nominal plant capacity)
MDF (Maximum Day Flow)	6.0 mgd (4165 gpm)
PHF (Peak Hour Flow)	10.0 mgd (6940 gpm)
Peak Instantaneous Flow	15.0 mgd (10400 gpm)
Treatment process	
Secondary and tertiary	Immersed Membrane Bioreactor (MBR)
Expected effluent quality	
BOD	<10 mg/L
TSS	<10 mg/L
Turbidity	<0.2 NTU daily average
Nitrate	<5 mg/L
Expected influent quality	
BOD	350 mg/L
TSS	400 mg/L
Ammonia	40 mg/L
TKN	50 mg/L
Nitrate	15 mg/L
Alkalinity	400 mg/L
TDS	1,200 mg/L
Temperature	16 degrees C
MBR process trains	
Number of process trains	4 train biological w/ 4 MBR tanks
Process train basins	De-oxygenation, anoxic, aeration, post-anoxic, membrane (all basins concrete)
Design SRT	20 days
Membrane system flow (initially installed)	4.0 mgd (average) 4.8 mgd (peak day)
Membrane system flow (w/ added membranes)	5.0 mgd (average) 6.6 mgd (peak day)
Control system	PLC
MBR De-oxygenation Basins	
Purpose	Reduce dissolved oxygen of recirculated mixed liquor
Quantity	1 per train (4 total)
Volume	75,400 gallons per basin (301,600 gallons total)
Dimensions	14 ft x 30 ft per basin
Operating water depth	24 ft
Mixed Liquor Suspended Solids	9,900 mg/L
HRT	1.45 hours (@ ADF 5.0 ,mgd)

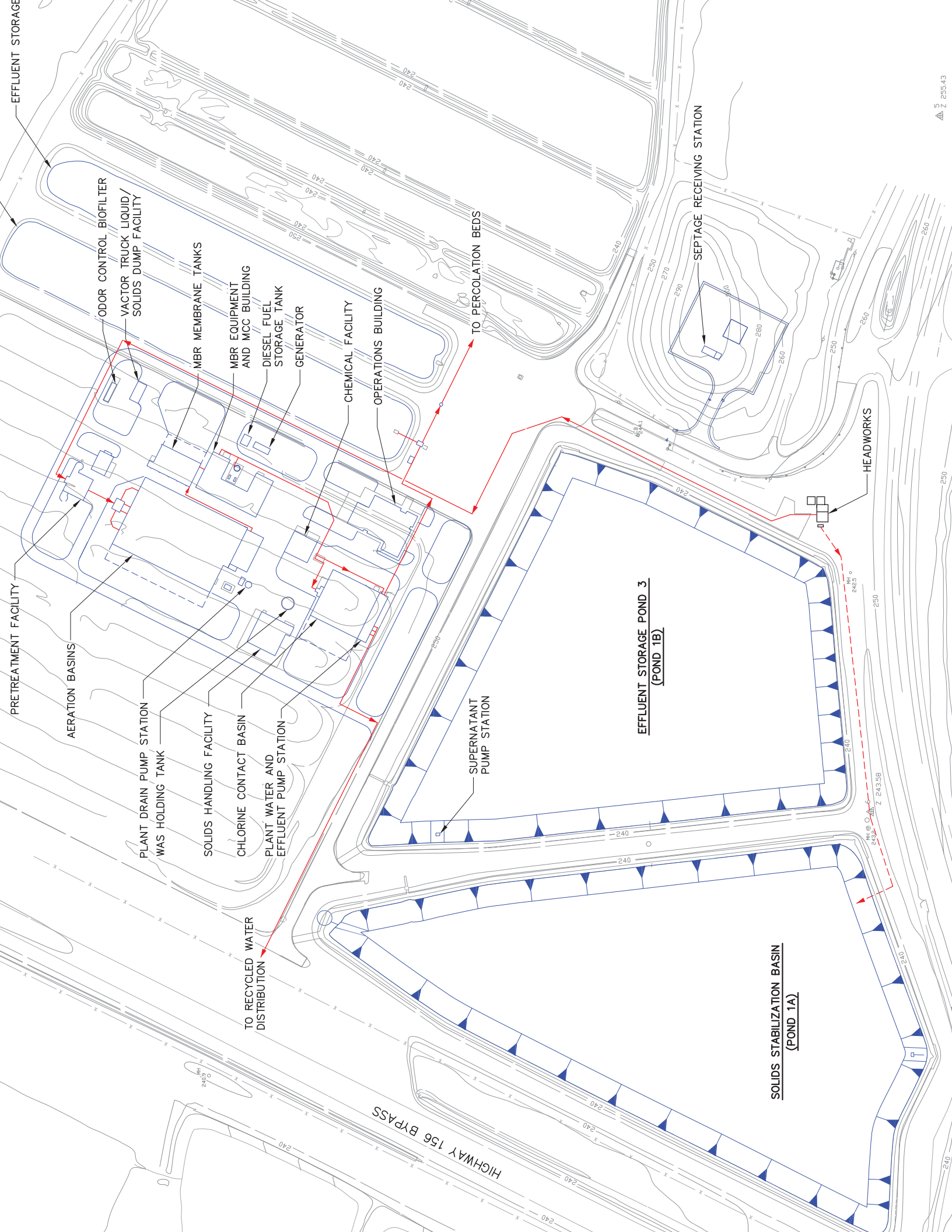
Parameter	Value
Anoxic basins	
Quantity	2 per train; 8 total anoxic basins
Volume	72,260 gallons per basin (578,080 gallons total)
Dimensions	14 ft x 30 ft per basin
Operating water level	23 ft
Mixed Liquor Suspended Solids	8,500 mg/L
HRT	2.77 hours (@ ADF 5.0 mgd)
Post-Anoxic basins	
Quantity	1 per train; 4 total post-anoxic basins
Volume	70,690 gallons per basin (282,760 gallons total)
Dimensions	15 ft x 30 ft per basin
Operating water level	21 ft
Mixed Liquor Suspended Solids	8,500 mg/L
HRT	1.35 hours (@ ADF 5.0 mgd)
Anoxic mixer	
Type	Submersible
Quantity	1 per anoxic basin (12 total)
Capacity	8 hp/ Mixer
Mixer power	96 hp total
Aeration basins	
Quantity	4 per train, 16 total
Volume	129,600 gallons per basin (2,073,600 gallons total)
Dimensions	27.5 ft x 30 ft x 21 ft SWD per basin
Operating water level	21 ft
Mixed Liquor Suspended Solids	8,500 mg/L
HRT	9.95 hours (@ ADF of 5.0 mgd)
SRT	20 days
Aerator type	Fine bubble diffusers
Process air flow required	1,459 scfm per basin (5838 scfm total)
Process air blowers	
Type	High speed turbocompressor
Quantity	4 total (3 duty, 1 standby)
Control	Continuous operation VFD drive
Capacity	2,774 scfm each (total 8,322 scfm)
Discharge pressure	11.6 psig
Motor power	150 hp each, 600 hp total
Membrane tank	
Quantity	4
Volume	45,930 gallons per basin (183,720 gallons total)
Dimensions	10 ft x 63 ft per basin
Operating water level	9.67 ft
Mixed Liquor Suspended Solids	9,900 mg/L
HRT	0.9 hours (@ ADF of 5.0 mgd)
Membrane aeration (scour air) blowers	
Type	High speed turbocompressors
Quantity	5 total (4 duty, 1 standby)
Control	Continuous operation
Capacity	3,461 scfm each (total 13,844 scfm)
Discharge pressure	5.1 psig
Motor power	100 hp each, 500 hp total

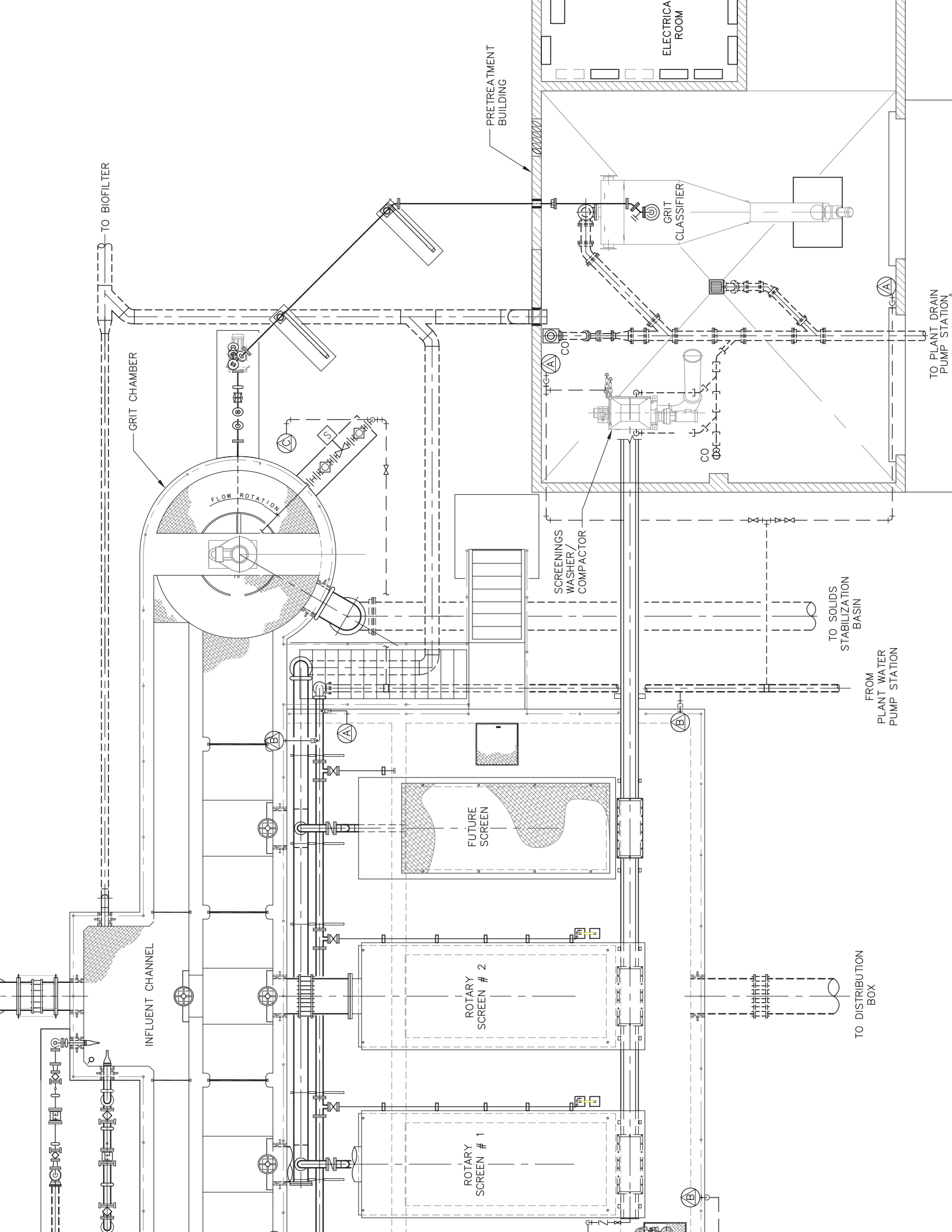
Parameter	Value
Membrane modules	
Type	Hollow fiber
Quantity of modules	280 per membrane train (1,120 total) (4.0 mgd membranes installed initially) Future 352 modules/train (1,408 total)
Quantity of cassettes (@ 48 modules per cassette, max)	6 cassettes per process train (30 total)
Spare space allowed (future cassettes)	3 spaces per train (10 total)
Pore size	0.04 micron
Surface area	340 square feet per module (380,800 square feet total) Future (478,420 square feet total)
Flux at average day flow	10.5 gfd (gallons/sq ft/day) (Future 10.44 gfd)
Flux at peak day flow	12.6 gfd (Future 12.5 gfd)
Permeate pumps	
Quantity	4 (4 duty, 1 standby shelf spare)
Type	Horizontal end suction centrifugal
Capacity	1,569 gpm each
TDH	39 ft
Motor power	25 hp each, 100 hp total
Control	VFD drive
Mixed liquor recirculation pumps	
Quantity	5 (4 duty, 1 spare)
Type	Axial flow pumps
Capacity	4,861 gpm each
TDH	10.5 ft
Motor Power	40 hp each, 160 hp total
Control	VFD drive
Backwash System	
Backwash pump quantity	2 (1 duty, 1 standby)
Capacity	2,040 gpm @ 27 ft TDH
Motor Power	20 hp each, 40 hp total
Tank type	Polyethylene
Tank quantity	1
Tank working capacity	6,340 gallons
Tank diameter	8.66 ft
Tank side wall height	16 ft
Maintenance clean hypochlorite design dose	500 mg/L
Chemical feed pump type	Air operated double diaphragm
Pump quantity	2
Hypochlorite solution strength	5% - 12% trade strength
Metering pump range	0-10,200 L/hr
Control strategy	Manual adjustment based on solution strength
Staging tank	
Tank type	Concrete
Tank quantity	1
Tank working capacity	4,000 gal
Tank dimensions	7 ft x 9 ft
Tank side wall height	9.5 ft

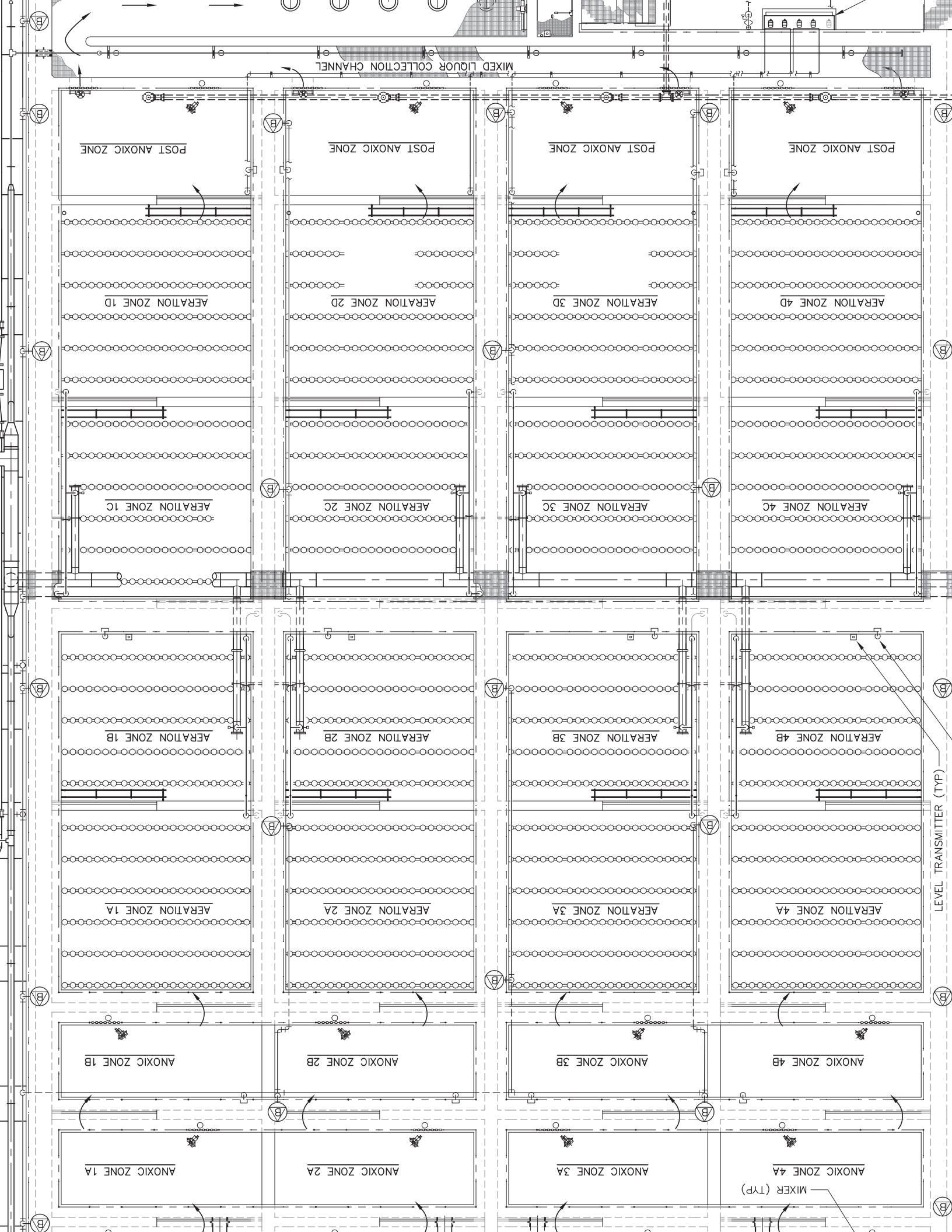
Parameter	Value
Chlorine contact basin	
Number of basins	2
Channel length	435 ft (per channel)
Channel width	8 ft (per channel)
Channel depth	9 ft
Number of passes	5 per basin
L/W ratio	54:1
Contact time	135 min @ ADF, 112 min @ MDF
Chlorine mixer	Submersible induction, 2 hp
Hypochlorite storage and handling facilities	
Storage tank type	High density cross-link polyethylene
Quantity tanks	1 (1 additional future)
Storage volume	8,000 gallons (nominal)
Tank diameter	12 ft
Tank sidewall height	11 ft
Hypochlorite solution strength	12.5% trade strength
Hypochlorite design dose	20 mg/L
Hypochlorite feed rate	42.3 gph (@ 5.0 mgd, 12.5% solution) (recycle)
Metering pump type	Peristaltic pump with variable speed input (4-20mA)
Quantity pumps	2 total (1 duty, 1 standby)
Metering pump range	0.43 – 51.6 gph
Control strategy	ORP/flow paced with chlorine residual backup
Purpose	Pump recycled water for plant water
Plant water pumping facilities	
Type	Vertical turbine
Quantity pumps	3 (2 duty, 1 standby)
Capacity	200 gpm @ 190 ft TDH each
Motor power	30 hp
Control	Variable speed control to maintain system pressure, with level switch turnoff
Flow measurement	Magnetic flow meter on pump station discharge line
Plant drain pump station	
Purpose	Collect process drainage and onsite sanitary sewerage and pump to pretreatment
Type	Submersible sewage pumps
Capacity	350 gpm @ 50' TDH each
Quantity	3 (2 duty, 1 standby)
Wet well	10 ft diameter
Depth	25 ft
Recycled Water Pump station	
Purpose	Pump effluent to offsite reuse areas
Type	Vertical turbine pumps
Capacity	1,750 gpm @ 260' TDH each
Quantity	3 (2 duty, 1 standby)
Wet well	10 ft x 30 ft x 15 ft
IWTP pump station (Low Head Recycled Water Pumps)	
Purpose	Pump effluent to offsite reuse and IWTP
Type	Vertical turbine pumps
Capacity	1,400 gpm @ 60' TDH each
Quantity	2 (1 duty, 1 standby)
Wet well	10 ft x 19 ft x 15 ft

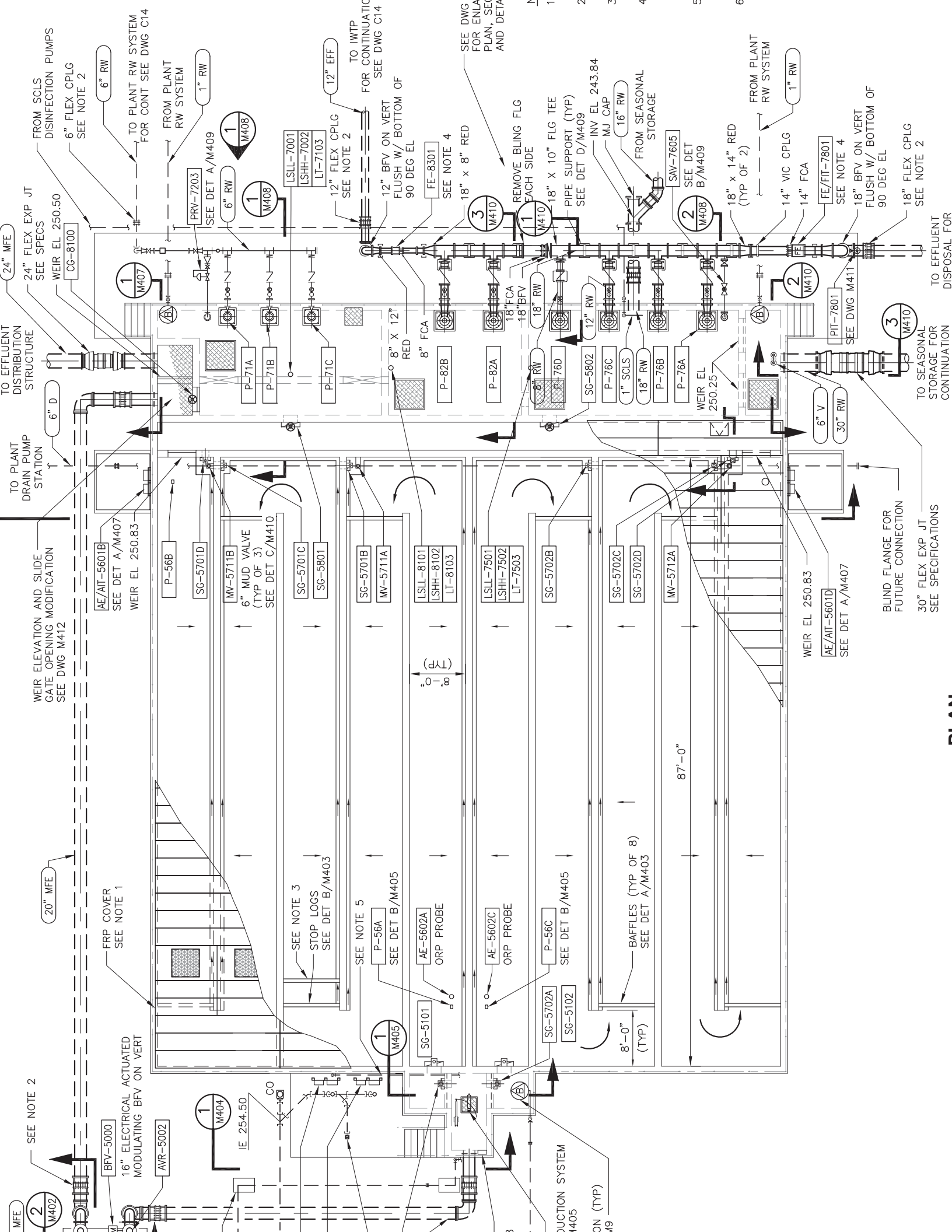
Parameter	Value
Recycled Water Storage	
Quantity	3 (Convert existing Ponds 3A, 3B and 1 B)
Volume	2-6 million gallons each, 1-44 million gallons
Inlet	Gravity overflow from effluent pump station/flow split structure
Outlet	Gravity to existing pump station

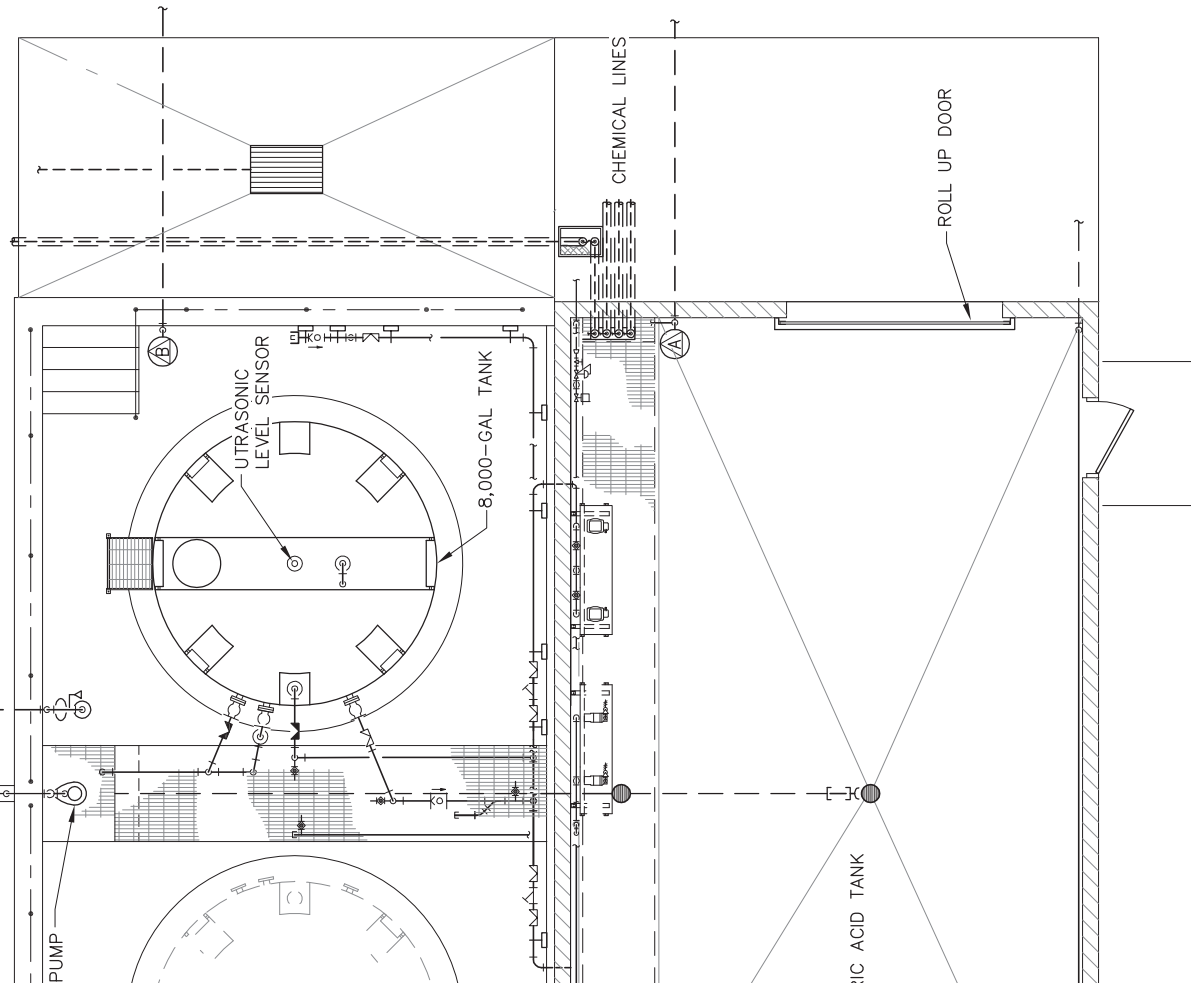




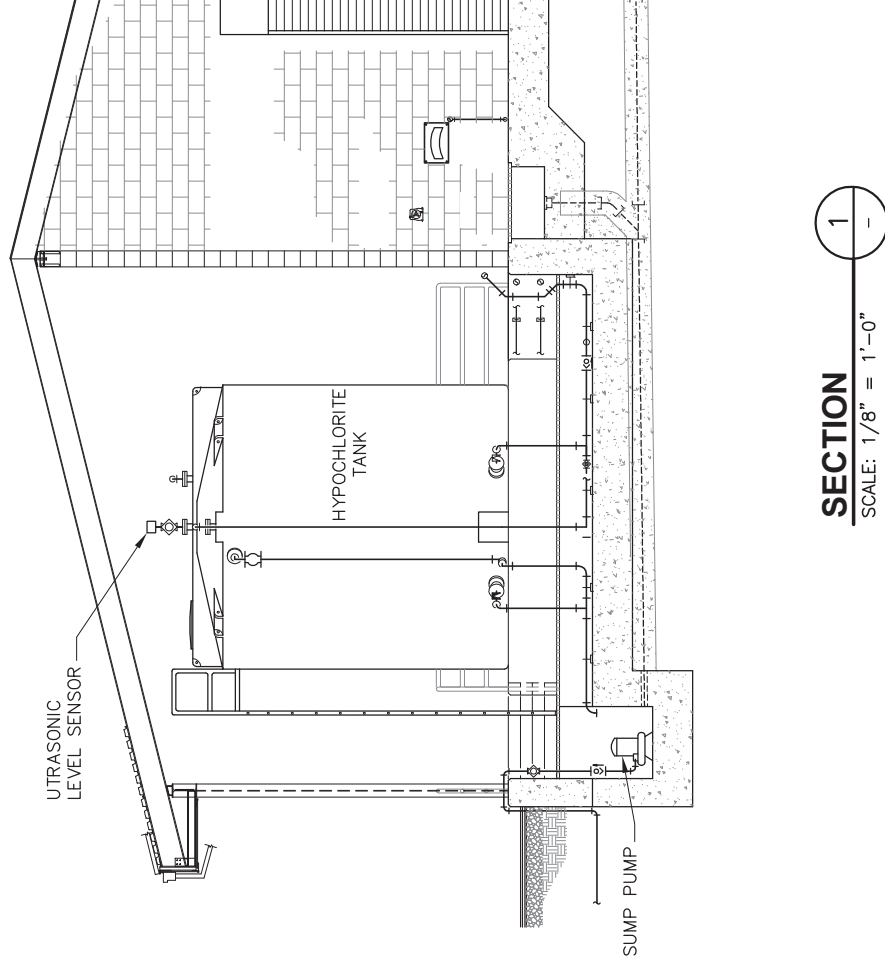








PLAN
SCALE: 1/8" = 1'-0"



SECTION
SCALE: 1/8" = 1'-0"

2.5.1 Pretreatment Facilities

Raw wastewater will be pumped from the existing influent lift station to a new plant pretreatment facility. The new pretreatment facility will consist of a grit chamber followed by fine screens. Figure 2-3 shows the plan layout for the pretreatment facilities.

2.5.2 Membrane Bioreactor

The MBR combines a suspended growth biological reactor for activated sludge treatment with submerged membranes for solids separation. The membranes are contained within the biological tanks and replace the clarifiers used in conventional activated sludge plants. The MBR process consists of an anoxic zone for denitrification, an aeration zone for soluble BOD reduction and nitrification, a post-anoxic zone for further denitrification and a membrane zone for solids separation. The process is designed with an SRT of 20 days, which should be adequate for nitrification/denitrification at the assumed minimum temperature of 16°C.

Each biological train is rated at 1.25 mgd. Based on the temperature monitoring it may be possible to re-rate the process trains up to a capacity of 1.5 mgd each. Expansion can be accomplished by adding biological trains and membrane trains. The design flow of 5.0 mgd meets the projected growth to the current planning horizon. Additional space for membranes will be provided in each membrane tank. It is important to note that the initial design has allowed for only enough membranes to handle 4.0 mgd ADF conditions. The City of Hollister determined that the full 5.0 mgd membrane capacity will not be required during the first five years of the Project (2008-2013). A decision to purchase the additional membranes at a later date was made as a cost savings measure.

2.5.2.1 MBR Influent Distribution

Screened wastewater will flow by gravity from the fine screens to the MBR influent distribution structure that will be used to distribute the wastewater to the MBR process trains. This structure will be covered with checkered plates for odor control. Air from the MBR influent distribution structure will be treated in the odor control biofilter.

The flow will be evenly distributed to each process train by providing a downward acting weir gate for each process train. The weir gates will be manually adjusted to provide a proper flow split to each train. The process trains could be isolated by raising or closing the weir gate. The influent will flow through pipes to each individual process train. The MBR influent distribution structure will be furnished with an emergency overflow weir. This weir will overflow in emergency conditions only. Any overflow will be diverted to the sludge stabilization basin.

2.5.2.2 Deoxygenation Zone

A deoxygenation (deox) zone is located upstream of the anoxic zone. This deox zone preconditions the recirculated mixed liquor from the MBR tanks. The MBR tanks add oxygen to the mixed liquor. Because of the strict nitrate limit anticipated, the deox zone was added to reduce the dissolved oxygen recirculated to the anoxic zone. Dissolved oxygen is reduced in the deox zone through endogenous respiration. The deox zone is sized to provide a hydraulic retention time (HRT) of 1.45 hours at a nominal plant flow rate of 5.0 mgd.

2.5.2.3 Anoxic Zone

The next treatment stage of the MBR process is the anoxic zone, which is operated as a completely mixed basin without any aeration. The dissolved oxygen in the anoxic basin will be purposely kept low, typically in the 0.1-0.2 mg/L range, to promote the oxidation of dissolved nitrates into nitrogen gas which is removed to the atmosphere, a process called denitrification. Mixing of the anoxic basins will be accomplished with submersible mixers.

MBR influent wastewater will be transferred in pipes to the anoxic basins. Recycled mixed liquor from the deox zone, which is rich in nitrates, will be mixed with the influent wastewater in the anoxic zone, thereby providing a carbon source for the denitrifying bacteria. This mixture of influent and recirculated mixed liquor along with the low dissolved oxygen level promotes the growth of the denitrifying bacteria. This denitrification process is capable of reducing the nitrate level to <5 mg/L nitrates (as N) to meet Basin Plan requirements.

The anoxic basin will be divided into two sections to reduce short-circuiting. The anoxic basin has a HRT of approximately 2.76 hours at 5.0 mgd. A recycle rate of approximately 6Q (six times the average influent flow rate) will be required to reduce the nitrates to the design level. The anoxic zones at the plant will be designed with four parallel trains to allow for continued operation when one basin is removed from service for maintenance.

2.5.2.4 Aeration Zone

The next stage of the MBR system is the aeration zone. Wastewater will flow by gravity from the anoxic basin to the aeration basin. The aeration zone will be equipped with fine bubble diffusers for mixing and oxygen transfer. The aeration zone is typically operated with a dissolved oxygen level of 2-3 mg/L. The aeration zone establishes an environment to promote a suspended biological growth that breaks down the soluble BOD by converting it into cellular biomass. The aeration zone is also designed to promote growth of nitrifying bacteria, which oxidize ammonia into nitrates – a process called nitrification.

Four aeration trains are designed to provide the ability to continue processing wastewater with one train out of service. The aeration basin will be designed with a HRT of approximately 10.3 hours at the nominal flow rate of 5.0 mgd. The aeration zone will operate with a mixed liquor suspended solids (MLSS) concentration of approximately 8,500 mg/L. Fine bubble, membrane disk diffusers will provide process air and mixing power to the aeration zone. Aeration air will be provided to the system by high-speed turbocompressors, which will be located in a blower building.

Three variable frequency drive (VFD) driven process blowers will be provided for the biological process requirements. One standby blower will be provided.

2.5.2.5 Post-Anoxic Zone

The aeration basins overflow into post-anoxic basins, which are designed to further enhance the denitrification process. The post-anoxic basins have a HRT of approximately 1.3 hours at 5.0 mgd. Each basin is equipped with a submersible mixer. A source of supplemental carbon, either methanol or ethanol, may be required during certain flow and loading conditions. An aboveground, fuel-type storage tank and gear pumps are used for storage and feed of the methanol. The denitrification process is highly dependent on wastewater temperatures and nitrogen loading. The supplemental carbon will probably not be required on a year round basis.

The flow exits the post-anoxic basin through slide gates into the mixed liquor collection channel. The mixed liquor is pumped from this channel to the membrane distribution channel. Six axial flow pumps (5 duty, 1 standby) will be used to provide a flow up to 6Q (influent flow). The mixed liquor recirculation pumps will be VFD driven for flow control. The pumps feed a 48-inch header pipe to transport the mixed liquor to the MBR tanks. A 40-inch magnetic flow meter will be provided to measure and control the recirculation rate.

2.5.2.6 Membrane Zone

The immersed membranes are located in a separate zone for each process train. The selected Zenon membrane configuration will consist of hollow fiber ultrafiltration membranes. These membranes have a nominal pore size of 0.04 microns. The process will have four membrane trains so that the plant flow could be maintained even with one membrane train out of service for cleaning or maintenance. The membrane basins will have inlet slide gates for isolation.

The immersed membranes separate the solids from the MBR effluent, called permeate. A slight vacuum will be applied to the permeate header by a centrifugal pump. Treated wastewater will then be drawn through the membrane and flow through the center of the membrane and through to connecting tubing to the permeate header. In turn, the permeate will flow through the pump to the downstream processes.

Ultrafiltration of the wastewater results in typical permeate turbidity values of less than 0.1 NTU. Operating data suggest that the membranes also reduce bacterial counts, potentially providing some degree of disinfection. Results at operating plants show permeate coliform counts typically less than 23 MPN/100-mL without additional disinfection, and often the MBR permeate coliform totals are less than 2.2 MPN/100-mL.

The mixed liquor in the membrane zone will be agitated by a coarse bubble diffuser system. The membrane agitation air provides a scouring velocity along the membranes required to prevent a buildup of solids at the surface of the membranes. Four (4) separate high-speed turbocompressors, which will be located in the blower building, will supply the membrane agitation air.

To maintain performance, the membranes require a “rest” period where flow through the membranes is stopped allowing the membranes to relax. This rest cycle is fully automated and is typically done on a cycle of 15-20 minutes on and 2 minutes of rest.

The membranes may need to be cleaned periodically through backwashing to prevent plugging or fouling of the membrane surfaces, which increases the transmembrane pressure through the membranes resulting in diminished permeate production. The method of backwashing is operator selected and is fully automated. Backwashing requires the use of permeate containing a high chlorine concentration to be pumped back through the membranes on a regular basis. Sodium hypochlorite will be used to dose stored permeate in a backwash tank. This solution will then be pumped back through the membranes in the reverse direction to the permeate flow. This backwash operation will occur approximately once or twice every day. Once or twice a year the membrane cassettes will be taken off-line and undergo a clean-in-place cycle. The clean-in-place begins by draining the membrane tank. The membranes are then backwashed with a chlorine solution, which fills the membrane tank. The membranes are allowed to soak. After soaking, the membrane tank is drained and refilled with mixed liquor and the membranes are placed back into service.

Solids will be retained in the membrane tank and will be continuously recycled back to the anoxic zone to maintain the biomass concentration in the system and also to undergo denitrification. This recycle flow is low in ammonia but high in nitrates. The nitrates in turn provide the denitrifying bacteria in the anoxic zone with an alternate oxygen source to promote denitrification of the wastewater. The recirculated mixed liquor from the membrane basins is typically high in dissolved oxygen. Since this dissolved oxygen will suppress the denitrification process it must be lowered. The deoxygenation basin receives the mixed liquor prior to the anoxic basin. The dissolved oxygen is lowered in the deoxygenation basin through endogenous respiration. The mixed liquor is kept in suspension by submersible mixers provided for each basin. From the deoxygenation basins, the mixed liquor flows by gravity into the anoxic basins.

2.5.2.7 MBR Process Equipment Area

The MBR process equipment, which includes the permeate pumps, backpulse pumps, backpulse tank, valves, flow meters and associated controls and instrumentation, will be located in a building adjacent to the MBR basins. This building is described in a later section.

2.5.3 Disinfection

Permeate (effluent) sent to the percolation beds for disposal will not be disinfected. The ultrafiltration membranes remove a large portion of the bacteria present. Coliform levels in the MBR effluent are typically less than 23 MPN/100mL. The EPA Design Manual for Land Treatment of Municipal Wastewater states that wastewater applied to rapid infiltration or percolation beds does not require

disinfection. Maintenance of Basin Plan standards for groundwater coliform should not require disinfection, and avoiding disinfection will minimize disinfection by products in the discharge.

Disinfection of the portion of the MBR effluent used for recycled water will be accomplished by chlorination. The chlorination system and contact basins will be designed to meet Title 22 disinfected tertiary recycled water requirements. With respect to Title 22, two conditions must be met:

1. The filtered wastewater has been disinfected by either:
 - (i) A chlorine disinfection process following filtration that provides a CT (the product of total chlorine residual and modal contact time measured at the same point) value of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow; or
 - (ii) A disinfection process that, when combined with the filtration process, has been demonstrated to inactivate and/or remove 99.999% of the plaque-forming units of F-specific bacteriophage MS2, or polio virus in the wastewater. A virus that is at least as resistant to disinfection as polio virus may be used for the purposes of the demonstration.
2. The median concentration of total coliform bacteria measured in the disinfected effluent does not exceed an MPN of 2.2 per 100-mL utilizing the bacteriological results of the last seven days for which analyses have been completed and the number of total coliform bacteria does not exceed an MPN of 23 per 100-mL in more than one sample in any 30 day period. No sample shall exceed an MPN of 240 total coliform bacteria per 100-mL.

2.5.3.1 Chlorine Contact Basins (CCB)

Permeate (effluent) for recycled water production will flow to the CCB, which will include a rapid-mix chamber, an inlet distribution channel, two chlorine contact basins, and an effluent wetwell. The CCB is shown in Figure 2-5. Permeate can be split between the two CCBs, or a portion of the flow can also bypass the CCB if necessary. The wetwells can be operated separately or in combination, because the divider wall in the wetwell will have an overflow weir and an interconnecting sluice gate.

Permeate will enter the rapid mix structure where sodium hypochlorite will be added through a polyvinyl chloride (PVC) pipe header. This structure will be equipped with an induction type chemical flash mixer. Flow will enter the mixing chamber through a pipe penetration. Sodium hypochlorite will then be injected into the flow stream through the mixer. The chemical flash mixer will evenly disperse the hypochlorite. The submersible induction mixer will be fabricated of stainless steel and will be mounted below the CCB cover. An access hatch will be provided for removal of the unit for maintenance.

After chlorine injection and mixing, the flow will continue through the CCB influent distribution channel. This channel will be equipped with slide gates to direct the flow to either one or both of the CCBs. The CCB will consist of a concrete basin with concrete baffle walls used to create a serpentine flow pattern. Each basin will be 8 feet wide with a side water depth of 9 feet and 2 feet of freeboard above the normal water level.

The CCBs each have five passes to create the serpentine flow and are further configured with a bypass pipe around the basin. The length of each pass is 87 feet. This layout with both basins in service will provide a minimum hydraulic detention time of 120 minutes at 5.5 mgd. With only one basin in service, the detention time for 2.8 mgd will be 120 minutes and 90 minutes for 3.75 mgd. The length to width ratio of each CCB is 54:1.

The CCB floors are constructed with a cross slope to a drainage channel that will allow the basin to be drained for maintenance. The water surface level in the CCB will be set by the use of a fixed weir on the outlet of each basin. Water flows over the fixed outlet weir into an effluent pump station wetwell for distribution to disposal or reuse. The contact basin will be covered to prevent degradation of chlorine

residual by sunlight and to prevent algae growth. The basin cover material is fiberglass. Access hatches are provided for maintenance and cleaning.

Sampling pumps will be used to withdraw samples from the CCB to the chlorine residual. The chlorine residual will be monitored at the influent end of the basin. The influent chlorine residual will be monitored to insure adequate chlorine has been added and will also be used to detect a chlorination system failure. The effluent chlorine residual will be monitored to insure compliance with Title 22 CT requirements for recycled water disinfection. The chlorine residual analyzers are mounted adjacent to the CCB. Signals from the analyzers are incorporated into the plant control system and Supervisory Control and Data Acquisition (SCADA) system. These signals are used to generate alarms to warn the operators of chlorination system failures or malfunctions.

Future expansion could be accomplished with the addition of a new CCB, chlorine injection systems, and chlorine analyzers.

2.5.3.2 Chemical Feed and Storage

A building for the chemical feed systems and chemical storage area will be required. A layout plan and section illustration is shown in Figure 2-6. Sodium hypochlorite will be used for the MBR backpulse and membrane cleaning operation as well as for effluent disinfection. The MBR manufacturer also requires the use of citric acid for membrane cleaning because of the hardness of the water. Additionally, methanol or ethanol will be required for supplemental carbon addition in the post-anoxic zone.

Chemical metering pumps will be provided for the effluent disinfection. Programmable logic controllers (PLCs) will control the speed of the pumps based on the CCB influent flow and the chlorine residual detected in the CCB. The pumps will be housed inside a concrete block building. The building will be heated to prevent freezing and equipped with ventilation fans to provide fresh air and minimal heat rejection. No air conditioning will be required for this building.

Sodium hypochlorite will be stored in a high-density, linear polyethylene (HDLPE) chemical storage tank. The tanks will be sized to provide approximately 8 days of storage at peak daily flow chlorine dosage. The storage area will be located outdoor under a canopy. An emergency spill containment area will be provided. The emergency spill containment area will also be included at the chemical metering facilities. A chemical loading station will be provided for chemical truck delivery. Emergency eyewash and shower will be provided inside the building and in the storage area. Hose bibs and hose racks for washdown will also be provided.

Methanol or ethanol will be stored in a double contained aboveground steel storage tank. Since methanol and ethanol are flammable, the storage and chemical pump area will be designated as a Class I, Division 2 area per NFPA 70 (National Electric Code). This will require special construction of electrical components to reduce the risk of fire and explosion. A containment area for the chemical pumps will be provided. All methanol/ethanol feed piping will be double contained. The methanol/ethanol tank will store approximately 3,000 gallons of chemical. The feed pumps will be magnetically driven gear pumps with explosion proof variable speed drives.

Manuals and training for the operation of the membrane bioreactor have been provided by the manufacturer (Zenon).

2.6 Plant Reliability Features

Sections 60333 through 60355 (Articles 8 through 10) of the Water Recycling Criteria define reliability requirements for wastewater treatment plants producing recycled water. The features that are incorporated at this plant to comply with these requirements are described below.

Article 8, §60333. Flexibility of Design

The plant has been designed to provide for efficient operation and maintenance and to permit the highest possible degree of treatment under varying circumstances. There are four process trains, each rated at 1.25 mgd. With additional temperature data it may be possible to re-rate these process basins to 1.5 mgd each. There are four membrane trains, each rated at 1 mgd as originally configured, expandable to 1.25 mgd per train with the addition of more membranes. The facility can treat the current ADWF of 2.6 mgd with one process train and one membrane train out of service, as initially configured.

The MBR influent distribution structure will be designed to overflow to the sludge storage basin (SSB) in case of emergency. A minimum of 16 million gallons of storage capacity will be available in the SSB for emergency conditions, providing the ability to divert and store more than 3 days of continuous average daily flow (at the design flow rate of 5.0 mgd) or more than 6 days of continuous average daily flow (at the current ADWF of 2.6 mgd). A decant pump station will be designed to allow pumping of the SSB contents back to the pretreatment facility for processing through the plant.

Table 2-3, which details the DWTP design parameters, also identifies the equipment incorporated in the unit process and the redundant elements in the process design. For example, the process air blowers, membrane air blowers and membrane permeate pumps all require one unit per process train (two in total), however, a third standby unit standby is provided to help assure plant reliability, flexibility and efficiency of operation.

Article 8, §60335. Alarms

The DWTP has a SCADA system that enables full monitoring and control of the plant. The SCADA system, which has battery power back-up, will provide the annunciation of the alarms required pursuant to section 60335, that is, the sounding of alarms for the following events:

- the loss of normal power
- failure of biological treatment process
- failure of a disinfection process
- failure of a filtration process

The existing SCADA system is Wonderware based and monitors and controls the operations of a variety of remote sites. Communication between the remote sites and the SCADA system is radio based and the controls at each site are PLC based using Tesco L2000 and Liq 4 PLCs.

The existing SCADA system consists of two sets of computers. The two SCADA locations are connected together by means of a Microwave Wide Area Network Link. The SCADA system utilizes Wonderware software and is currently set up to communicate with the Tesco L2000 and Liq 4 PLCs. As part of the new plant, the existing set of computers located at the City Yard would be relocated to the new plant Operations Building.

The plant upgrades will include instrumentation and control systems to allow the plant operators the ability to monitor and control the operations associated with the new plant processes, as well, as provide for new SCADA-based control and monitoring of portions of the existing interim plant that will be incorporated into the new plant such as the influent and effluent pump stations.

The plant upgrades will consist of two major control systems – the packaged MBR control system and the general plant facilities control system. There will be an MBR Main Control Panel and a Plant Main Control Panel. Each system will be PLC based and each system will include an operator interface panel that allows an operator to monitor and control the associated plant processes. The two main control panels will be situated next to each other and will allow an operator to view operations associated with each system simultaneously from the Operator Interface Panels located in each of the Main Control Panels.

In order to maintain the existing communication protocols currently in place at the City, the Plant Main Control Panel will utilize a new Tesco L2000 PLC with a Wonderware based Operator Interface Panel. The new Tesco L2000 PLC will be the plant's primary connection point to the existing City-wide SCADA system. The MBR Main Control Panel will be required to provide a PLC capable of Modicon Modbus communication, preferably a Modicon PLC and will also be equipped with a Wonderware based Operator Interface Panel. This would allow for a Modicon Modbus communication network that would provide for a connection for the existing Influent Pump Station PLC, the existing Effluent Pump Station PLC, the new MBR PLC and any other new PLCs associated with any pre-packaged control systems such as the fine screens. The Modbus communication network would be then connected to the new Plant Tesco L2000 PLC. This configuration would provide the City's SCADA system direct access to all of the various plant PLCs.

With an existing city-wide Wonderware based SCADA system, the new Operator Interface Panels should also utilize Wonderware based software. The Operator Interface Panels would allow for local operation of the new plant independent from the city-wide SCADA system but would also simplify the integration of the Operator Interface Panel Displays into the City-wide SCADA system. With this approach, plant operations would then be performed locally at the operator interface panels or from any of the existing SCADA workstations. The existing SCADA system would be updated to monitor and control the new plant processes and allow the existing SCADA system to handle data logging, data monitoring and report generation for the new plant.

Currently the DWTP is staffed 9 hours per day during the week and checked once per day on the weekend. An operator is on call the rest of the time. It is expected that during start-up the new MBR plant may be staffed 16-24 hours per day, but will transition to 8 to 12 hours per day as the system is debugged and operating experience is gained. A certified operator will be on-call whenever it is not staffed, aided by an extensive SCADA system with phone dial out and remote computer access. The initial contact person for DWTP alarms will be Dennis Rose, City of Hollister Wastewater Supervisor, at telephone 831-524-1880.

Article 8, §60337. Power Supply

The DWTP has a loss of power supply alarm and a standby power source, as required under 22 CCR §60337(a). Backup power for the entire WWTP is supplied by two generators (2500 kW and 1500 kW) which are located in the electrical yard area southeast of the MBR building. The backup power supply operates automatically when power is lost for a preset amount of time.

Article 9, §60339. Primary Treatment

This section of the Code is not applicable to the Hollister DWTP because it applies to facilities that produce primary effluent for reuse. The Hollister DWTP will produce only disinfected tertiary treated effluent for reuse. Primary and secondary treatment both occur in the MBR tanks.

Article 10, §60341. Emergency Storage or Disposal

The MBR influent distribution structure will be designed to overflow to the SSB in case of emergency. A minimum of 16 million gallons of storage capacity will be available in the SSB for emergency conditions, providing the ability to divert and store more than 3 days of continuous average daily flow (at the design flow rate of 5.0 mgd) or more than 6 days of continuous average daily flow (at the current ADWF of 2.6 mgd). A decant pump station will be designed to allow pumping of the SSB contents back to the pretreatment facility for processing through the plant.

All of the pumping facilities are connected to the standby power source for the facility. All of the necessary sensors, instruments, and valves to enable fully automatic diversion of untreated or partially treated wastewater to the emergency storage facility have been installed.

Article 10, §60343. Primary Treatment

The provisions of §60343 require multiple primary treatment units, or standby primary units, or long term storage. Primary treatment is combined with secondary treatment and filtration in an MBR treatment plant. The design provides multiple primary/secondary treatment units by virtue of providing four MBR process trains. The DWTP has sufficient capacity for at least one MBR process train to remain out of service, in standby mode, at all influent flow loadings up to the rated 5.0 mgd design flow.

Article 10, §60345. Biological Treatment

Primary treatment is combined with secondary (biological) treatment in an MBR treatment plant. The design provides multiple biological treatment units by virtue of providing four process trains and four membrane trains. The WWTP has sufficient capacity for at least one MBR process train to be taken out of service, at all influent flow loadings up to the rated 5.0 mgd design flow. The biological treatment process is monitored with alarms described under Article 8, §60335, above.

Article 10, §60347. Secondary Sedimentation

Secondary sedimentation is not required in an MBR facility because the microfiltration membranes separate the solids from the treated effluent.

Article 10, §60349. Coagulation

The Code requirements for reliability and redundancy features for the uninterrupted feed of coagulant are not applicable to the DWTP because coagulant is not used in the membrane microfiltration process. Coagulation facilities are not required for microfiltration processes because the small pore size of the membranes provides an absolute barrier to solids particles larger than an approximate size of 0.04 microns.

Article 10, §60351. Filtration

Primary treatment is combined with secondary treatment and filtration in an MBR treatment plant. The reliability features for filtration required under this section of the Code are met by the same design features cited in §60343 and §60345 above. The facility provides four process trains. As originally outfitted with an initial membrane capacity of 1 mgd per train, only three process trains will be required to treat the current average daily dry weather flow of 2.6 mgd. The MBR trains are sized to produce the peak day flow with one MBR train out of service. Together with the alarms described under §60335, above, the requirements of §60351 are met. In addition, the design provides short term retention (storage) in the SSB adequate for more than 6 days of influent at the current ADWF of 2.6 mgd.

Article 10, §60353. Disinfection

The DWTP will use liquid sodium hypochlorite for chlorine disinfection of the MBR permeate to produce Disinfected Tertiary Recycled Water. The two CCBs are designed to provide the following reliability features for the uninterrupted disinfection feed:

- Standby chemical feed pumps
- Automatic residual control of disinfection dosage
- On-line continuous automatic measuring and recording of chlorine residual
- Alarms

Sampling pumps will be used to withdraw samples from the CCB to the chlorine residual/ oxidation-reduction potential (ORP) monitoring equipment. The chlorine residual and ORP will be monitored at the influent end of the basin. The influent chlorine residual and ORP will be monitored to insure adequate chlorine has been added and will also be used to detect a chlorination system failure. The effluent chlorine residual will be monitored to insure compliance with Title 22 CT requirements for recycled water disinfection. The chlorine residual analyzers will be mounted adjacent to the CCB. Signals from the

analyzers will be incorporated into the plant control system and SCADA system. These signals will be used to generate alarms to warn the operators of chlorination system failures or malfunctions.

2.7 Supplemental Water Supply

The Hollister DWTP includes recycled water storage that will provided supplemental water during periods when recycled water is not being produced. The available water in storage will depend on the time of year. Future recycled water customers may also have backup groundwater supplies or potable water backup. The City's rules and regulation, construction inspection program, cross connection testing program, and user self monitoring reporting are targeted at preventing cross connections from occurring. The initial use areas (Airport and Brigantino Park) will not have any onsite supplemental supplies.

Any supplemental water sources will be protected with double check valves, reduced pressure principle backflow devices, or by air gap connections, as required under Sections 7601-7604 of Title 17 of the California Code of Regulations. Specific sources of supplemental water supply and the methods of cross-connection control will vary for each recycled water use area and are to be further defined in use area applications and plans that will be submitted to DPH for review.

2.8 Monitoring and Reporting

Monitoring and reporting for the treatment facility and distribution system will be performed by the City in compliance with Title 22 regulations, Waste Discharge Requirement (WDR), and the MRP. Required testing will be performed by the City's lab.

The major instruments at the DWTP associated with recycled water production and compliance are shown in Table 2-4 along with location, sampling frequency, and calibration/maintenance frequency. The location for these instruments ensures that any non-spec water is detected prior to entry into the recycled water facilities, thereby protecting the integrity of the recycled water store or in the system. This approach will allow the use of stored recycled water while the cause of off-spec water is corrected.

Sampling for total coliform concentrations will be accomplished through daily grab samples at the end of the chlorine contact basin per Title 22 CCR §60343.

Table 2-4: Instrumentation List for Recycled Water Production

Instrument	Location	Sampling Frequency	Calibration or Maintenance Frequency
Magnetic Flow Meter	Influent Pump Station	Continuous	Annually
Turbidity Meters	Membrane Permeate	Continuous	Daily ¹
Magnetic Flow Meter	Membrane Permeate Line	Continuous	Annually
Chlorine Analyzer	Chlorine Basin Influent	Continuous	Daily ¹
Chlorine Analyzer	Chlorine Basin End	Continuous	Daily ¹
Magnetic Flow Meter	Distribution Pipeline	Continuous	Annually
Level indication	RW Pump Station Wet Well	Continuous	Annually
Refrigerated Composite Sampler	Chlorine Basin Effluent	Flow Based	Per vendor specification

Footnotes:

1. Turbidity meters and chlorine analyzers are checked daily and calibrated as needed.

Distribution and use site monitoring will include cross connection testing, user self monitor reporting, and user reporting on facility modification logs.

2.9 Contingency Plan

Section 60323 of the California Code of Regulations requires that this report contain a contingency plan designed to prevent untreated or inadequately treated wastewater from being delivered to the use area(s). As discussed below, the monitoring system together with the SCADA, alarm system and automated control valves will prevent inadequately treated wastewater from leaving the plant and being delivered to the recycled water use areas.

Section 60323 requires a list of conditions that would require an immediate diversion of the recycled water to take place. The following conditions would indicate the possibility that the plant is producing inadequately treated effluent. The occurrence of any of the following conditions would result in shutting down the MBR permeate pumps and the effluent pumps, closing the CCB influent flow and diverting undisinfected permeate to the onsite percolation beds:

- If the turbidity of the filtered water exceeds 0.5 NTU;
- If the calculated CT falls below 450 mg-minutes/L;
- If an effluent total coliform bacteria result in any sample is greater than an MPN of 23 per 100 mL, or if the 7-day median concentration of total coliform is greater than an MPN of 2.2 per 100 mL; (this is not an automatic function)
- If the chlorine residual falls below 0.5 mg/L or rises above 8.0 mg/L; or
- If complete failure of a unit process causes the treatment process to not meet the process train requirements of the Water Recycling Criteria. (this is not an automatic function)

In all cases where a shutdown is required, an investigation will be conducted to determine the cause of the incident, and the recycled water will not be redirected to the equalization basin until the problem has been corrected and the effluent conditions have returned to acceptable levels.

Section 60323 requires a plan for the disposal or treatment of inadequately treated effluent and a description of diversion procedures. In cases of inadequate treatment, diversion is accomplished by the MBR influent distribution structure, which is designed to overflow to the SSB in case of emergency. A minimum of 16 million gallons of storage capacity will be available in the SSB for diversion of influent raw wastewater. A decant pump station pumps the SSB supernatant back to the pretreatment facility for processing through the plant, when diversion conditions are remedied.

Chapter 3 Recycled Water Transmission and Distribution

This chapter is intended to provide the information required by Section 3.0 in the *Guidelines for the Preparation of an Engineering Report* (DHS, March 2001). The sections below provide an overview of the system transmission and distribution system. Detailed plans of the recycled water pipelines to be constructed are included in Appendix D.

3.1 Recycled Water Transmission and Distribution System Overview

The Project includes approximately 9 miles of distribution pipelines as seen in Figure 3-1 and will be supplied by a recycled water distribution pump station located at the DWTP. The distribution pump station consists of four vertical turbine pumps in a wet well that discharges into the distribution header pipe. The pump station wet well can be fed directly from the chlorine contact basin or from the DWTP recycled water storage pond.

In its initial phase, the Project will have two separate recycled water use sites (Airport and Brigantino Park). Additional information on the recycled water use sites can be found in Appendix C. Transmission pipeline sizes vary from 12 to 20 inches. The system has a total delivery capacity of about 3,300 gallons per minute (gpm).

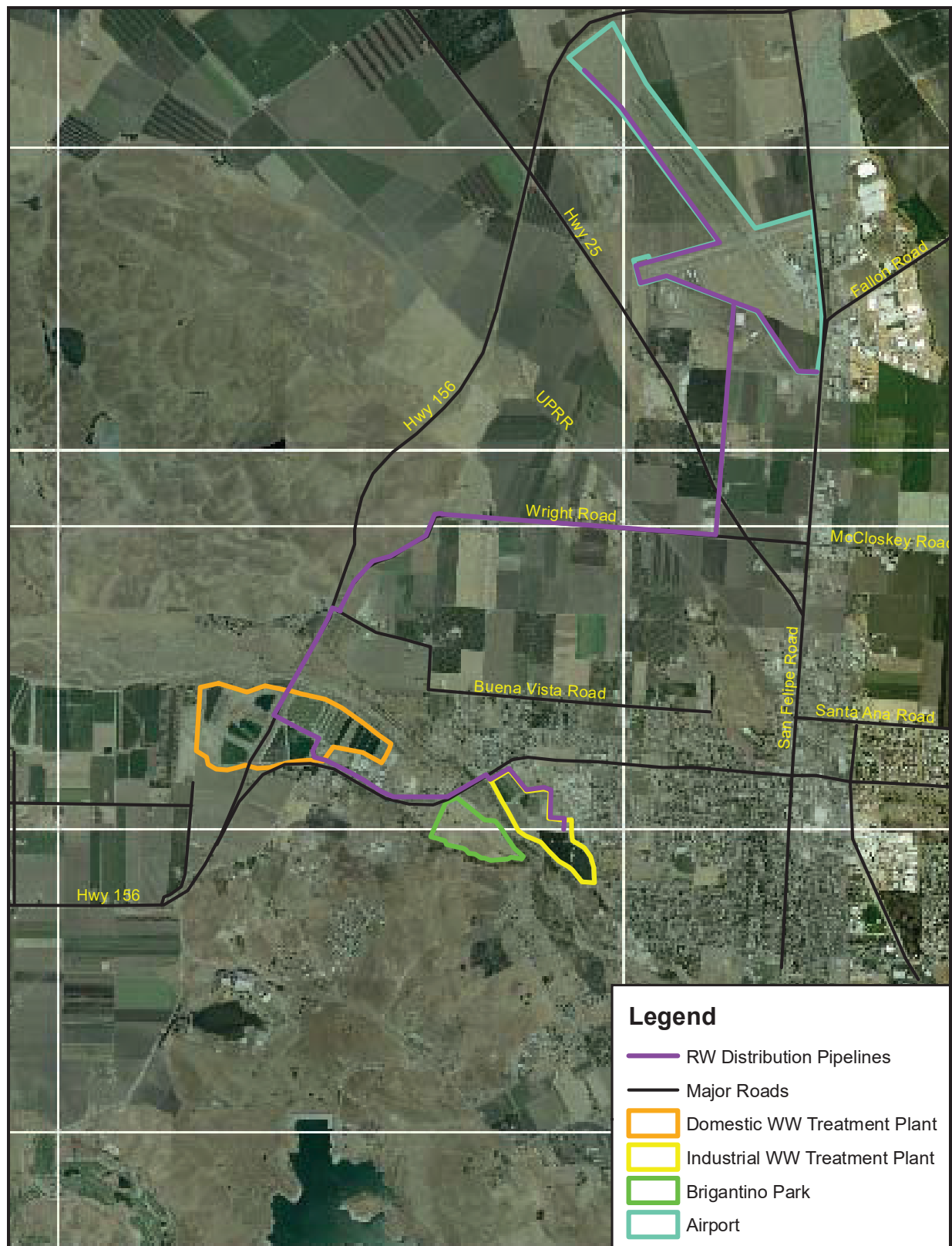
Future recycled water customers along the alignments could be served off of the recycled water transmission pipelines through new laterals and/or turnouts. Additional transmission mains are also being considered to serve other areas.

All proposed pipelines and appurtenances are being designed and will be installed in accordance with guidelines and regulations pertaining to the distribution of recycled water in California including the following:

- Guidelines for Distribution of Non-potable Water from the California-Nevada Section – American Water Works Association (AWWA)
- Disinfected Tertiary Recycled Water Guidelines: On-site Facility Retrofitting from California-Nevada Section – AWWA
- California Waterworks Standards from CCR – Title 22
- Regulations Relating to Cross-connections from Title 17, Chapter 5, Group 4
- Manual of Cross-connections Control/Procedures and Practices from DHS

Appendix D includes plans of the transmission lines for the recycled water system with a table describing the separation requirements.

Figure 3-1: Map of Hollister Recycled Water Distribution System



3.2 Recycled Water Pipeline Design Criteria

The Project must comply with the requirements of Title 22. Project design criteria have been established and utilized in the design of the transmission and distribution system. Key design criteria are summarized in Table 3-1 and have been incorporated into the pipeline designs. These criteria will also be used in the design of any future connections. Compliance with this criteria is described in the table in Appendix D include with the recycled water pipeline drawings.

Table 3-1: Recycled Water Pipeline Design Criteria

Horizontal Separation for Parallel Construction	
Pipe Separation	Construction Requirements
Greater than 10 feet separation	No special construction required
Between 4 to 10 feet separation	Must meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • Welded steel • HDPE pipe with fusion welded joints • Fused PVC
Less than 4 feet separation	Not allowed
Vertical Separation at Crossings	
Pipe Separation	Construction Requirements
Less than 4 inches below Potable	Not allowed
4 to 12 inches below potable	A full standard pipe length must be centered below the crossing and must also meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • Welded steel • HDPE pipe with fusion welded joints • Fused PVC
12 inches or greater below potable	No special construction required
Less than 4 inches above potable	Not allowed
4 inches or greater above potable	A full standard pipe length must be centered over the crossing and must also meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • Welded steel • HDPE pipe with fusion welded joints • Fused PVC

Notes:

1. Guidance Memo No. 2003-02: Guidance Criteria for the Separation of Water Mains and Non-Potable Pipelines, Department of Health Services, October 16, 2003.

3.3 Recycled Water Pipeline Materials

Pipeline materials used for the recycled water distribution system include PVC, ductile iron (DI), and high density polyethylene (HDPE). HDPE or DI pipe is utilized for trenchless crossings within a steel casing. The requirements for pipe materials, as stipulated in the project specifications, are summarized in Table 3-2.

Table 3-2: Recycled Water Pipe Material Uses and Standards

Pipe Material	Uses	Fittings	Standards	Additional Requirements
PVC	Preferred material for pressure less than 150 psi	Ductile Iron	AWWA C900 and C905	Pipe shall be purple (Pantone No. 512) and colored at the time of extrusion.
Ductile Iron	Restrained joint lengths, trenchless crossing carrier pipe	Restrained Ductile Iron	AWWA C150 and C151	Pipe will have purple (Pantone No. 512) polyethylene vinyl wrap (AWWA C105).
HDPE	Trenchless crossing	Butt-Fused	AWWA C906	Pipe shall have purple stripe.

3.4 Recycled Water Pipeline Installation

Recycled water pipelines will be installed using open trench construction where possible. Where open trench construction is not possible (crossing of highways and railroad tracks), a trenchless construction method such as bore and jack will be used to install a casing, through which the recycled water pipeline is installed.

The contractor will be made responsible for verifying the location of all existing crossing and parallel (within 3 feet) utilities in advance of construction.

Care will be taken during installation to prevent cross-connections between the existing potable and the new recycled water systems.

3.5 Recycled Water Pipeline Identification

All recycled water pipeline facilities will be appropriately identified, following the AWWA California-Nevada Section's Guidelines for Distribution of Non-Potable Water (1992). As stipulated in the design construction contract documents, underground warning tape shall be placed 12 inches above buried pipe. The warning tape will be purple (Pantone 512) and read "Recycled Water – Do Not Drink – No Tomar." Detection wire shall also be installed for all non-metallic pipes to identify recycled water facilities.

3.6 Distribution System Valves and Appurtenances

Valve covers will have a recognizable inscription cast on the top surface or name plate which reads "Recycled Water – Do Not Drink – No Tomar." All exposed appurtenances will have a warning tag reading "Recycled Water – Do Not Drink – No Tomar."

Chapter 4 Recycled Water Use Areas

This chapter is intended to provide information on the City of Hollister Recycled Water Use Program (Program) to meet the requirements of Section 4.0 in the *Guidelines for the Preparation of an Engineering Report* (DHS, March 2001). The Program is tailored to establish rules and requirements for users to meet Title 22 for use areas. This Program shall facilitate issuance of a MRP from the RWQCB. Detailed information on the initial recycled water users (Hollister Airport and Brigantino Park) are included in Appendix C. This chapter includes the following information:

- Overview of recycled water program,
- Planned recycled water uses,
- Use area design,
- Inspection and monitoring, and
- Employee training.

4.1 Recycled Water Use Area Program Overview

The City will be implementing the Program to incorporate users into the recycled water project. The Program includes various materials that will help ensure that the Regulations of Title 22 are complied with at all use sites. The City's Program materials include:

- Recycled Water Use Manual and Rules of Service (Appendix A)
- User Application for Recycled Water Service
- Recycled Water Plan Review Checklist
- Recycled Water Standard Details
- Construction Inspection Checklist
- Cross Connection Certification
- Recycled Water User Permit
- Recycled Water Annual Self Monitoring Report
- Site Modification Log

Copies of the draft Program materials are included in Appendix B.

4.2 General Recycled Water Use Information

The City of Hollister will be the responsible party for the distribution system up to the service connections (i.e. turnout, water meter, etc.) to the Customer/User. The application process for recycled water service and use includes an application where users will identify the following use area information and characteristics:

- Type of land use
- Type of reuse proposed
- Customer name and designated onsite user supervisor
- Use area containment measures
- Map showing areas of use, public access, surrounding land, location and details of wells, location and type of signage
- Description of employee or public access

- Where potable and recycled water lines exist, a description of cross connection control procedures

This information shall be supplemented with additional information as required for specific uses described in the subsequent sections below.

The City will review applications for completeness and conformance with recycled water requirements. Following City review, the City shall then submit the application package to the DPH for final review of the use site.

4.3 Irrigation

The long-term vision for the City is to use recycled water for a variety of irrigation uses as allowed per Section 60304 of the California Code of Regulations Title 22. Near-term use of recycled water would only be for golf course, park, and airport irrigation. The list of long-term irrigation uses includes:

- Food crops, including all edible root crops, where the recycled water comes into contact with the edible portion of the crop
- Parks and playgrounds
- School yards
- Unrestricted and restricted access golf courses
- Cemeteries
- Freeway landscaping
- Ornamental nursery stock and sod farms where access by the general public is not restricted
- Pasture for animals producing milk for human consumption
- Non-edible vegetation with controlled access
- Orchards, vineyards, non-food bearing trees

Irrigation use sites will be required to submit an application and use area plans to the City for review prior to receiving recycled water service. The City's application for recycled water service (see Appendix B) requires information from the irrigation user including:

- Description of what will be irrigated
- Method of irrigation (e.g. spray, flood, drip)
- Location of domestic water supply facilities in or adjacent to use area
- Description of site drainage
- List of drinking water fountains, bleachers, and designated outdoor eating areas
- Proposed schedule for irrigation
- Public warning signs location and wording (if different from the City standard sign)

All irrigation use sites will be required to protect against ponding of recycled water and overspray of recycled water onto public use facilities.

4.4 Landscape Impoundments

The City is planning to utilize recycled water for landscape impoundments as allowed per Section 60305 of the California Code of Regulations Title 22.

Use sites using recycled water in a landscape impoundment will be required submit an application and use area plans, as well as, a procedure to assure that recycled water is not discharged from the impoundment due to seasonal storm events or the collection of runoff. This could include reducing the use of recycled

water to coincide with the start of the wet season. The City's application for recycled water service (see Appendix B) requires information from the landscape impoundment users including:

- Description of the degree of public access to the impoundment
- Description of site drainage

4.5 Other Planned Recycled Water Uses

There are other recycled water uses allowed per Sections 60304 through 60307 of the California Code of Regulations Title 22. The following uses are planned as part of the Hollister recycled water:

- Nonstructural fire fighting
- Backfill consolidation around non-potable piping
- Soil compaction
- Mixing concrete
- Dust control on roads and streets
- Cleaning roads, sidewalks and outdoor work areas
- Flushing sanitary sewers

Use sites using recycled water for one of the uses listed above will be required to submit an application and use area plans to the City for review prior to receiving recycled water service.

4.6 Use Area Design

This section describes how the recycled water use sites will be designed, operated, and maintained to comply with applicable recycled water use criteria and regulations.

4.6.1 Signage

All recycled water use sites that are accessible to the public will be posted with signs conforming to the requirements of Section 60310 (g) of the California Code of Regulations Title 22. All signs will be in English and Spanish. In order to notify non-readers and those who speak neither English nor Spanish, all signs will also include a graphic indicating that water is non-potable. Signs will be posted at all use areas that are accessible to the public and will be clearly visible to the public. Proposed project signage is shown in Appendix B as part of the recycled water standard details. Signs will be at least 4 inches high by 8 inches wide.

In addition to signs, new recycled water piping, equipment and appurtenances will be colored purple as described in **Error! Reference source not found. - Error! Reference source not found..**

4.6.2 Use Area Containment Measures, Drainage, and Ponding Minimization Measures

All recycled water use sites will comply with the City's recycled water use requirements as well as the requirements of Section 60310 of the California Code of Regulations Title 22. Specifically, all recycled water use sites will:

- Preclude irrigation with tertiary recycled water within 50 feet of any domestic water supply well, unless requirement in Section 60310 (a) of the California Code of Regulations allowing for closer irrigation are met;
- No impoundment of disinfected tertiary recycled water shall occur within 100 feet of any domestic water supply well;
- Confine irrigation runoff to the recycled water use areas, unless the runoff does not pose a public health threat and is authorized by the regulatory agency;
- Perform irrigation in a manner that minimizes ponding or runoff of recycled water;

- Ensure that no spray, mist, or runoff enters any dwellings, designated outdoor eating areas, or food handling facilities; and
- Ensure that all drinking water fountains are protected from contact with recycled water spray, mist, or runoff.

4.6.3 Minimizing Public Contact

It is expected that the use sites corresponding to the recycled water uses allowed under the City's Program will have varying degrees of public access. All drinking water fountains, designated outdoor eating areas, bleachers and other public gathering areas located at use sites will be protected against overspray of recycled water. Use sites with public access will also be restricted from having hose bibs on the piping system utilizing recycled water. As discussed in Section 4.6.1, all use sites will have signs posted that are clearly visible to the public and indicate that the irrigation water is non-potable.

4.6.4 Cross Connection Control

4.6.4.1 Distribution System

Because the recycled water system will be new, there are no cross-connections in the system. As described in Section 3.4, special care will be taken during installation to ensure that cross-connections between the potable and recycled water systems do not occur.

In general, cross-connection control monitoring will be the responsibility of the City. Monitoring will be mainly comprised of observation of the pipeline alignment to detect any illegal connections. Comparing customer meter records to the recycled water and City supply meters will also be done to detect flow imbalances, which could be an indication of a leak or illegal connection. Cross-connection monitoring activities will be reported as required in the MRP.

4.6.4.2 Recycled Water Use Site System

Inspection and testing for user site cross-connections will be the responsibility of the City and User Supervisors. The City's cross-connection control program for use site systems will include training of User Supervisors concerning the testing and monitoring process and the importance of preventing cross-connections. Cross-connection control measures for user sites are detailed in the City's RW Use Handbook. Cross-connection testing will be completed by a Cross-Connection Control Specialist certified through the California-Nevada Section of the AWWA. The Cross Connection Certification Form (see Appendix B) will be completed and filed by the Cross-Connection Control Specialist before beginning recycled water service.

4.7 Use Area Testing, Inspections, and Monitoring

4.7.1 Construction Inspections and Testing During Start-Up

The City and its contractors will be responsible for construction inspections to ensure that new construction meets the criteria for recycled water systems and design requirements for the Project. Following construction, a final inspection of new facilities will be conducted by the City and its contractors. The final inspection will make certain that the system is in compliance with the MRP, including requirements for system identification, signage, backflow installations, and site containment. As part of the final inspection process, the City will conduct cross-connection testing to verify system separation in accordance with all applicable guidelines. Where appropriate, the City will also conduct user site cross-connection tests to confirm separation of on-site domestic facilities and the recycled water system.

4.7.2 Ongoing Monitoring, Testing and Reporting Responsibilities

The City will be responsible for monitoring and reporting activities as outlined in the MRP and WDR. The City will regularly monitor use area conditions and will also rely on User Supervisors to notify the City of any violations of permit requirements. User Supervisors will be trained by the City concerning potential problems and measures for preventing problems.

In the event that a non-compliance event is discovered by the User Supervisors, they will immediately contact the City. If the event does not endanger public health or the environment, City staff will prepare an on-site observation report of the violation. In the event that a non-compliance event occurs that may endanger public health or the environment, the City will verbally notify the RWQCB and DPH within 24 hours. Upon RWQCB or CDPH staff request, a written report will be submitted and will contain a description of the non-compliance event, its cause, and any corrective actions taken.

The City's testing and inspection activities as part of the recycled water program will include:

- Cross-connection tests, performed every five years
- Annual testing of backflow assemblies
- On-site observation reports
- Tracking of pumped flow and total flow used at turnouts
- Tracking of distribution system changes

4.8 Training of Distributor and Users

4.8.1.1 Training for City of Hollister Staff

The Recycled Water Supervisor identified by the City (see Section 2.3.2) will be trained on recycled water use criteria and on implementation procedures for the programs outlined in the MRP. In addition, at least one City staff person will be certified in cross-connection control to carry out the City's cross connection control program (see Section 4.6.4).

4.8.1.2 Training for Use Area Employees

The City will offer quarterly training to the User Supervisors (see Section 2.3.3). Each site receiving recycled water must have a User Supervisor certified through the City's training program. Training shall include:

- Procedures for handling and use of recycled water;
- Rules and regulations stipulated in the MRP and in the City's RW Use Handbook; and
- Basic cross-connection and backflow principles and procedures.

User Supervisors will be responsible for training other site employees on a continual basis.

References

American Water Works Association. *Guidelines for Distribution of Nonpotable Water*. California-Nevada Section of the American Water Works Association. 1992.

American Water Works Association. *Guidelines for On-Site Retrofit of Facilities Using Disinfected Tertiary Recycled Water*. California-Nevada Section of the American Water Works Association. 1997.

California Department of Health Services. *Guidance Memo No. 2003-02: Guidance Criteria for the Separation of Water Mains and Non-Potable Pipelines*. 2003 October 16.

California Department of Health Services. *Guidelines for the Preparation of an Engineering Report for the Production, Distribution and Use of Recycled Water*. March 2001.

California Department of Health Services. *Manual of Cross-connections Control/Procedures and Practices*.

California Department of Health Services. *The Purple Book*. June 2001.

Regulations Relating to Cross-connections from Title 17, Chapter 5, Group 4

Appendix A - Draft Recycled Water Use Manual and Rule of Service Handbook



Recycled Water Use Manual and Rules of Service

Draft Report

Prepared by:
RMC
Water and Environment

February 2008

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List of Abbreviations

AWWA	American Water Work Association
DPH	Department of Public Health

Chapter 1 Introduction

The City of Hollister's Recycled Water Use Manual and Rules of Service is intended to:

- Assist new customers through the application, design and construction process in order to receive a Recycled Water Use Permit
- Serve as a reference source for customers and their Site Supervisors regarding the proper operation and maintenance of their recycled water system

In order to receive and use recycled water from the City, customers must follow the rules as set forth in this document.

This document was written to be in compliance with the existing codes, laws, statues and regulations of the State of California and local governing bodies concerning the currently-approved use of recycled water. It is not intended to void any of these regulations. The manual is also not intended to void standard industry practices as presented in the American Water Work Association (AWWA) California-Nevada Section's *Guidelines for Distribution of Nonpotable Water* or *Guidelines for the On-site Retrofit of Facilities using Disinfected Tertiary Recycled Water*. However, since legal and regulatory requirements can change without the knowledge of the City, the City assumes no liability for errors in this manual. If any section, subsection, clause or phrase of the rules and regulations presented in this document is determined to be invalid, the remaining portions of these rules and regulations shall remain in effect.

Amendments to this document may be made periodically to comply with State regulations or to meet new City standards. If amendments are made, existing users will be notified of the new rules of service as well as the effective date for compliance with the new rules.

Questions regarding the information presented herein should be directed to the City's Recycled Water Supervisor. The Recycled Water Supervisor oversees all programs and facilities related to the use of recycled water distribution by the City and will coordinate with the appropriate City departments to resolve customer inquiries. Table 1-1 provides contact information for the current Recycled Water Supervisor as well as an emergency contact number.

Table 1-1: City Contacts

Contact	Mailing Address	Phone	Fax
Dennis Rose	375 Fifth Street Hollister, CA 95023	(831) 524-1880	(831) 636-4349
<u>Emergency:</u>		(XXX) XXX-XXXX	

Chapter 2 Obtaining a Recycled Water Use Permit

2.1 Approved Uses

Each recycled water customer must have a permit from the City prior to receiving recycled water. The State of California regulates the use of recycled water, as directed under the California Code of Regulations, Titles 17 and 22. Local authorities, at their discretion, can require or specify what sites and/or uses of recycled water are to be utilized in their service area so long as it complies with State requirements.

In the City of Hollister's service area, recycled water may be used for the following applications, all of which have been approved by the California Department of Public Health (DPH) under the City's Master Permit (Order No. XXXXX):

- Irrigation
 - Food crops, including all edible root crops, where the recycled water comes into contact with the edible portion of the crop
 - Parks and playgrounds
 - School yards
 - Unrestricted and restricted access golf courses
 - Cemeteries
 - Freeway landscaping
 - Ornamental nursery stock and sod farms where access by the general public is not restricted
 - Pasture for animals producing milk for human consumption
 - Non-edible vegetation with controlled access
 - Orchards, vineyards, non-food bearing trees
- Landscape Impoundment
- Nonstructural Fire Fighting
- Other Uses
 - Backfill consolidation around non-potable piping
 - Soil compaction
 - Mixing concrete
 - Dust control on roads and streets
 - Cleaning roads, sidewalks and outdoor work areas
 - Flushing sanitary sewers

The City reserves the right to disallow a proposed use on the grounds of safety, public health, technical feasibility or other concerns, for which the City's judgment shall be final. At its discretion, the City may also set forth site specific requirements as conditions for the permitted use.

2.2 Application

The first step in obtaining a Recycled Water Use Permit is to submit an Application for Recycled Water Service.¹ Copies of the application form are available from the City upon request. A completed application form will include:

- **Property Information:** This should include a brief description of the property, including the type of land uses within the proposed recycled water use area.
- **Applicant:** Applicants must either be the owner or an authorized representative of the property to be served.
- **Site Owner:** If the site owner is the same as the applicant, this section can be left blank.
- **Developer/Design Engineer Contact:** If applicable, provide contact of person that can respond to technical questions on the proposed recycled water uses and site development.
- **User Supervisor:** Each recycled water use site owner/operator will appoint a User Supervisor to be the main point of contact with the City regarding recycled water use. The User Supervisors will be responsible for ensuring compliance with the programs and procedures for the use of recycled water, as outlined in the Recycled Water Use Permit and the *Recycled Water Use Manual and Rules of Service*. The User Supervisor must be certified through the City's training program; if the User Supervisor has not received certification at the time of application, the certification number can be left blank; however the User Supervisor must be certified in order for the site to receive a Recycled Water Use Permit. In addition to the standard contact information, an emergency phone number shall be provided for the User Supervisor. The User Supervisor must be reachable at all times in case of an emergency.
- **Proposed Recycled Water Use:** Recycled water is approved for irrigation, landscape impoundments, fire suppression and other uses listed in Section 2.1.
- **Recycled Water Demand Estimates:** The customer's best estimate should be provided to assist the City in evaluating the adequacy of the distribution system.
- **Use Area Containment Measures:** For irrigation projects this could include irrigating at agronomic rates to avoid runoff, constructing slopes to maintain drainage on-site, and/or installing containment beams. For landscape impoundments, water levels should be reduced when heading into the wet season to prevent rain induced overflows.
- **Employee or Public Access:** Potential access by employees or the public should be described.
- **Plans:** The plans should show the proposed recycled water use areas and areas to be excluded; recycled water infrastructure including the location and size of service connections, meters and backflow devices; a brief description of all special construction requirements, if applicable; areas of public access; surrounding land uses; construction details of wells in or near the use area; and location and type of signage that will be used. The design information presented in Section 2.3 should be considered in development of the plans.
- **Cross-Connection Control Procedures:** Where both potable and recycled water lines exist, a description of cross-connection control procedures shall be submitted.
- **Additional Attachments:** Other site specific information may be required depending on the proposed recycled water use:
 - Location of domestic water supply facilities in or adjacent to use area (for irrigation projects)

¹The Recycled Water Use Permit designates approval for recycled water service. Additional permits may be necessary for the construction of the recycled water facilities. The customer is responsible for applying for and obtaining other permits as may be necessary for the construction of the recycled water facilities.

- Description of Site Drainage (for irrigation and landscape impoundment projects)
- List of outdoor drinking fountains, bleachers, and designated outdoor eating areas (for irrigation projects)
- Signatures: The application shall be signed by the appointed User Supervisor as well as the owner or an authorized representative of the property to be served.

Upon receipt of a completed application, the City will review the information submitted to determine whether the proposed use conforms to the City's Rules of Service and whether the City has the necessary capacity to serve the applicant. If revisions to the proposed use are required or if additional information is needed, the City will contact the applicant and request that the necessary corrections be made or that the supplemental information be submitted. If the application meets the City's standards, the City will then forward a copy of the plans to DPH for review and approval. Following approval by DPH, the City shall approve use of recycled water per the applicant's plans.

2.3 Design Requirements

A summary of the basic design requirements for customers' facilities is provided below:

- No Cross-Connections: No cross-connections are allowed between the recycled water system and the potable water system.
- Backflow Prevention:
 - Potable System: In order to protect the public drinking water system from accidental cross-connections, a reduced pressure principal backflow prevention device is required at all meters that supply potable water to a site where recycled water is present. Additionally, for sites where the potable water is used as a back-up to the recycled water system, there must be an air gap separation between the two systems.
 - Non-Potable Wells: For agricultural wells that are connected to the customer's recycled water distribution system, a double check valve backflow prevention device is required on the well.
 - Recycled Water System: In most cases, backflow prevention devices will not be required on the recycled water service. Backflow prevention would be required on the customer's recycled water system if the City determines that there is a condition on-site that would threaten the integrity of the recycled water distribution system. Examples of potential threats to the recycled water distribution system are chemical fertilizer injection systems that are directly installed on the irrigation system and recycled water impoundments. In these cases a reduced pressure principal backflow prevention device shall be upstream of the potential threat.
- Pipe Separation: The customer's recycled water system shall be installed to meet the following separation requirements from all potable water pipelines.
 - Horizontal Pipe Separation: All recycled water service laterals and meters should be at least 10 feet from the nearest potable water facility, including pipelines, meters and hydrants. If a ten-foot parallel spacing is not practical, a separation of at least 4 feet may be allowed subject to the special construction conditions outlined in Table 2-1. Under no circumstances is horizontal separation of less than four feet or construction in the same trench as potable water facilities allowed.
 - Vertical Pipe Separation: Where a buried constant-pressure recycled water pipeline crosses a buried potable water pipeline, it should be a minimum of 4 inches below the potable water pipeline. Constant-pressure recycled water pipelines are allowed over potable water pipelines with a minimum of 4 inches vertical separation if a full standard pipe length is centered over the crossing or the recycled water pipeline is installed in a

pipe sleeve which extends a minimum of 10 feet on either side of the potable water piping. The centering of a full standard pipe length over the crossing guarantees that the recycled water pipe crosses the potable water line at a point that is halfway between subsequent joints on the recycled water line; this ensures that joints on the recycled water line are located as far as possible from the potable water line. Intermittently pressurized irrigation laterals may be located a minimum of 4 inches above potable water pipelines without sleeving.

Table 2-1: Pipe Separation Requirements

Horizontal Separation for Parallel Construction	
Pipe Separation	Construction Requirements
Greater than 10 feet separation	No special construction required
Between 4 to 10 feet separation	Must meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • Welded steel • HDPE pipe with fusion welded joints • Fused PVC
Less than 4 feet separation	Not allowed
Vertical Separation at Crossings	
Pipe Separation	Construction Requirements
Less than 4 inches below Potable	Not allowed
4 to 12 inches below potable	A full standard pipe length must be centered below the crossing and must also meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • Welded steel • HDPE pipe with fusion welded joints • Fused PVC
12 inches or greater below potable	No special construction required
Less than 4 inches above potable	Not allowed
4 inches or greater above potable	A full standard pipe length must be centered over the crossing and must also meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • Welded steel • HDPE pipe with fusion welded joints • Fused PVC

- **Pipe Depth** – The customer’s on-site recycled water piping must be designed to provide a minimum of 12-36 inches of cover from top of the pipe to finished pavement. The minimum cover depends on whether the pipes are under constant or intermittent pressure and the pipe size. These minimum cover requirements are summarized in Table 2-2.

Table 2-2: Pipe Depth Requirements

Pipe Type	Minimum Cover (in)
Intermittent Pressure, All Sizes	12
Constant Pressure, 2.5" diameter and smaller	18
Constant Pressure, 3-10" diameter	24
Constant Pressure, 12" diameter and larger	36

- **Pipe Identification:**
 - New piping: All new recycled water piping below or above grade shall be purple colored pipe with the words "CAUTION — RECYCLED WATER" printed on the pipe or on marking tape. Buried recycled water piping shall be laid so that the wording faces upwards.
 - Existing piping: All existing piping above grade must be labeled with purple identification tape with the imprinted words "CAUTION — RECYCLED WATER," or replaced by purple colored pipe with the required wording. Additionally, where feasible existing piping which is uncovered during construction should be marked with the words "CAUTION — RECYCLED WATER."
- **Recycled Water Warning Tags, Stickers, Labels:** All recycled water appurtenances including valve boxes, quick couplers, recycled water storage tanks, air/vacuum relief valves, pressure reducing valves, pumps, backflow prevention devices and system controller boxes must be labeled with purple, weatherproof warning tags, stickers or other labels. The tags, stickers or labels must include the words "WARNING – RECYCLED WATER — DO NOT DRINK" or similar wording and "AVISO — AGUA IMPURA — NO TOMAR."
- **Potable Water Identification Tags, Stickers, Labels:** At sites where both recycled water and potable water systems are used, all potable water meter and above grade water devices including backflow prevention devices and hose bibs must be labeled with blue, weatherproof identification tags, stickers or other labels. The tags, stickers or labels must include the words "POTABLE WATER" and "AQUA PARA TOMAR."
- **Signage:** Advisory signs must be posted within the recycled water use area in locations where they can easily be seen. Lettering on the sign must be at least ½-inch tall and include the words "RECYCLED WATER" and "DO NOT DRINK – NO BEBER." At a minimum signs should be posted at entrances to the site and around outdoor eating areas. Signs should be spaced no further than 1,000 feet apart.
- **Quick Coupling Valves:** Quick coupling valves used on the recycled water system must be made specifically for recycled water use. The covers on all quick coupling valves must be permanently attached and made of purple rubber or vinyl with the words "RECYCLED WATER" imprinted on the locking cover.
- **No Hose Bibbs:** Hose bibbs are not allowed on the recycled water system. For retrofit systems, hose bibbs can be replaced by below grade quick coupling valves with locking caps.
- **No Runoff:** The recycled water system must be designed to prevent runoff outside the approved use area.
- **No Ponding:** Except for permitted landscape impoundments, the recycled water system must be configured and operated so that ponding does not occur.
- **No Windblown Spray outside the Approved Use Area;** Recycled water irrigation systems must be configured such that windblown spray will remain within the approved use area.

- Drinking Fountains, Outdoor Eating Areas, Bleachers: Drinking fountains, outdoor eating areas, bleachers and similar facilities must be located in areas that will not be subject to spray from sprinklers. The facilities may be protected by modifying the irrigation system, relocating the protected facilities, or installing a shielding device.
- Protection of Aquifers: No irrigation with recycled water shall take place within a horizontal distance of 50 feet of any domestic water supply well. All irrigation systems shall be designed and located accordingly. In addition, all recycled water impoundments must be horizontally separated by a distance of at least 100 feet from any domestic water supply well.

2.4 Construction

Provided that all permits for construction have been obtained and the City approves the plans submitted with the Application for Recycled Water Service, construction of the recycled water systems may begin.

As part of the construction process for new facilities a temporary potable water connection will be required to perform cross-connection and coverage testing. The customer must obtain approval from the City prior to constructing a jumper (temporary connection) to an on-site potable system. A City inspector must be on-site when a jumper to the potable water system is made and removed. Jumpers connecting to a recycled water supply are prohibited at all times. If there are no potable water facilities in the vicinity of the project, the City at its discretion may eliminate the need for cross-connection testing, and coverage testing will be waived until recycled water is available to the site.

Because the City is responsible for inspecting the system during and after construction to ensure compliance with the approved plans, the applicant must notify the City prior to the start of construction and provide the City with updated construction schedules.

All conceptual or major changes from the approved plans must be approved by the City before implementation.

2.5 Inspection

The City will maintain and update a Construction Inspection Checklist throughout the construction process. The elements of the Construction Inspection Checklist are:

- Certified User Supervisor Designated: The City must verify that the designated User Supervisor has been certified through the City's annual training program.
- Utilities Horizontal and Vertical Separation Verified: Through the City's on-site inspections during the construction of the system, inspectors will be able to verify adherence to the pipe separation requirements specified in Section 2.3.
- Advisory Signs, Tags and Labeling in Accordance with DPH Approved Plans: The City must check that the location and type of signage and warning labels match the plans as approved by DPH.
- Additional DPH Requirements Completed: The City is responsible for ensuring that additional requirements resulting from DPH's review have been incorporated.
- Cross-Connection Test Passed: The customer must conduct a cross-connection test to ensure that the recycled water system is not connected to the potable water system. The test must be performed by a certified AWWA cross-connection control specialist provided by the City. In addition to coordinating with the City to complete the cross-connection test, the customer must submit a map showing all potable connections that will be removed from the recycled water system prior to connecting to the meter.
- Coverage Test Passed: The customer shall schedule a coverage test walk through to demonstrate to the City that there is not excessive overspray, runoff or ponding and that no contact occurs

between recycled water and drinking water fountains, outdoor eating areas or bleachers. If adjustments to the system are required that can not be made during the site walk, the City will issue a punch list and the customer must address each item on the list prior to rescheduling the coverage test.

- **Meter Installation:** Following the final construction inspection and successful completion of the cross-connection and coverage tests, the City will coordinate with the customer on installation of the recycled water meter(s).
- **On-site System Connected to Recycled Water:** Prior to connecting to recycled water system the City shall verify that all potable water connections, as identified on the customer's map, have been removed.
- **Record Drawings Submitted:** The customer is required to submit record drawings. The drawings must include all changes from the original plans.

2.6 Permit

Once the City has verified that each item on the Inspection Checklist has been completed, a Recycled Water Use Permit will be issued to the customer. Recycled water service is conditioned upon requirements which may be attached to the permit and proper operation and maintenance of the system as discussed in Chapter 3.

If the property is transferred to a new owner or tenant, the customer must notify the City in writing within 30 days to transfer the permit.

If the customer's recycled water system is found to be in violation of the rules of service, the City will direct the customer to mitigate for these violations. A site inspection will be scheduled after a reasonable mitigation period to ensure compliance. Failure to comply will result in termination of recycled water service until improvements are made.

Chapter 3 Operating and Maintaining a Recycled Water System

3.1 User Supervisor Responsibilities

As part of the application and permitting process, each recycled water use customer will appoint a User Supervisor to be the main point of contact with the City regarding recycled water use. The User Supervisors will be responsible for ensuring compliance with City and State regulations for the use of recycled water. The User Supervisor will be responsible for:

- Ensuring that all use site facilities that serve recycled water are maintained and operated in accordance with the Recycled Water Use Permit and this manual.
- Operating and controlling the irrigation systems in a manner to prevent human consumption of recycled water, to control and limit runoff, and to prevent contamination of nearby domestic supply wells, as applicable.
- Ensuring that all personnel are educated in practices and procedures for working with recycled water.
- Installing and maintaining signs around the use site.
- Preventing cross-connection between recycled water facilities and domestic water facilities.
- Notifying, preparing and submitting reports to the City when there are system failures that cause unauthorized discharges.
- Notifying the City for its approval of any proposed modification or additions to facilities.

In the event that a customer wishes to make personnel changes to the User Supervisor position, the City must be notified in writing, and the newly designated User Supervisor must attend the City's training program with 120 days of the personnel change.

3.2 Maintenance

In order to keep the recycled water system in compliance with the City's rules and regulations, the User Supervisor shall perform the following preventive maintenance activities:

- Perform regular inspections of the entire recycled water system including sprinkler heads, drip irrigation system emitters, spray patterns, piping and valves, pumps, storage facilities and controllers.
- Immediately repair all broken sprinkler heads, faulty spray patterns, leaking pipes or valves, or any other noted condition that violates the recycled water use requirements.
- Check all recycled water identification signs, tags, stickers, and above grade pipe markings for their proper placement and legibility. Replace damaged, unreadable, or missing signs, tags, stickers, and pipe markings.
- Check spray patterns to eliminate ponding, runoff and wind blown spray conditions. If evidence of ponding or runoff is noted, affected areas should be indicated on a sketch and sprinkler heads should be adjusted to prevent further ponding or runoff. Evidence of mosquitoes breeding within ponding should be noted and immediately eliminated.
- Establish and maintain an accurate record keeping system of all inspections, modifications and repair work. Records of repairs and system upgrades should be included in a Facilities Modification Log and submitted to the City with the Annual Self-monitoring Report.
- Coordinate with the City to complete a cross-connection test every five years.
- Coordinate with the City to complete annual testing of backflow assemblies.

3.3 Hours of Irrigation

Hours of irrigation with recycled water must be maintained to minimize potential public contact with recycled water. The following guidelines should be followed:

- For areas that are easily accessed by the general public, irrigation must take place during the periods of least use by the general public, generally between the hours of 9 pm and 7 am.
- For areas where public access is restricted or prohibited, such as highway medians, commercial nurseries, etc., irrigation may occur at all times.
- Irrigation of areas accessed by the general public may be performed at other times provided that the irrigation system is operated manually and is supervised (someone present at all times) to avoid inadvertently exposing any members of the general public.
- Irrigators should take into account that irrigated areas might need a reasonable dry-out period before the area is accessed by the general public.

3.4 System Modifications

The User Supervisor must obtain approval from the City before modifying an approved recycled water system. This includes converting any piping used for recycled water back to potable water, such as switching from a recycled water system to a backup potable water system.

Emergency modifications necessary to prevent contamination, damage or a public health hazard can be made without the prior approval of the City. However, immediately after the modification has been implemented and the hazard has been abated, the User Supervisor shall notify the City of the emergency modifications and file a written report.

3.5 Personnel Training

The User Supervisor is responsible for training all personnel involved with recycled water so they are familiar with the rules and regulations. At a minimum, the training program should convey the following information:

- The City's recycled water, although highly treated, is non-potable and must never be used for human consumption.
- Regulations prohibit ponding, windblown spray and runoff of recycled water.
- Working with non-potable recycled water is safe if common sense is used and appropriate regulations are followed.
- State law prohibits a connection between the recycled water and the potable water systems.

3.6 Annual Self Inspection Report

An annual self-inspection must be conducted at each use site while the recycled water system is in use. The User Supervisor must conduct this inspection and provide a report of the inspection to the City using the City's Annual Self Inspection Report form. The original report must be submitted to the City with a copy kept on-site by the User Supervisor.

The following questions must be answered as part of the annual inspection report:

- Is there evidence of recycled water runoff from the site? If yes, the User Supervisor must submit a sketch showing the affected area(s) and estimated volume of runoff.

- Is there an odor due to recycled water at the site? If yes, the User Supervisor must provide a description of the apparent source, characterization, direction of travel, and any public use areas or off-site facilities affected by the odors.
- Is there evidence of recycled water ponding, and/or evidence of mosquitoes breeding due to ponded water?
- Is there evidence of leaks or breaks in the system piping or tubing?
- Are advisory signs, tags and above-ground pipe markings in good condition and properly posted to inform public that water is recycled?
- Is there evidence of plugged, broken or otherwise faulty drip irrigation system emitters or spray irrigation sprinklers on the site?
- In the past year or since the last annual site inspection report, were there any modifications of the piping or the recycled water system as approved by DPH? If yes, the User Supervisor must submit a Facilities Modification Log.
- In the past year or since the last annual site inspection report, has the site owner, tenant or user supervisor changed?
- What corrective actions are being taken to correct any problems noted in the report?

3.7 Emergency Procedures

3.7.1 Disaster Response

Following a major earthquake, flood, fire or other incident which could likely damage a customer's recycled or potable water systems, the User Supervisor should inspect the potable and recycled water systems for damage as soon as it is safe to do so. If either system appears damaged, both the potable and recycled water systems should be shut off at their points of connection. If the User Supervisor cannot inspect the site and damage is expected, then both water systems should be shut off at their points of connection. The Supervisor should immediately contact the City for further instruction.

3.7.2 Unauthorized Discharge

An unauthorized discharge is defined as any amount of recycled water that leaves the designated use site. If a system failure results in an unauthorized discharge, every effort should be made to contain the recycled water and prevent it from entering the storm drain system. The User Supervisor is responsible for immediately notifying the City of the unauthorized discharge, and then preparing and submitting a report estimating the volume of discharge, identifying the cause of the system failure and discussing the corrective measures taken.

In the event that an unauthorized discharge occurs that may endanger public health or the environment, the City will verbally notify the RWQCB and DPH within 24 hours.

3.7.3 Cross-Connection

If a cross-connection occurs or is suspected of having occurred on the customer's premises, the User Supervisor must immediately implement the following Emergency Cross-Connection Response Plan:

- Immediately following discovery of the cross-connection, shut down the recycled water supply to the facility.
- Notify the City and DPH of the incident by telephone immediately. The City will notify the local health department. Within 24 hours the customer's verbal notifications must be followed by a written notice to the City that includes an explanation of the nature of the cross-connection, date and time discovered, steps taken to mitigate the cross-connection and the contact information of the person reporting the cross-connection.

- Keep the potable system pressurized and post "Do Not Drink" signs at all potable water fixtures and outlets.
- Provide bottled water for employees until the potable water system is deemed safe to drink.
- Coordinate with the City to collect water samples from the potable water system and perform a 24-hour bacteriological analysis. If the bacteriological analysis is positive, chlorinate the potable water system maintaining a chlorine residual of at least 50 mg/l for 24 hours, flush the potable water system after the 24-hour chlorination period and perform standard bacteriological analysis. The City shall assist with the sampling and laboratory analyses at the customer's expense.
- Coordinate with the City to conduct a cross-connection test to verify that all cross-connections were eliminated.
- Follow additional procedures outlined by the City, DPH and local health department.

After final approval has been obtained from DPH and the local health department, the City will bring the recycled water system back into service and inform the customer to remove the "Do Not Drink" signs from all potable water fixtures and outlets.

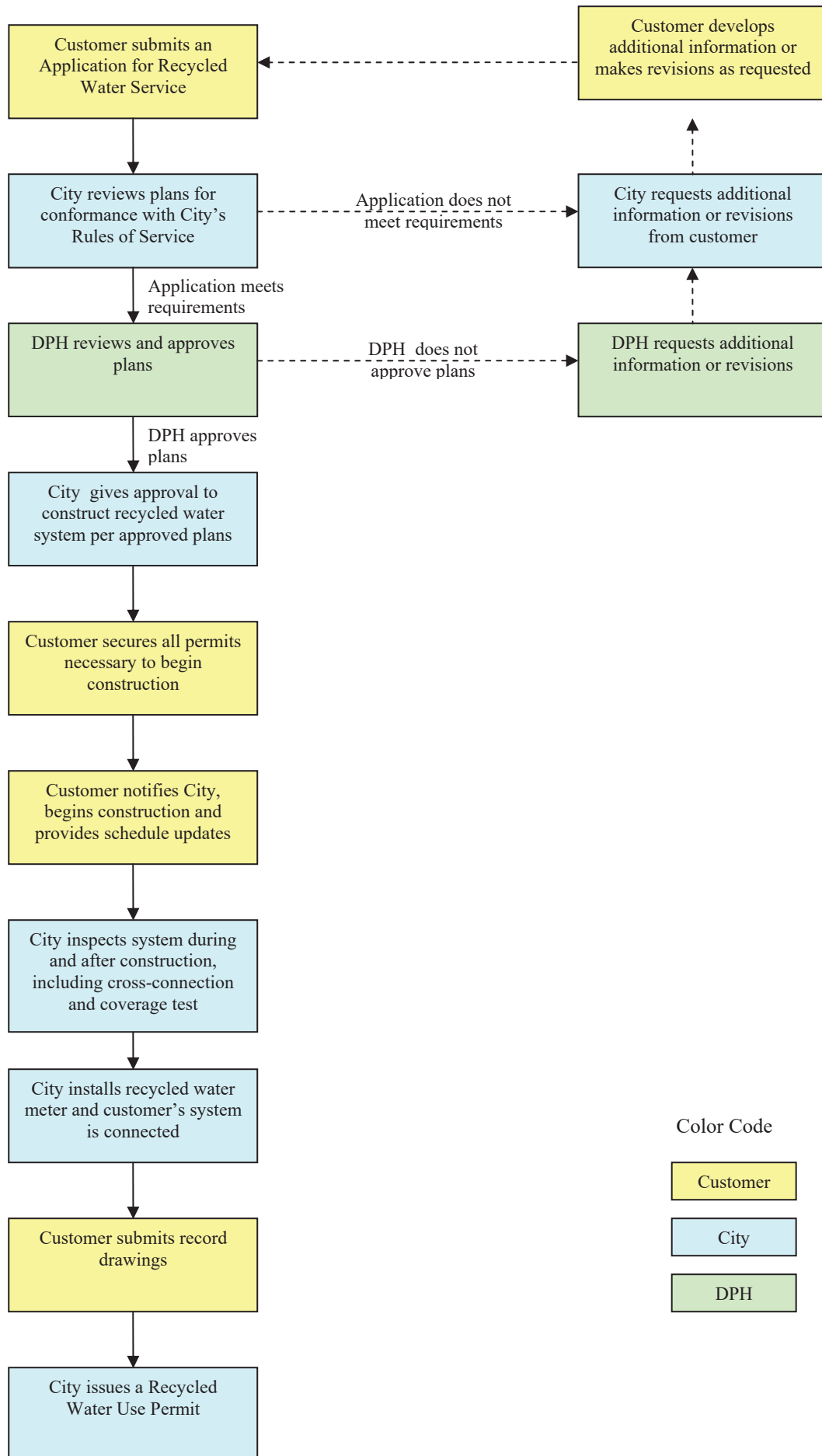
Appendix A - Definitions

Whenever the following terms occur in this document, the intent and meaning shall be interpreted as follows:

AIR GAP	A physical separation between the free flowing discharge end of a water supply pipeline and an open or non-pressure receiving vessel. An approved air gap must be at least twice the diameter of the water supply pipe measured vertically above the overflow rim of the vessel, and in no case less than one inch.
APPROVED USE	An application of recycled water in a manner, and for a purpose, designated in a Recycled Water Use Permit issued by the City and in compliance with all applicable Regulatory Agency requirements.
APPROVED USE AREA	A site with well-defined boundaries designated on the approved Site Drawings, to receive recycled water for an approved use and acknowledged by all applicable regulatory Agencies.
CROSS-CONNECTION	Any physical connection between any part of a water system used or intended to supply water for drinking purposes and any source or system containing water or substance that is not or cannot be approved for human consumption. This includes direct piping between the two systems, regardless of the presence of valves, backflow prevention devices, or other appurtenances.
CUSTOMER	Any person, persons or firm including any public utility, municipality or other public body or institution issued a Recycled Water Use Permit by the City. They may be the owner, tenant, or property manager as appropriate.
INSPECTOR	Any person authorized by the City, the local authorities or the local or state health agencies to perform inspections on or off the customer's site before construction, during construction, after construction and during operation.
LANDSCAPE IMPOUNDMENT	A body of recycled water used for aesthetic enjoyment or which otherwise serves a function not intended to include public contact.
NONPOTABLE WATER	Water that has not been treated for human consumption in conformance with the latest edition of the United States Public Health Service Drinking Water Standards, the California Safe Drinking Water Act, or any other applicable standards.
ON-SITE	Designates or relates to all irrigation facilities downstream of the recycled water meter.
OVERSPRAY	The spray of recycled water outside of the approved irrigation area.
OWNER	Any holder of legal title, contract purchaser, or lessee under a lease with an unexpired term of more than one (1) year, for property for which recycled water service has been requested or established.
POINT OF CONNECTION	This is the point where the customer's system ties to the City's system. This is usually at the water meter.
PONDING	Unauthorized retention of recycled water on the surface of the ground or other natural or manmade surface for a period following the cessation of an approved recycled water use activity.
POTABLE WATER	Water that is authorized for human consumption according to the latest edition of the California Safe-Drinking Water Act, or other applicable standards.
PUBLIC	Any person or persons other than the site owner or employees who may come in contact with facilities and/or areas where recycled water is approved for use.
RECYCLED WATER (or	Water that meets California Administration Code Title 22,

NONPOTABLE RECYCLED WATER)	Division 4 of the Environmental Health Water Reclamation Criteria and is approved for purposes other than human consumption. For the purpose of these rules and regulations, “recycled water” refers to “Nonpotable recycled water.”
RECYCLED WATER USE PERMIT	A permit issued to the customer as required by DPH and the RWQCB that outlines monitoring, self-inspection, reporting, and site-specific requirements.
RUNOFF SERVICE	Recycled water that is allowed to drain outside the approved use area. The furnishing of recycled water to a customer through a metered connection to the onsite facilities.
STANDARD PIPE LENGTH	A section of pipe 18 to 20 feet in length that has no joints.
UNAUTHORIZED DISCHARGE	Any release of recycled water that violates the rules and regulations of City or applicable Federal, State or local statutes, regulations, ordinances, contracts or other requirements.
USER SUPERVISOR	The responsible person designated by the customer to provide liaison with the City. This person must have the authority to carry out any requirements of the City, must be responsible for the operation and maintenance of the recycled water system, and must prevent potential violations.
VIOLATION	Noncompliance with any condition of the Recycled Water Use Permit by any person, action or occurrence, intentional or unintentional.

Appendix B - Flow Chart



Appendix B - Recycled Water Program Material

Table of Contents

- Recycled Water Use Manual and Rules of Service Handbook (See Appendix A)
- User Application for Recycled Water Service
- Plan Review Checklist
- Recycled Water Standard Details
- Construction Inspection Checklist
- Cross Connection Certification
- Recycled Water Use Permit
- Annual Self Monitoring Report
- Facilities Modification Log

**CITY OF HOLLISTER
APPLICATION FOR RECYCLED WATER SERVICE**

PROPERTY INFORMATION			
Site Name			
Site Address		City	Zip
Site Facilities Manager	Phone	APN	Acreage
Description of Property			
APPLICANT			
Name			
Contact Name		Title	
Address			
City		Zip	
Phone	Fax	Email	
SITE OWNER (IF DIFFERENT FROM APPLICANT)			
Name			
Contact Name		Title	
Address			
City		Zip	
Phone	Fax	Email	
DEVELOPER/DESIGN CONTACT			
Name			
Contact Name		Title	
Phone	Fax	Email	
USER SUPERVISOR			
Name		Certification Number	
Organization		Title	
Address			
City		Zip	
Phone	Fax	Email	
Emergency Phone Number			

PROPOSED RECYCLED WATER USES	
☐ Irrigation	☐ Landscape Impoundments
☐ Fire Suppression	☐ Other
Briefly describe the proposed use checked above. Include types of plants to be irrigated and method of irrigation:	
RECYCLED WATER DEMAND ESTIMATES	
Estimated Annual Use	☐ _____ AF ☐ _____ CCF ☐ _____ Gallons
Peak Use In Gallons/Minute (GPM)	
Hours of Use	
Days of Use	
Period of Use	☐ Use According to Evapotranspiration Needs ☐ Year-Round
USE AREA CONTAINMENT MEASURES	
Briefly describe the measures to be used:	
EMPLOYEE OR PUBLIC ACCESS	
Briefly describe the degree of potential access by employees or the public:	
ATTACHMENTS	
☐ Plans ☐ Cross-connection control procedures ☐ Additional attachments:	
USER'S RECYCLED WATER SUPERVISOR	APPLICANT
I have read and understand the City of Hollister Recycled Water Use Manual and Rules of Service. I will operate the recycled water system in compliance with all conditions of the Permit to Use Recycled Water. Print _____ Signature _____ Date _____	I designate the named person as the User's Recycled Water Supervisor in accordance with the City of Hollister Recycled Water Use Manual and Rules of Service. I am a principal owner of this site or a duly authorized representative and certify that the information contained in this application is true and correct to the best of my knowledge. Print _____ Signature _____ Date _____

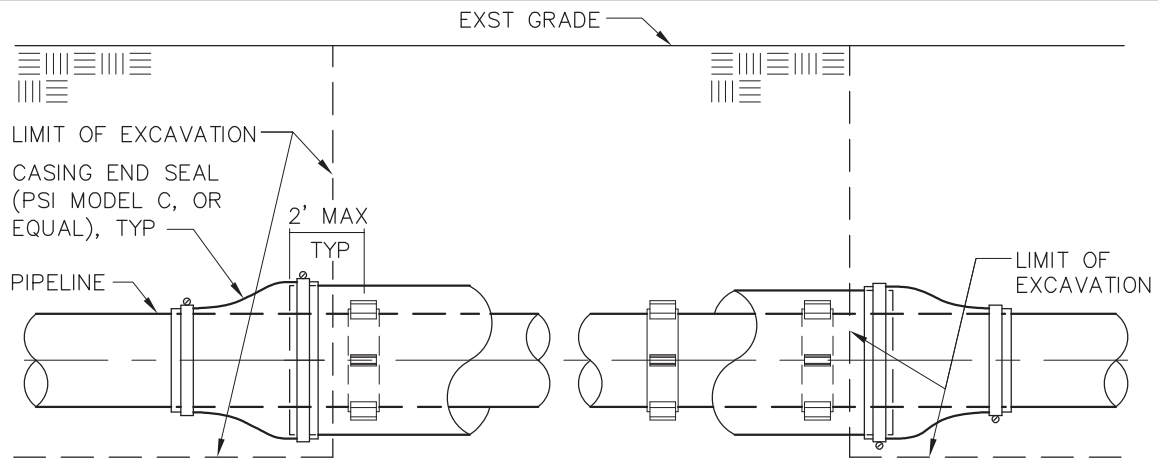
PLAN REVIEW CHECKLIST

Site Name _____

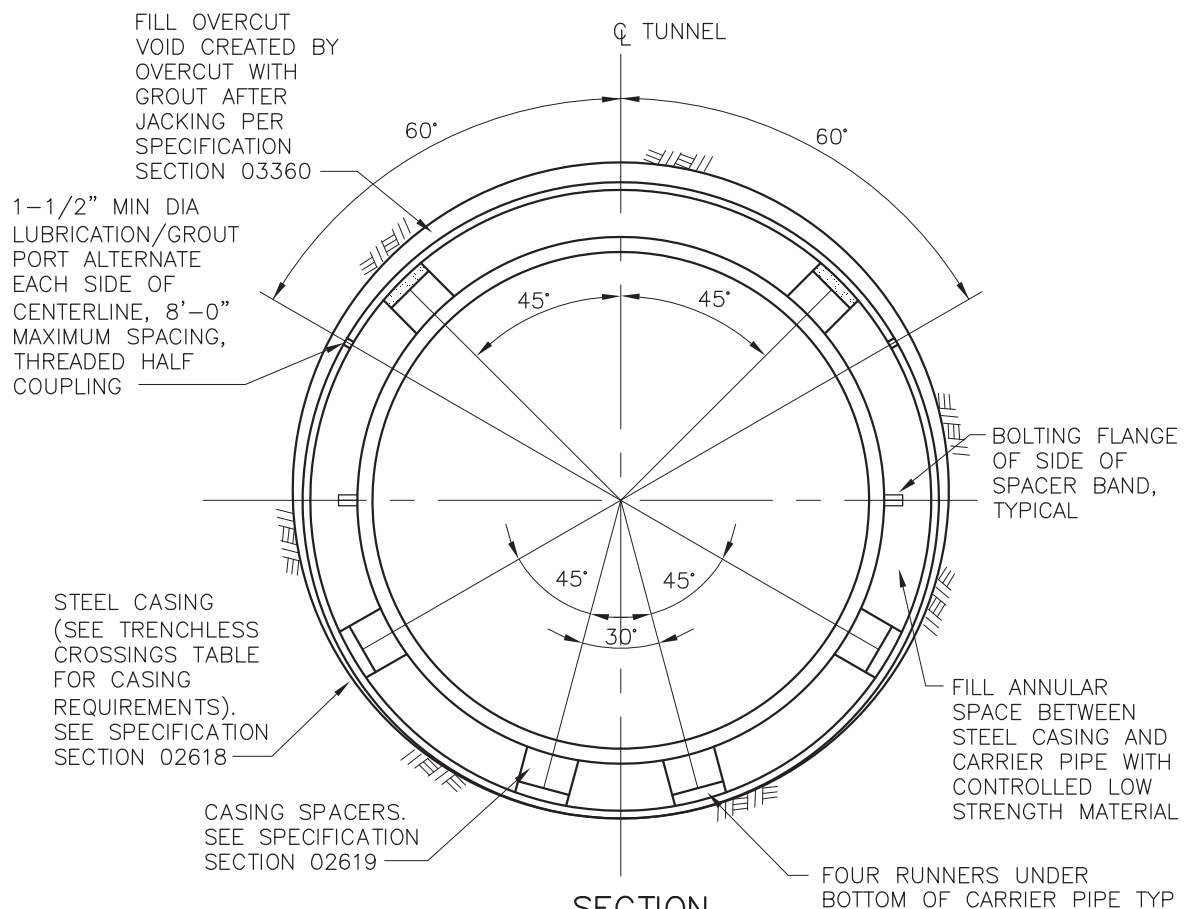
Site Address _____

- ☐ Show proposed use area and areas to be excluded
- ☐ Show site/piping plan and details of connection points (including location and size of all water meters)
- ☐ Show the proper backflow prevention devices, as applicable
- ☐ Show the location of all water pipelines (including potable and well lines) crossing the site. If space does not permit this information to be placed on the irrigation plans, then a separate site or utility plan can be used to show this information. Exception for an existing irrigation system converting to recycled water: Although it may not be possible to show the location of all water pipelines at this type of site, all locations where future recycled water piping must be separated from the potable water piping must be clearly indicated on the plans.
- ☐ Indicate proper pipe separation between potable water and recycled water pipelines.
- ☐ Specify special construction requirements
- ☐ Identify public access points
- ☐ Show outdoor drinking fountains, bleachers, and designated outdoor eating areas
- ☐ Identify surrounding land uses
- ☐ Show the location of any wells, lakes, ponds, reservoirs, or other water impoundments located on the site or within 100 feet of the site, and indicate the type of water source.
- ☐ Show location and type of signage to be used
- ☐ Obtain Department of Public Health approval

COMMENTS:



ELEVATION



SECTION

NOTES:

1. FOR INSTALLATION OF CASING, SEE SPECS.
2. CASING THICKNESS AT UPRR CROSSING SHALL BE 7/16".

CASING DETAIL

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

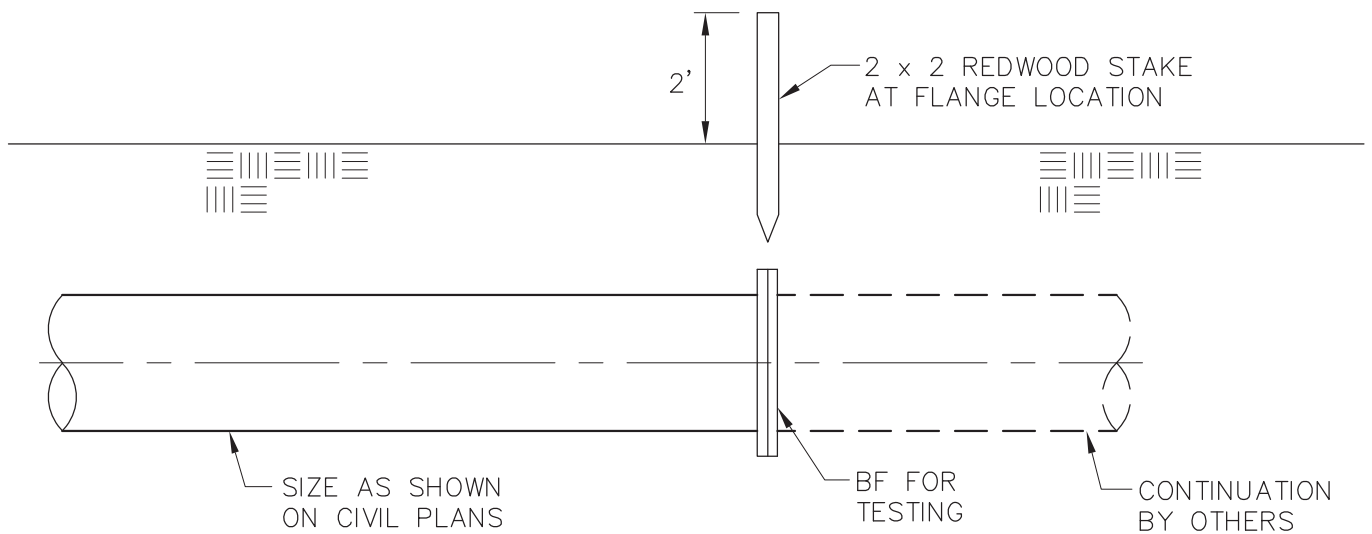
DATE

REVISED

DETAIL

RW-1

SHEET NO 1 OF 1



NOTES:

1. CONTRACTOR SHALL PROVIDE SUFFICIENT TEMPORARY THRUST RESTRAINT FOR TESTING.

TYPICAL PIPE END OF CONTRACT DETAIL

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

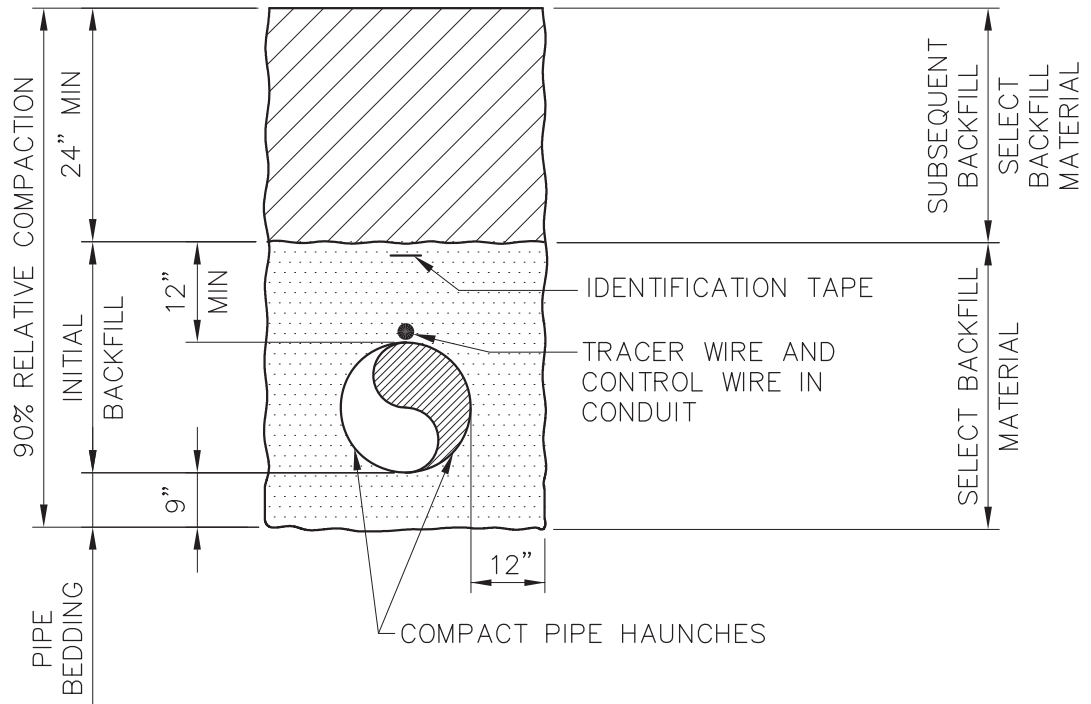
DATE

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DETAIL

RW-2

SHEET NO 1 OF 1



PIPE TRENCH BACKFILL - UNPAVED SURFACE

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

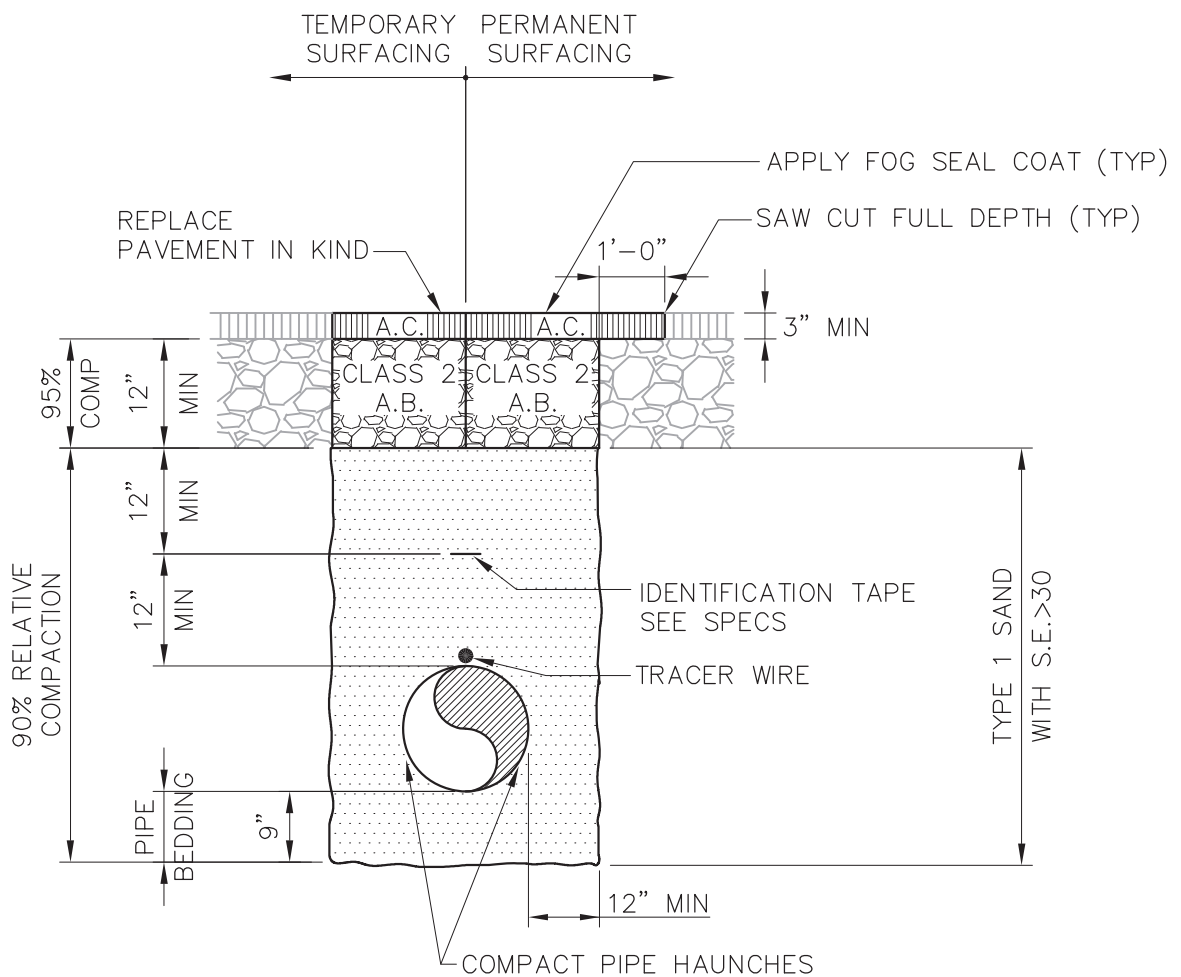
DATE

REVISED

DETAIL

RW-3

SHEET NO 1 OF 1



PIPE TRENCH BACKFILL - PAVED SURFACE

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

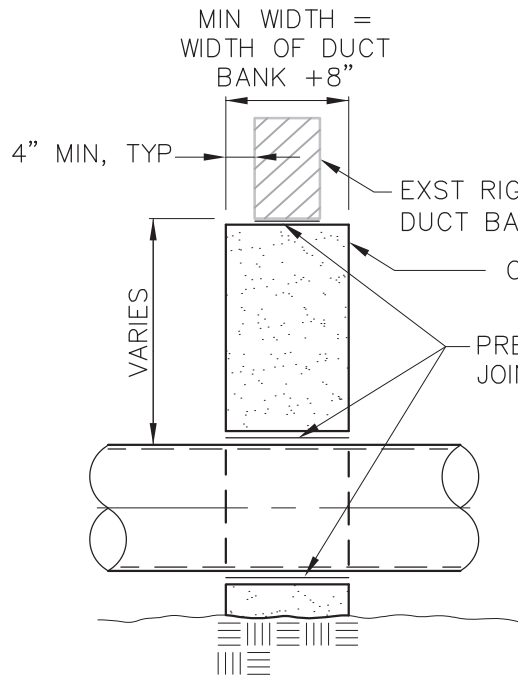
DATE

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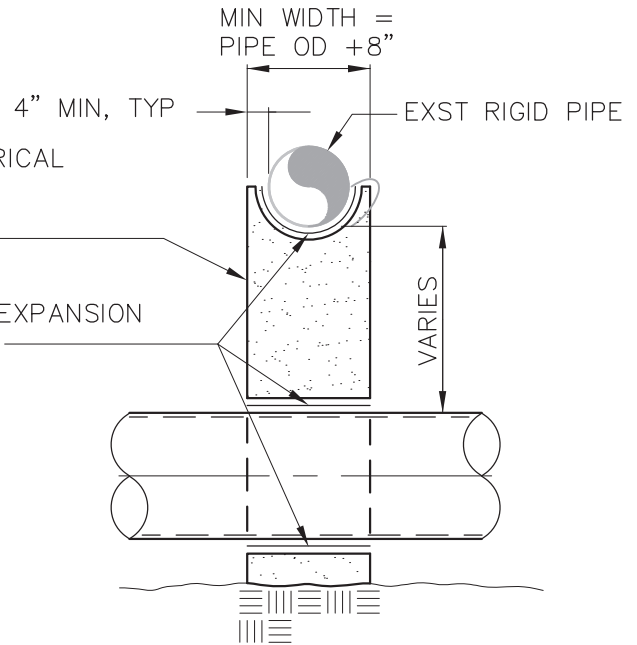
DETAIL

RW-4

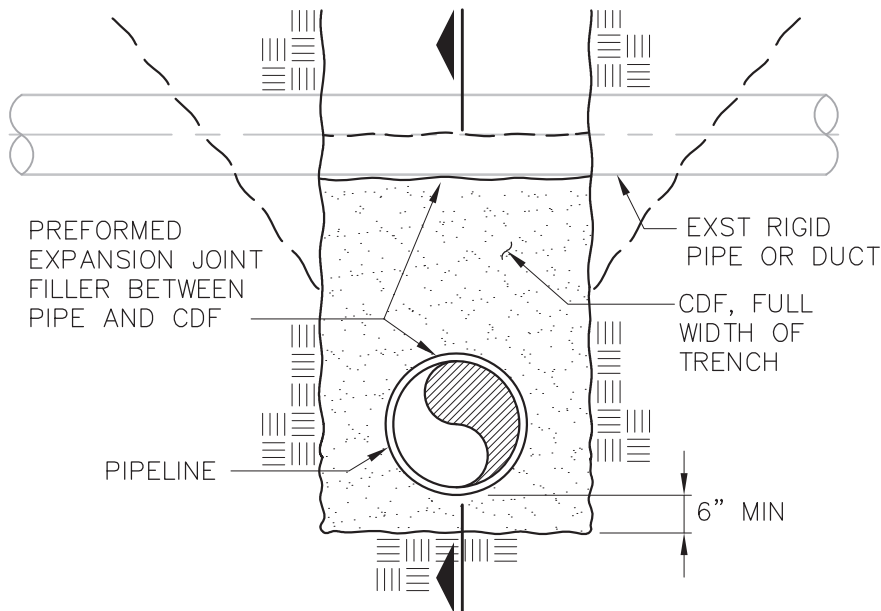
SHEET NO 1 OF 1



SECTION OF DUCT BANK CROSSING



SECTION OF PIPE CROSSING



UTILITY CROSSING ELEVATION

NOTES:

1. SUBMIT SHOP DRAWING ON EXPANSION JOINT FILLER MATERIAL.
2. BACKFILL TO BE BROUGHT UP UNIFORMLY ON BOTH SIDES OF UTILITY SUPPORT.
3. EXISTING PIPE OR DUCT SHALL BE FIRMLY SUPPORTED DURING INSTALLATION OF NEW PIPE AND UTILITY SUPPORT.
4. FORM THE SIDES OF UTILITY SUPPORT WHICH CROSS TRENCH.
5. RIGID PIPE SHALL BE DEFINED AS CONCRETE PIPE OR OTHER RIGID MATERIAL.

UNDERGROUND UTILITY SUPPORT DETAIL

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

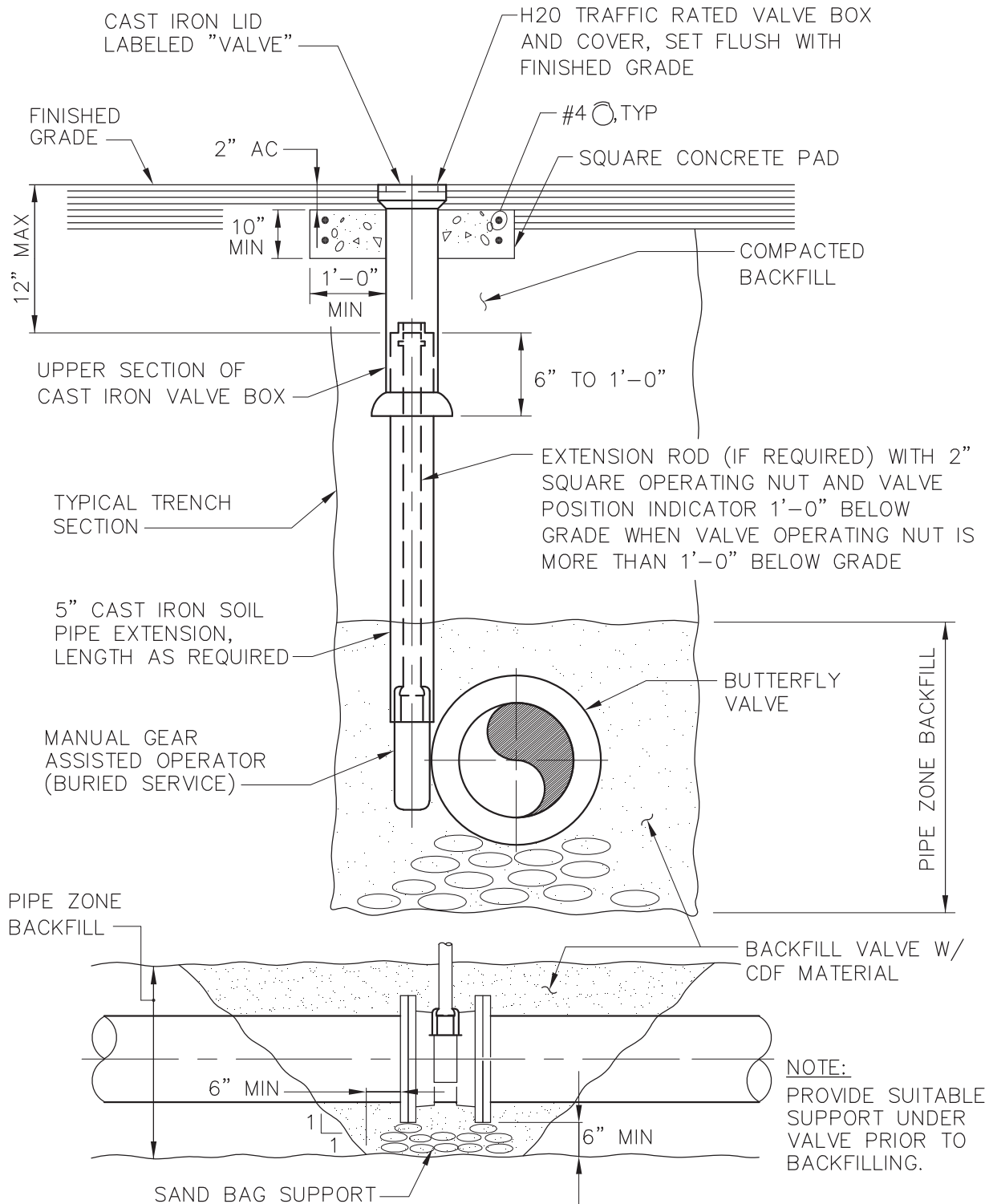
DATE

REVISED

DETAIL

RW-5

SHEET NO 1 OF 1



BURIED BUTTERFLY VALVE BOX

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

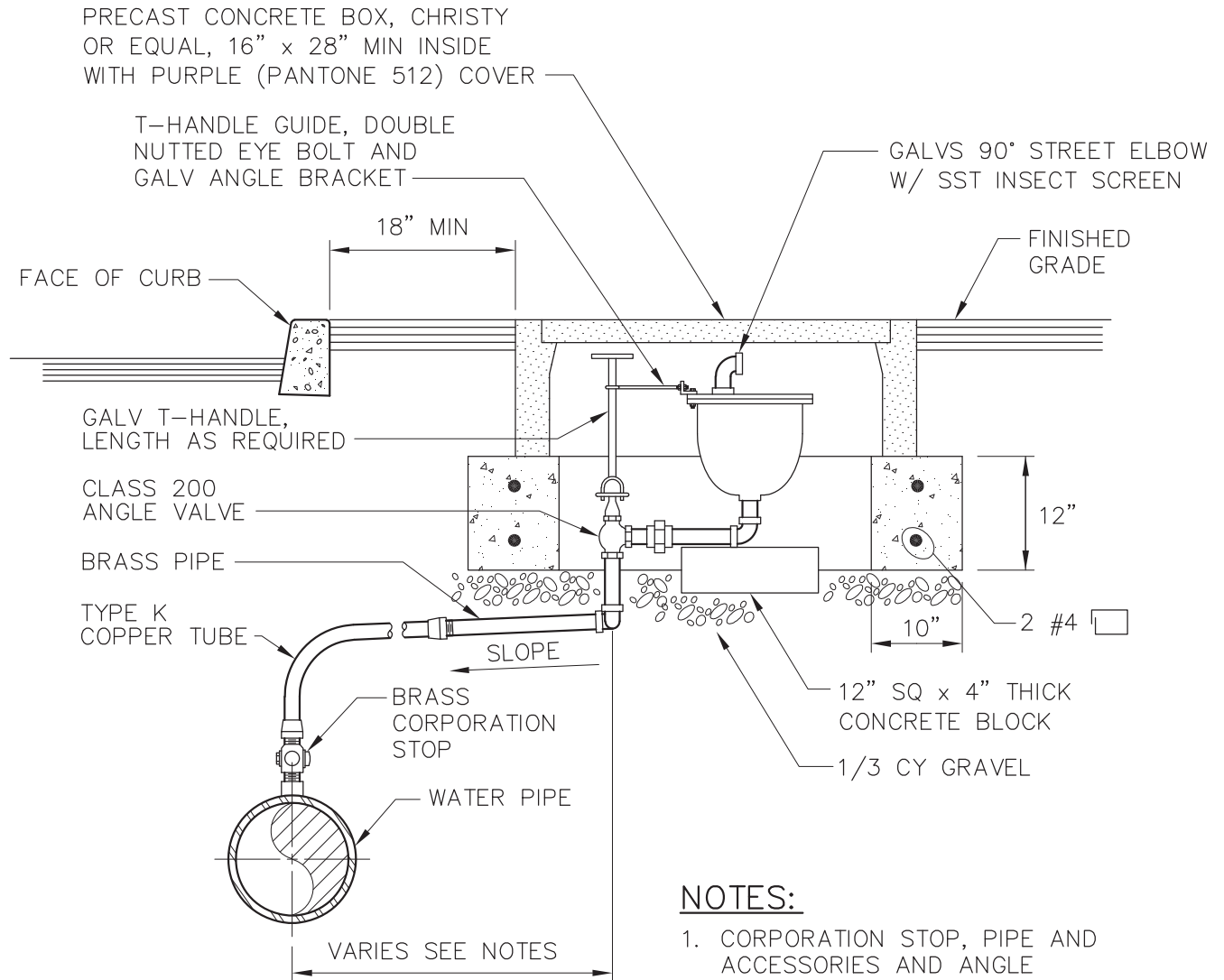
DATE

REVISED

DETAIL

RW-6

SHEET NO 1 OF 1



NOTES:

1. CORPORATION STOP, PIPE AND ACCESSORIES AND ANGLE VALVE TO BE SAME DIAMETER AS COMBINATION AIR RELEASE VALVE INLET.
2. ALL EXPOSED SURFACES SHALL BE PAINTED PANTONE 512 PURPLE.
3. COMBINATION AIR RELEASE VALVE BOX TO BE PLACED AT CLOSEST SIDEWALK TO PIPE.
4. SIDEWALK REPAIR SHALL BE PER CITY STANDARDS.

2" COMBINATION AIR RELEASE VALVE ASSEMBLY

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

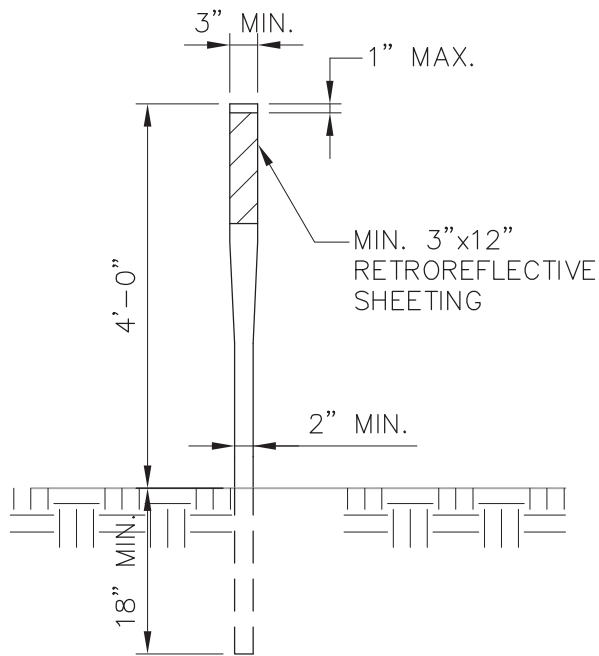
DATE

REVISED

DETAIL

RW-8

SHEET NO 1 OF 1



NOTES:

1. INSTALL DELINEATOR AT ALL UNPAVED PIPELINE SURFACE FEATURES INCLUDING CAV'S, LOCATOR STATIONS, AND VALVES.

CALTRANS CLASS 1 FLEXIBLE PLASTIC POST MARKER

CITY OF HOLLISTER

STANDARD PLANS

APPROVED: _____

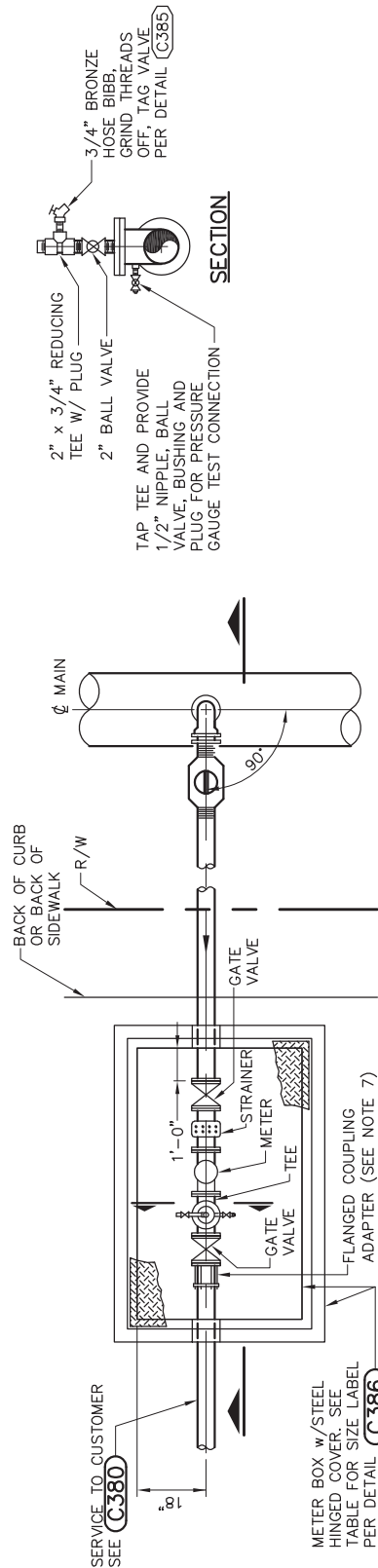
DATE

REVISED

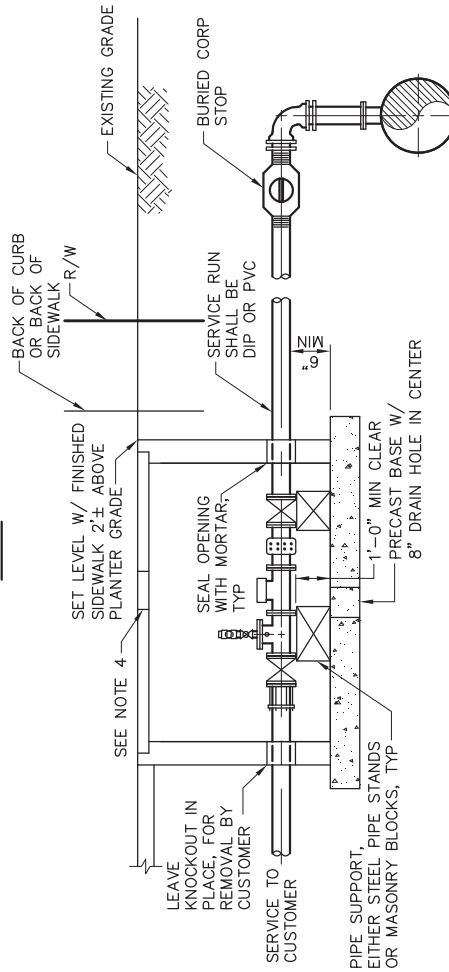
DETAIL

RW-9

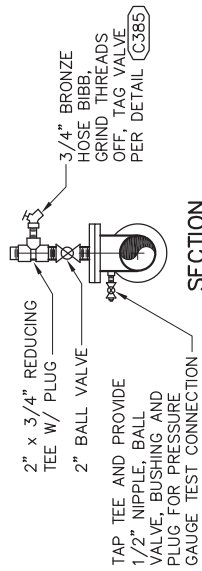
SHEET NO 1 OF 1



PLAN



ELEVATION



NOTES:

1. PAINT INSIDE OF BOX PURPLE (PANTONE 512).
2. METER AND VALVES SHOULD BE LABELED OR TAGGED TO INDICATE RECYCLED WATER.
3. METER BOX SIZE MAY BE REDUCED TO ACCOMMODATE SPECIFIC REQUIREMENTS OF RETAILER AND CUSTOMER.
4. METER READING LID SHALL BE DIRECTLY OVER THE METER REGISTER. LID OPENING SHALL BE CUT AFTER THE COMPLETION OF THE METER INSTALLATION.
5. FOR CONNECTIONS TO MAIN LINE WHERE MAIN LINE IS LESS THAN 8" IN DIAMETER, A SADDLE SHALL BE USED.
6. MINIMUM VAULT SIZE: 6'-6" LENGTH, 4'-0" WIDTH, 4'-6" DEPTH.
7. IF SERVICE RUN IS DIP INSTALL INSULATING FLANGED COUPLING ADAPTER.

BURIED SERVICE CONNECTION 2-INCH AND SMALLER

CITY OF HOLLISTER STANDARD PLANS

APPROVED: _____

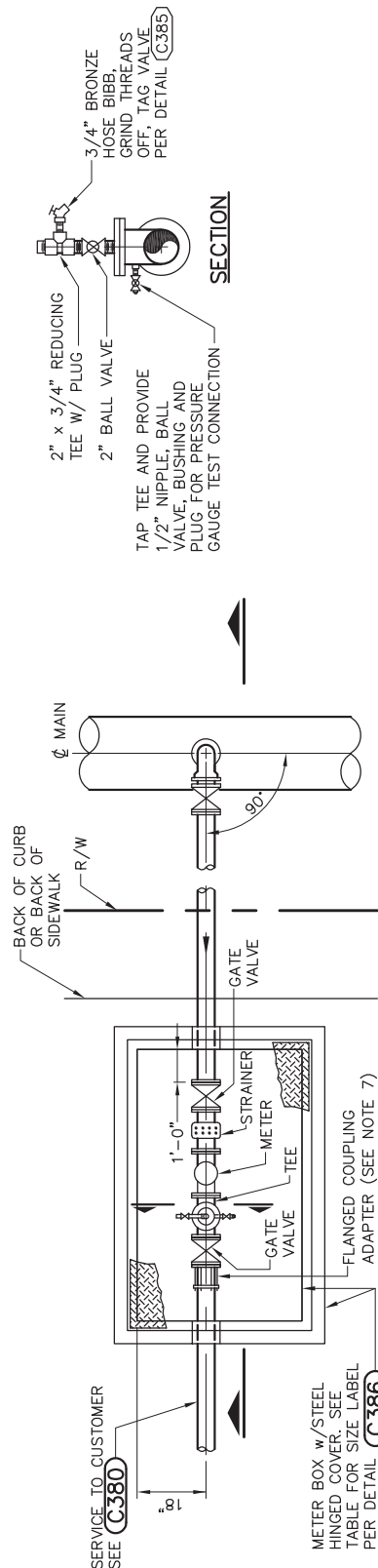
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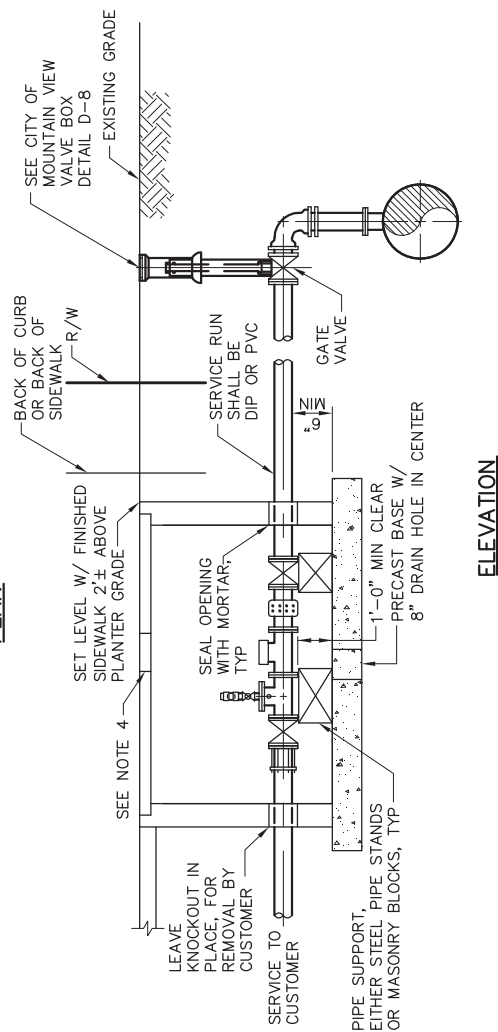
DETAIL

RW-10

SHEET NO 1 OF 1



- NOTES:**
1. PAINT INSIDE OF BOX PURPLE (PANTONE 512).
 2. METER AND VALVES SHOULD BE LABELED OR TAGGED TO INDICATE RECYCLED WATER.
 3. METER BOX SIZE MAY BE REDUCED TO ACCOMMODATE SPECIFIC REQUIREMENTS OF RETAILER AND CUSTOMER.
 4. METER READING LID SHALL BE DIRECTLY OVER THE METER REGISTER. LID OPENING SHALL BE CUT AFTER THE COMPLETION OF THE METER INSTALLATION.
 5. FOR CONNECTIONS TO MAIN LINE WHERE MAIN LINE IS LESS THAN 8" IN DIAMETER, A SADDLE SHALL BE USED.
 6. MINIMUM VAULT SIZE: 6'-6" LENGTH, 4'-0" WIDTH, 4'-6" DEPTH.
 7. IF SERVICE RUN IS DIP INSTALL INSULATING FLANGED COUPLING ADAPTER.



BURIED SERVICE CONNECTION LARGER THAN 2-INCH

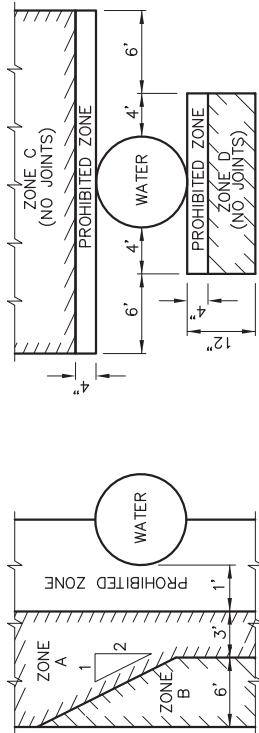
CITY OF HOLLISTER STANDARD PLANS

APPROVED: _____

DATE	
REVISED	
DETAIL	RW-11

SHEET NO 1 OF 1

SITUATION: LOCATION OF NEW RECYCLED WATER LINES TO EXISTING DOMESTIC WATER LINES.



PARALLEL CONSTRUCTION

IF ANY RECYCLED WATER PIPELINES ARE TO BE CONSTRUCTED WITHIN ANY OF THE ABOVE INDICATED ZONES, SPECIAL CONSTRUCTION SHALL BE REQUIRED AS DESCRIBED BELOW.

ZONE **RECYCLED WATER**

- A** DO NOT LOCATE ANY PARALLEL RECYCLED WATER LINES IN THIS AREA WITHOUT STATE AND LOCAL HEALTH DEPARTMENT APPROVAL.
- B** USE CLASS 200 DIP, CLASS 200 PVC – AWWA C900, WELDED STEEL, HDPE OR FUSED PVC PIPE
- C** USE CLASS 200 DIP, CLASS 200 PVC – AWWA C900 OR WELDED STEEL, HDPE OR FUSED PVC PIPE
- D** USE CLASS 200 DIP OR CLASS 200 PVC – AWWA C900 OR WELDED STEEL, HDPE OR FUSED PVC PIPE

PERPENDICULAR CROSSING

GENERAL NOTES

- NO PIPE JOINTS SHALL BE PERMITTED WITHIN ZONE D. IT IS THE INTENT OF THESE SPECIFICATIONS THAT NO JOINTS SHALL OCCUR WITHIN ZONE C. IF THAT CANNOT BE ACCOMPLISHED, THE NEW PIPELINE SHALL BE ENCASED IN CONCRETE FOR THE FULL LENGTH OF ZONE C.
- ALL DIP MUST HAVE HOT DIP BITUMINOUS COATING AND ALL PVC SHALL BE CLASS 200 PER AWWA C900 OR EQUIVALENT.
- SEWER FORCE MAINS SHALL NOT BE PERMITTED IN ZONES A THROUGH D.
- THIS CRITERIA DOES NOT APPLY FOR A RECYCLED WATER LINE CROSSING ANOTHER RECYCLED WATER LINE.
- THE CONSTRUCTION CRITERIA APPLIES TO HOUSE LATERALS THAT CROSS ABOVE A PRESSURE WATER MAIN BUT NOT TO THOSE HOUSE LATERALS THAT CROSS BELOW A PRESSURE WATER MAIN.

BASIC SEPARATION STANDARDS

- PARALLEL CONSTRUCTION: THE HORIZONTAL DISTANCE BETWEEN PRESSURE DOMESTIC WATER AND RECYCLED WATER MAINS AND SEWER LINE SHALL BE AS SHOWN BELOW.
- PERPENDICULAR CONSTRUCTION (CROSSING): PRESSURE WATER MAINS SHALL BE AT LEAST ONE FOOT ABOVE SANITARY SEWER AND RECYCLED WATER LINES WHERE THESE LINES MUST CROSS.
- SPECIAL PROVISIONS: ALTERNATIVE CONSTRUCTION CRITERIA WHERE THE BASIC SEPARATION STANDARDS CANNOT BE ATTAINED ARE SHOWN ABOVE.

STANDARD CLEARANCES TO SEWERS & DOMESTIC WATER MAINS

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

DATE	
REVISED	
DETAIL	RW-12

SHEET NO 1 OF 1



FRONT



BACK

NOTE:

TAG BACKGROUND SHALL BE PURPLE
(PANTONE 512) WITH BLACK LETTERING.

WARNING TAG

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

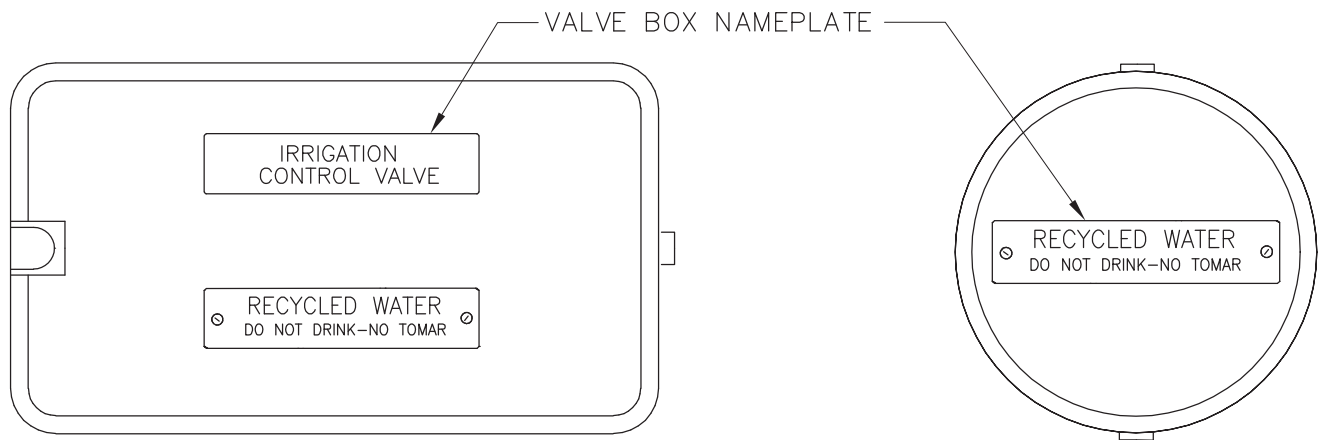
DATE

REVISED

DETAIL

RW-13

SHEET NO 1 OF 1



REFERENCE: T. CHRISTY ENTERPRISES, INC.
CATALOG PART NO. 3800 OR EQUAL

IDENTIFICATION OF IRRIGATION BOX COVERS/LIDS W/WARNING SIGN

CITY OF HOLLISTER

STANDARD PLANS

APPROVED: _____

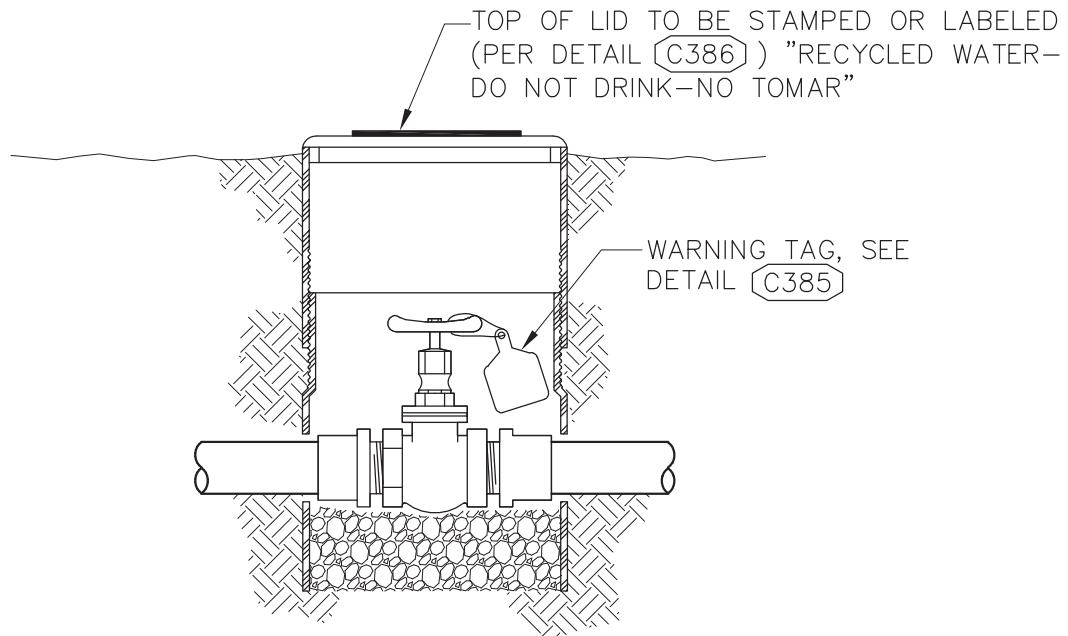
DATE

REVISED

DETAIL

RW-14

SHEET NO 1 OF 1



IDENTIFICATION OF ISOLATION GATE VALVE 3" OR SMALLER W/WARNING TAG

CITY OF HOLLISTER STANDARD PLANS

APPROVED: _____

DATE

REVISED

DETAIL

RW-15

SHEET NO 1 OF 1

TYPICAL SIGN WORDING



5/16"Ø x 2" BOLT WITH WASHERS (TYP FOR CORNERS OF LARGE SIGN ONLY)

LARGE SIGN: ATTACH 2x4 STIFFENERS TO 4x4 POST WITH 5/16"Ø x 4" BOLTS WITH WASHERS. COUNTERSINK HEAD OF BOLT TO ALLOW SIGN TO BE FLUSH MOUNTED TO STIFFENERS (TYP OF 2)

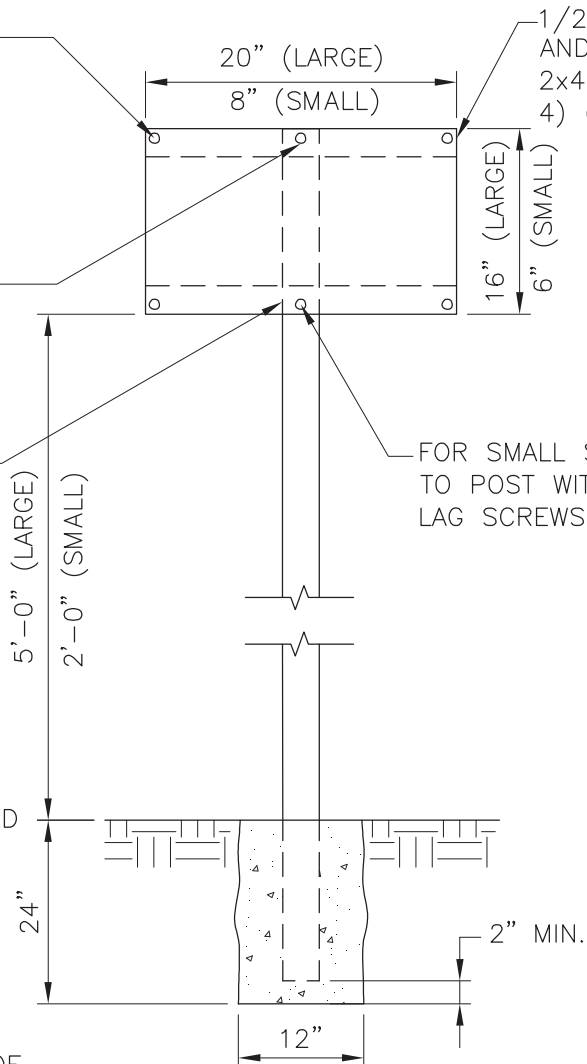
NOTCH 4x4 POST AS REQUIRED TO MOUNT 2x4 STIFFENERS FLUSH WITH FACE OF POST (TYP OF 2) (LARGE SIGN ONLY)

1/2" CHAMFER AT SIDE AND BACK EDGES OF 2x4 STIFFENER (TYP OF 4) (LARGE SIGN ONLY)

FOR SMALL SIGN: ATTACH SIGN TO POST WITH 5/16"Ø x 2" LAG SCREWS (TYP OF 2)

NOTES:

1. 4"x4" POST AND STIFFENERS SHALL BE REDWOOD OR TREATED DOUGLAS FIR (STATE OF CALIF. SPEC NO. 56-2.02B).
2. FOOTING SHALL BE 24" IN DEPTH, IN CONCRETE.
3. 5/16" Ø x 1/2" ZINC PLATED STEEL BOTH WITH VANDAL PROOF NUTS.



RECYCLED WATER ADVISORY SIGN

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

DATE

REVISED

DETAIL

RW-16

SHEET NO 1 OF 1

RECYCLED WATER COLOR: PURPLE



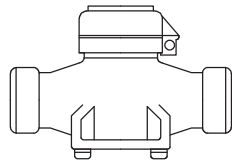
FRONT



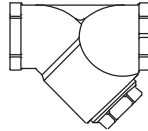
BACK



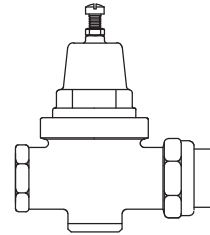
SEE DETAILS C385 AND C386 FOR
MORE INFORMATION



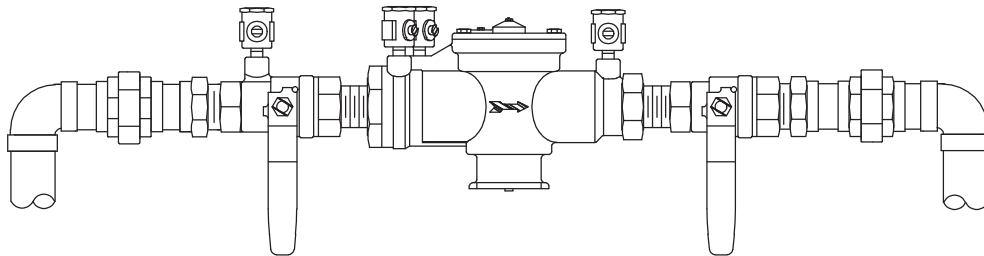
FLOWMETER



STRAINER



REGULATOR



BACKFLOW PREVENTER

IDENTIFICATION OF WATER CONTROL DEVICES W/WARNING TAG

CITY OF HOLLISTER

STANDARD PLANS

APPROVED: _____

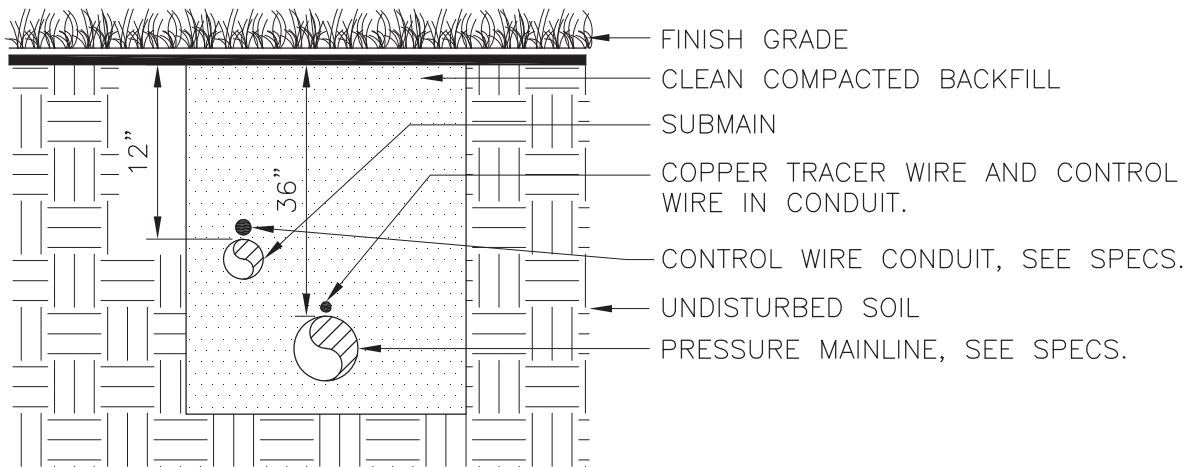
DATE

REVISED

DETAIL

RW-17

SHEET NO 1 OF 1



IRRIGATION PIPE INSTALLATION

CITY OF HOLLISTER

STANDARD PLANS

APPROVED: _____

DATE

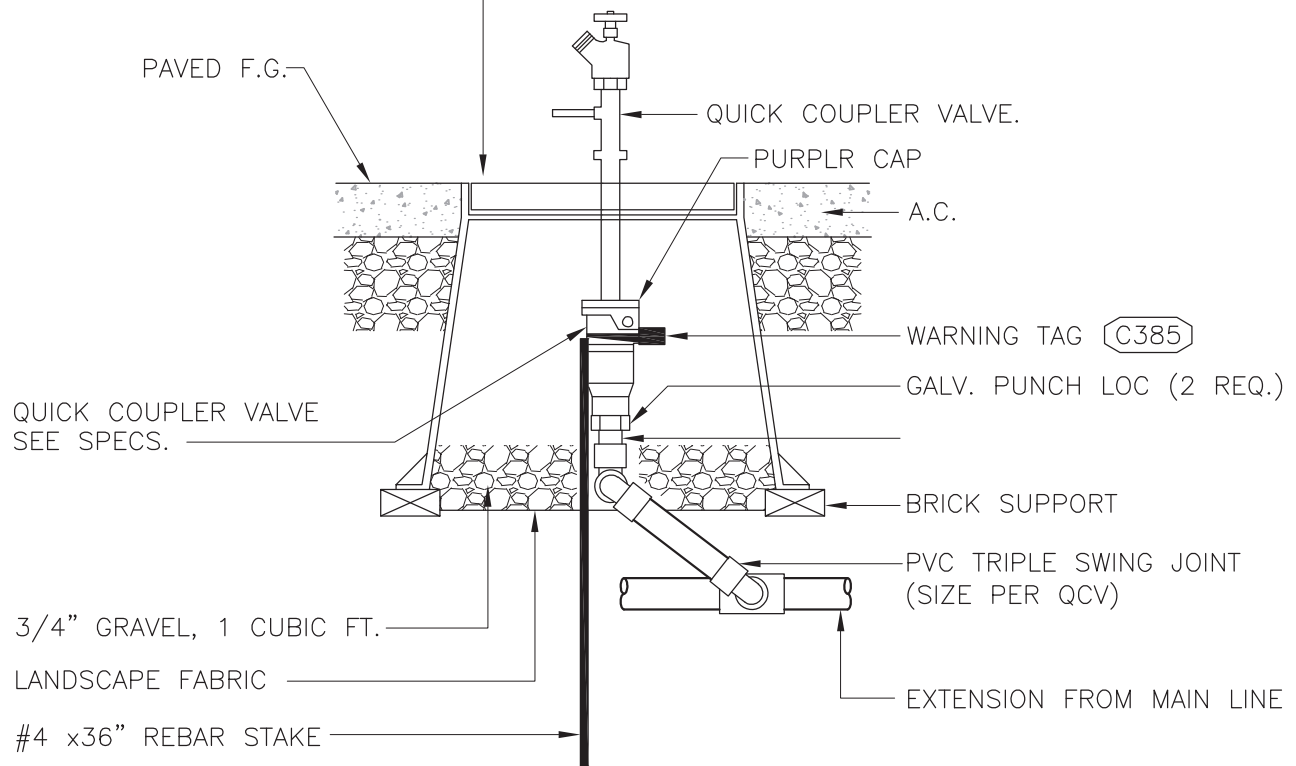
REVISED

DETAIL

RW-18

SHEET NO 1 OF 1

PURPLE RECYCLED WATER PLASTIC
ROUND VALVE BOX WITH BOLT DOWN
COVER, USE STAINLESS STEEL BOLT
NUT AND WASHER HEAT BRAND "QCV"
ONTO LID



NOTE:

1. USE TEFLON TAPE ON ALL THREADED FITTINGS TYPICAL.

3/4" QUICK COUPLER VALVE

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

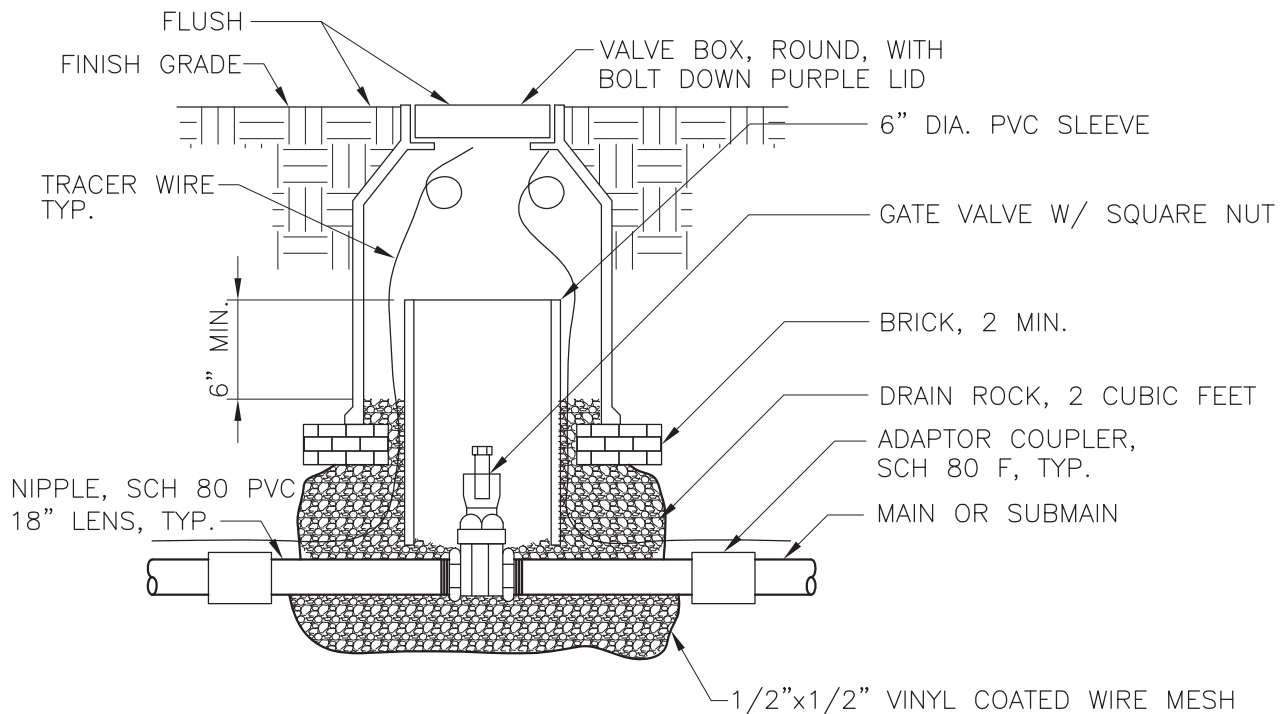
DATE

REVISED

DETAIL

RW-19

SHEET NO 1 OF 1



NOTES:

1. GATE VALVE SIZE SHALL MATCH PIPE SIZE.
2. DECODER(S) FOR ADJACENT CONTROL VALVES SHALL BE LOCATED WITHIN THE LATERAL GATE VALVE BOX.
3. ALL WIRING OUTSIDE THE VALVE BOX AND WIRE MESH SHALL BE WITHIN 1" PVC CONDUIT (NOT SHOWN) FOR PROTECTION FROM RODENTS.
4. A 24-INCH WIRE LOOP SHALL BE INCORPORATED INTO EACH RUN OF WIRE TO ALLOW THE DECODER TO BE REMOVED FROM THE BOX WITHOUT DISCONNECTING THE WIRES.
5. ALL WIRE CONNECTIONS SHALL BE MADE WITH WATERPROOF WIRE CONNECTORS PER DET C403.

4", 6" OR 8" GATE VALVE

CITY OF HOLLISTER
STANDARD PLANS

APPROVED: _____

DATE

REVISED

DETAIL

RW-20

SHEET NO 1 OF 1

CONSTRUCTION INSPECTION CHECKLIST

Site Name _____

Site Address _____

INITIAL	TASK	DATE
_____	1. Certified Site Supervisor Designated	_____
	_____ Name (Please Print Clearly) _____ Certificate No.	
_____	2. Utilities Horizontal and Vertical Separation Verified	_____
_____	3. Advisory Signs, Tags and Labeling in Accordance with DPH Approved Plans	_____
_____	4. Additional DPH Requirements Completed	_____
_____	5. Cross Connection Test Passed	_____
	☐ Completed form attached	
	☐ Map attached showing all potable connections to be removed from recycled water system prior to connection to meter	
_____	6. Coverage Test Passed	_____
	☐ Map attached showing all potable connections to be removed from recycled water system prior to connection to meter	
_____	7. Meters Installed	_____
	_____ Meter Number _____ Size	
	_____ Meter Number _____ Size	
	_____ Meter Number _____ Size	
_____	8. On-site System Connected to Recycled Water Meter	_____
	☐ Removal of all potable water connections to the recycled water system verified	
_____	9. Record Drawings Submitted	_____

COMMENTS:

CROSS CONNECTION CERTIFICATION FORM

Site Name: _____

Site Meter Location: _____

Site Address: _____

Site City & Zip Code: _____

Retailer: _____

Service Number: _____

Date Test Conducted: _____

Other Attendees at the Test:

Name	Company	Phone Number
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

I, _____ AWWA Cross-Connection Specialist # _____
(print name)

after carefully reviewing the systems and conducting the test as per California UPC Appendix J, and City of Hollister Rules and Regulations, find no indication of a cross-connection between the Recycled Water system and Potable system at the above indicated location:

Signed: _____ Date: _____ Time: _____

Company: _____ Phone: _____

Address: _____

**CITY OF HOLLISTER
RECYCLED WATER USE PERMIT**

Site Name _____

Site Address _____

License # _____ Effective Date _____

In accordance with all terms and conditions of the City of Hollister's Recycled Water Use Manual and Rules of Service, all City codes, ordinances, resolutions, policies, standards and with any and all applicable provisions of Federal and/or State laws or regulations as may currently exist, or may be amended or updated from time to time and any conditions specific to this site. Permission is hereby granted to

Owner:

Address:

Phone #:

Email Address:

Site Supervisor Name:

Phone #:

Email Address:

to use recycled water at the site as shown on the attached customer connection drawing.

Recycled water customer understands all the conditions of this license and agrees to comply with all the conditions of this license. Failure to comply with the requirements of this license may be grounds for administrative action, license revocation, or disconnection of service.

The Licensee, is hereby given authorization to use recycled water provided that:

- The Licensee prohibits cross-connections between the recycled water piping and potable water piping.
- All maintenance personnel are receiving education/training on a continuous basis, of the presence of recycled water, and the fact that it is not approved for drinking purposes.
- The use of recycled water for irrigation shall be performed during periods that minimize human contact.
- The Licensee must monitor and maintain the system to minimize equipment and material failure. Broken sprinkler heads, leaks, unreliable valves, etc. should be repaired/replaced as soon as they become apparent.
- Submit a proposal for any changes to the onsite system for the City approval prior to construction, unless required on an emergency basis.

- Ensure that recycled water shall be applied at a rate that does not exceed the infiltration rate of the soil.
- Maintain records of repairs, system upgrades, types and quantities of materials utilized in the irrigation system, employee training records, and all applicable permit documents.
- The Licensee must submit **Annual Self-Monitoring Reports and Facilities Modification Logs** by fax to (831) 636-4310 or by mail to the following address:

City of Hollister
375 Fifth Street
Hollister, CA 95023
Attn: Recycled Water Supervisor

The City shall:

- Have the right and responsibility to suspend or terminate recycled water service in the event that the above conditions are not being adhered to.
- Have the right to perform periodic inspections of the site to assure compliance.

Protection of Public Health

The City reserves the right to take any action necessary with respect to the operation of the Licensee's recycled water system to safeguard the public health.

Water Quality

City reserves the right to provide the Licensee either recycled water or other non-potable water as may be in the City's best interest. City warrants that water quality shall meet all State health requirements for irrigation purposes.

Fee and Charges

The Licensee shall pay bi-monthly user charges in accordance with the rates established by the City. The City shall periodically review the user charges and may make modifications as may be appropriate. The Licensee will be responsible for payment of all licensing and infraction fees.

Period of Service

City shall provide non-potable water for as long as the Licensee complies with all City requirements and for as long as the City may be permitted by regulatory agencies.

The Licensee is advised that there may be periods when non-potable water is unavailable due to system maintenance. The Licensee will be notified in advance of these occurrences.

User Supervisor

A person who the customer or property owner has designated to be a liaison with the City shall be available to the City at all times, shall have the authority to carry out any requirements of the City, and shall be responsible for the installation, operation and maintenance of the recycled water system and also prevention of potential hazards. This person is a representative of the property owner.

Certification Statement/Signature Section

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, and accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and/or disconnection of recycled water service.”

Print (Owner or Authorized Representative)

Signature

Date

“I have reviewed the application package and approve of providing recycled water service in accordance with provisions contained in this recycled water permit.”

Print (City Representative)

Signature

Date

ANNUAL SELF MONITORING REPORT

Site Name _____

Site Address _____

License # _____

	Yes	No
1. Is there evidence of recycled water runoff from the site? If yes, please sketch affected area on a separate sheet and estimate runoff volume.	☐	☐
2. Is there an odor due to recycled water at the site? If yes, please provide a description of the apparent source, characterization, direction of travel, and any public use areas or off-site facilities impacted by the odor.	☐	☐
3. Is there evidence of ponding of recycled water, and/or evidence of mosquitoes breeding due to ponded water?	☐	☐
4. Is there evidence of leaks or breaks in the system pipelines or tubing?	☐	☐
5. Are advisory signs, tags and above-ground pipe marketing in good condition and properly posted to inform public that water is recycled?	☐	☐
6. Is there evidence of plugged, broken or otherwise faulty drip irrigation system emitters or spray irrigation sprinklers on the site?	☐	☐
7. Has a cross-connection test been performed on the system within the past five years? If potable water is not available on site, please skip this question.	☐	☐
8. Have the backflow assemblies passed annual testing? If not applicable, please skip this question.	☐	☐
9. In the past year or since the last annual site inspection report, have there been any modifications of the piping for the recycled water system? If yes, please complete the Facilities Modification Log.	☐	☐
10. In the past year or since the last annual site inspection report, has the site owner, tenant or user supervisor changed? If yes, please fill out the Change of Information on the back of the form.	☐	☐
11. Describe actions being taken to correct the problems noted above		

12. Additional Comments:

Change of Information

Site Owner

NAME		
CONTACT NAME	TITLE	
ADDRESS		
CITY	ZIP	
PHONE	FAX	EMAIL

User Supervisor

NAME		CERTIFICATION NUMBER
ORGANIZATION	ADDRESS	
CITY	ZIP	
PHONE	FAX	EMAIL
EMERGENCY PHONE NUMBER		

I certify that the information in this report, to the best of my knowledge, is correct and true.

_____ User Supervisor Signature	_____ User Supervisor Certification Number	_____ Date of Inspection
------------------------------------	---	-----------------------------

FACILITIES MODIFICATION LOG

Site Name _____

Site Address

License # _____

[illegible]

Appendix C - Recycled Water Use Area Applications and Site Plans

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- Select Brigantino Park Drawings
- Select Airport Irrigation Project Drawings

Appendix D - Transmission Line Plans

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- [Select DWTP Project Drawings](#)
- [Select IWTP Pipeline Project Drawings](#)
- [Select Airport Pipeline Project Drawings](#)



Recycled Water Use Manual and Rules of Service

CITY OF HOLLISTER WATER RECLAMATION FACILITY

June 2010

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Appendix A -	Definitions
Appendix B -	Flow Chart

List of Abbreviations

AWWA	American Water Work Association
DPH	Department of Public Health

Chapter 1 Introduction

The City of Hollister's Recycled Water Use Manual and Rules of Service is intended to:

- Assist new customers through the application, design and construction process in order to receive a Recycled Water Use Permit
- Serve as a reference source for customers and their Site Supervisors regarding the proper operation and maintenance of their recycled water system

In order to receive and use recycled water from the City, customers must follow the rules as set forth in this document.

This document was written to be in compliance with the existing codes, laws, statues and regulations of the State of California and local governing bodies concerning the currently-approved use of recycled water. It is not intended to void any of these regulations. The manual is also not intended to void standard industry practices as presented in the American Water Work Association (AWWA) California-Nevada Section's *Guidelines for Distribution of Nonpotable Water* or *Guidelines for the On-site Retrofit of Facilities using Disinfected Tertiary Recycled Water*. However, since legal and regulatory requirements can change without the knowledge of the City, the City assumes no liability for errors in this manual. If any section, subsection, clause or phrase of the rules and regulations presented in this document is determined to be invalid, the remaining portions of these rules and regulations shall remain in effect.

Amendments to this document may be made periodically to comply with State regulations or to meet new City standards. If amendments are made, existing users will be notified of the new rules of service as well as the effective date for compliance with the new rules.

Questions regarding the information presented herein should be directed to the City's Recycled Water Supervisor. The Recycled Water Supervisor oversees all programs and facilities related to the use of recycled water distribution by the City and will coordinate with the appropriate City departments to resolve customer inquiries. Table 1-1 provides contact information for the current Recycled Water Supervisor as well as an emergency contact number.

Table 1-1: City Contacts

Contact	Mailing Address	Phone	Fax
Dennis Rose	375 Fifth Street Hollister, CA 95023	(831) 524-1880	(831) 636-4349
Water Reclamation Facility	2690 San Juan Hollister Road Hollister, CA 95023	(831) 636-4350	(831) 636-4351
<u>Emergency:</u>		911	

Chapter 2 Obtaining a Recycled Water Use Permit

2.1 Approved Uses

Each recycled water customer must have a permit from the City prior to receiving recycled water. The State of California regulates the use of recycled water, as directed under the California Code of Regulations, Titles 17 and 22. Local authorities, at their discretion, can require or specify what sites and/or uses of recycled water are to be utilized in their service area so long as it complies with State requirements.

In the City of Hollister's service area, recycled water may be used for the following applications, all of which have been approved by the California Department of Public Health (DPH) under the City's Master Permit (Order No. R3-2008-0069):

- Irrigation
 - Food crops, including all edible root crops, where the recycled water comes into contact with the edible portion of the crop
 - Parks and playgrounds
 - School yards
 - Unrestricted and restricted access golf courses
 - Cemeteries
 - Freeway landscaping
 - Ornamental nursery stock and sod farms where access by the general public is not restricted
 - Pasture for animals producing milk for human consumption
 - Non-edible vegetation with controlled access
 - Orchards, vineyards, non-food bearing trees
- Landscape Impoundment
- Nonstructural Fire Fighting
- Other Uses
 - Backfill consolidation around non-potable piping
 - Soil compaction
 - Mixing concrete
 - Dust control on roads and streets
 - Cleaning roads, sidewalks and outdoor work areas
 - Flushing sanitary sewers

The City reserves the right to disallow a proposed use on the grounds of safety, public health, technical feasibility or other concerns, for which the City's judgment shall be final. At its discretion, the City may also set forth site specific requirements as conditions for the permitted use.

2.2 Application

The first step in obtaining a Recycled Water Use Permit is to submit an Application for Recycled Water Service.¹ Copies of the application form are available from the City upon request. A completed application form will include:

- **Property Information:** This should include a brief description of the property, including the type of land uses within the proposed recycled water use area.
- **Applicant:** Applicants must either be the owner or an authorized representative of the property to be served.
- **Site Owner:** If the site owner is the same as the applicant, this section can be left blank.
- **Developer/Design Engineer Contact:** If applicable, provide contact of person that can respond to technical questions on the proposed recycled water uses and site development.
- **User Supervisor:** Each recycled water use site owner/operator will appoint a User Supervisor to be the main point of contact with the City regarding recycled water use. The User Supervisors will be responsible for ensuring compliance with the programs and procedures for the use of recycled water, as outlined in the Recycled Water Use Permit and the *Recycled Water Use Manual and Rules of Service*. The User Supervisor must be certified through the City's training program; if the User Supervisor has not received certification at the time of application, the certification number can be left blank; however the User Supervisor must be certified in order for the site to receive a Recycled Water Use Permit. In addition to the standard contact information, an emergency phone number shall be provided for the User Supervisor. The User Supervisor must be reachable at all times in case of an emergency.
- **Proposed Recycled Water Use:** Recycled water is approved for irrigation, landscape impoundments, fire suppression and other uses listed in Section 2.1.
- **Recycled Water Demand Estimates:** The customer's best estimate should be provided to assist the City in evaluating the adequacy of the distribution system.
- **Use Area Containment Measures:** For irrigation projects this could include irrigating at agronomic rates to avoid runoff, constructing slopes to maintain drainage on-site, and/or installing containment beams. For landscape impoundments, water levels should be reduced when heading into the wet season to prevent rain induced overflows.
- **Employee or Public Access:** Potential access by employees or the public should be described.
- **Plans:** The plans should show the proposed recycled water use areas and areas to be excluded; recycled water infrastructure including the location and size of service connections, meters and backflow devices; a brief description of all special construction requirements, if applicable; areas of public access; surrounding land uses; construction details of wells in or near the use area; and location and type of signage that will be used. The design information presented in Section 2.3 should be considered in development of the plans.
- **Cross-Connection Control Procedures:** Where both potable and recycled water lines exist, a description of cross-connection control procedures shall be submitted.
- **Additional Attachments:** Other site specific information may be required depending on the proposed recycled water use:
 - Location of domestic water supply facilities in or adjacent to use area (for irrigation projects)

¹The Recycled Water Use Permit designates approval for recycled water service. Additional permits may be necessary for the construction of the recycled water facilities. The customer is responsible for applying for and obtaining other permits as may be necessary for the construction of the recycled water facilities.

- Description of Site Drainage (for irrigation and landscape impoundment projects)
- List of outdoor drinking fountains, bleachers, and designated outdoor eating areas (for irrigation projects)
- Signatures: The application shall be signed by the appointed User Supervisor as well as the owner or an authorized representative of the property to be served.

Upon receipt of a completed application, the City will review the information submitted to determine whether the proposed use conforms to the City's Rules of Service and whether the City has the necessary capacity to serve the applicant. If revisions to the proposed use are required or if additional information is needed, the City will contact the applicant and request that the necessary corrections be made or that the supplemental information be submitted. If the application meets the City's standards, the City will then forward a copy of the plans to DPH for review and approval. Following approval by DPH, the City shall approve use of recycled water per the applicant's plans.

2.3 Design Requirements

A summary of the basic design requirements for customers' facilities is provided below:

- No Cross-Connections: No cross-connections are allowed between the recycled water system and the potable water system.
- Backflow Prevention:
 - Potable System: In order to protect the public drinking water system from accidental cross-connections, a reduced pressure principal backflow prevention device is required at all meters that supply potable water to a site where recycled water is present. Additionally, for sites where the potable water is used as a back-up to the recycled water system, there must be an air gap separation between the two systems.
 - Non-Potable Wells: For agricultural wells that are connected to the customer's recycled water distribution system, a double check valve backflow prevention device is required on the well.
 - Recycled Water System: In most cases, backflow prevention devices will not be required on the recycled water service. Backflow prevention would be required on the customer's recycled water system if the City determines that there is a condition on-site that would threaten the integrity of the recycled water distribution system. Examples of potential threats to the recycled water distribution system are chemical fertilizer injection systems that are directly installed on the irrigation system and recycled water impoundments. In these cases a reduced pressure principal backflow prevention device shall be upstream of the potential threat.
- Pipe Separation: The customer's recycled water system shall be installed to meet the following separation requirements from all potable water pipelines.
 - Horizontal Pipe Separation: All recycled water service laterals and meters should be at least 10 feet from the nearest potable water facility, including pipelines, meters and hydrants. If a ten-foot parallel spacing is not practical, a separation of at least 4 feet may be allowed subject to the special construction conditions outlined in Table 2-1. Under no circumstances is horizontal separation of less than four feet or construction in the same trench as potable water facilities allowed.
 - Vertical Pipe Separation: Where a buried constant-pressure recycled water pipeline crosses a buried potable water pipeline, it should be a minimum of 4 inches below the potable water pipeline and should meet the requirements of Table 2-1. Constant-pressure recycled water pipelines are allowed over potable water pipelines with a minimum of 4 inches vertical separation and should meet the requirements of Table 2-1 or the recycled

water pipeline is installed in a pipe sleeve which extends a minimum of 10 feet on either side of the potable water piping. Intermittently pressurized irrigation laterals may be located a minimum of 4 inches above potable water pipelines without sleeving.

Table 2-1: Pipe Separation Requirements

Horizontal Separation for Parallel Construction	
Pipe Separation	Construction Requirements
Greater than 10 feet separation	No special construction required
Between 4 to 10 feet separation	Must meet one of the following requirements: • Ductile iron pipe with compression joints • Class 200 PVC – AWWA C900 • HDPE pipe with fusion welded joints • Fused PVC
Less than 4 feet separation	Not allowed
Vertical Separation at Crossings	
Pipe Separation	Construction Requirements
Less than 4 inches below Potable	Not allowed
4 to 12 inches below potable	The recycled water line shall have no joints within 4 ft from either side of the water main, shall have a standard pipe segment centered over the pipe being crossed, and must also meet one of the following requirements: • Ductile iron pipe with mechanical joints • Class 200 PVC – AWWA C900 • HDPE pipe with fusion welded joints • Fused PVC
12 inches or greater below potable	No special construction required
Less than 4 inches above potable	Not allowed
4 inches or greater above potable	The recycled water line shall have no joints within 10 ft from either side of the water main, shall have a standard pipe segment centered over the pipe being crossed, and must also meet one of the following requirements: • Ductile iron pipe (continuous section) • Class 200 PVC – AWWA C900 • HDPE pipe with fusion welded joints • Fused PVC

- **Pipe Depth** – The customer's on-site recycled water piping must be designed to provide a minimum of 12-36 inches of cover from top of the pipe to finished pavement. The minimum cover depends on whether the pipes are under constant or intermittent pressure and the pipe size. These minimum cover requirements are summarized in Table 2-2.

Table 2-2: Pipe Depth Requirements

Pipe Type	Minimum Cover (in)
Intermittent Pressure, All Sizes	12
Constant Pressure, 2.5" diameter and smaller	18
Constant Pressure, 3-10" diameter	24
Constant Pressure, 12" diameter and larger	36

- **Pipe Identification:**
 - New piping: All new recycled water piping below or above grade shall be purple colored pipe with the words "CAUTION — RECYCLED WATER" printed on the pipe or on marking tape. Buried recycled water piping shall be laid so that the wording faces upwards.
 - Existing piping: All existing piping above grade must be labeled with purple identification tape with the imprinted words "CAUTION — RECYCLED WATER," or replaced by purple colored pipe with the required wording. Additionally, where feasible existing piping which is uncovered during construction should be marked with the words "CAUTION — RECYCLED WATER."
- **Recycled Water Warning Tags, Stickers, Labels:** All recycled water appurtenances including valve boxes, quick couplers, recycled water storage tanks, air/vacuum relief valves, pressure reducing valves, pumps, backflow prevention devices and system controller boxes must be labeled with purple, weatherproof warning tags, stickers or other labels. The tags, stickers or labels must include the words "WARNING – RECYCLED WATER — DO NOT DRINK" or similar wording and "AVISO — AGUA IMPURA — NO TOMAR."
- **Potable Water Identification Tags, Stickers, Labels:** At sites where both recycled water and potable water systems are used, all potable water meter and above grade water devices including backflow prevention devices and hose bibs must be labeled with blue, weatherproof identification tags, stickers or other labels. The tags, stickers or labels must include the words "POTABLE WATER" and "AQUA PARA TOMAR."
- **Signage:** Advisory signs must be posted within the recycled water use area in locations where they can easily be seen. Lettering on the sign must be at least ½-inch tall and include the words "RECYCLED WATER" and "DO NOT DRINK – NO BEBER." At a minimum signs should be posted at entrances to the site and around outdoor eating areas. Signs should be spaced no further than 1,000 feet apart.
- **Quick Coupling Valves:** Quick coupling valves used on the recycled water system must be made specifically for recycled water use. The covers on all quick coupling valves must be permanently attached and made of purple rubber or vinyl with the words "RECYCLED WATER" imprinted on the locking cover.
- **No Hose Bibbs:** Hose bibbs are not allowed on the recycled water system. For retrofit systems, hose bibbs can be replaced by below grade quick coupling valves with locking caps.
- **No Runoff:** The recycled water system must be designed to prevent runoff outside the approved use area.
- **No Ponding:** Except for permitted landscape impoundments, the recycled water system must be configured and operated so that ponding does not occur.
- **No Windblown Spray outside the Approved Use Area;** Recycled water irrigation systems must be configured such that windblown spray will remain within the approved use area.

- Drinking Fountains, Outdoor Eating Areas, Bleachers: Drinking fountains, outdoor eating areas, bleachers and similar facilities must be located in areas that will not be subject to spray from sprinklers. The facilities may be protected by modifying the irrigation system, relocating the protected facilities, or installing a shielding device.
- Protection of Aquifers: No irrigation with recycled water shall take place within a horizontal distance of 50 feet of any domestic water supply well. All irrigation systems shall be designed and located accordingly. In addition, all recycled water impoundments must be horizontally separated by a distance of at least 100 feet from any domestic water supply well.

2.4 Construction

Provided that all permits for construction have been obtained and the City approves the plans submitted with the Application for Recycled Water Service, construction of the recycled water systems may begin.

As part of the construction process for new facilities a temporary potable water connection will be required to perform cross-connection and coverage testing. The customer must obtain approval from the City prior to constructing a jumper (temporary connection) to an on-site potable system. A City inspector must be on-site when a jumper to the potable water system is made and removed. Jumpers connecting to a recycled water supply are prohibited at all times. If there are no potable water facilities in the vicinity of the project, the City at its discretion may eliminate the need for cross-connection testing, and coverage testing will be waived until recycled water is available to the site.

Because the City is responsible for inspecting the system during and after construction to ensure compliance with the approved plans, the applicant must notify the City prior to the start of construction and provide the City with updated construction schedules.

All conceptual or major changes from the approved plans must be approved by the City before implementation.

2.5 Inspection

The City will maintain and update a Construction Inspection Checklist throughout the construction process. The elements of the Construction Inspection Checklist are:

- Certified User Supervisor Designated: The City must verify that the designated User Supervisor has been certified through the City's annual training program.
- Utilities Horizontal and Vertical Separation Verified: Through the City's on-site inspections during the construction of the system, inspectors will be able to verify adherence to the pipe separation requirements specified in Section 2.3.
- Advisory Signs, Tags and Labeling in Accordance with DPH Approved Plans: The City must check that the location and type of signage and warning labels match the plans as approved by DPH.
- Additional DPH Requirements Completed: The City is responsible for ensuring that additional requirements resulting from DPH's review have been incorporated.
- Cross-Connection Test Passed: The customer must conduct a cross-connection test to ensure that the recycled water system is not connected to the potable water system. The test must be performed by a certified AWWA cross-connection control specialist provided by the City. In addition to coordinating with the City to complete the cross-connection test, the customer must submit a map showing all potable connections that will be removed from the recycled water system prior to connecting to the meter.
- Coverage Test Passed: The customer shall schedule a coverage test walk through to demonstrate to the City that there is not excessive overspray, runoff or ponding and that no contact occurs

between recycled water and drinking water fountains, outdoor eating areas or bleachers. If adjustments to the system are required that can not be made during the site walk, the City will issue a punch list and the customer must address each item on the list prior to rescheduling the coverage test.

- **Meter Installation:** Following the final construction inspection and successful completion of the cross-connection and coverage tests, the City will coordinate with the customer on installation of the recycled water meter(s).
- **On-site System Connected to Recycled Water:** Prior to connecting to recycled water system the City shall verify that all potable water connections, as identified on the customer's map, have been removed.
- **Record Drawings Submitted:** The customer is required to submit record drawings. The drawings must include all changes from the original plans.

2.6 Permit

Once the City has verified that each item on the Inspection Checklist has been completed, a Recycled Water Use Permit will be issued to the customer. Recycled water service is conditioned upon requirements which may be attached to the permit and proper operation and maintenance of the system as discussed in Chapter 3.

If the property is transferred to a new owner or tenant, the customer must notify the City in writing within 30 days to transfer the permit.

If the customer's recycled water system is found to be in violation of the rules of service, the City will direct the customer to mitigate for these violations. A site inspection will be scheduled after a reasonable mitigation period to ensure compliance. Failure to comply will result in termination of recycled water service until improvements are made.

Chapter 3 Operating and Maintaining a Recycled Water System

3.1 User Supervisor Responsibilities

As part of the application and permitting process, each recycled water use customer will appoint a User Supervisor to be the main point of contact with the City regarding recycled water use. The User Supervisors will be responsible for ensuring compliance with City and State regulations for the use of recycled water. The User Supervisor will be responsible for:

- Ensuring that all use site facilities that serve recycled water are maintained and operated in accordance with the Recycled Water Use Permit and this manual.
- Operating and controlling the irrigation systems in a manner to prevent human consumption of recycled water, to control and limit runoff, and to prevent contamination of nearby domestic supply wells, as applicable.
- Ensuring that all personnel are educated in practices and procedures for working with recycled water.
- Installing and maintaining signs around the use site.
- Preventing cross-connection between recycled water facilities and domestic water facilities.
- Notifying, preparing and submitting reports to the City when there are system failures that cause unauthorized discharges.
- Notifying the City for its approval of any proposed modification or additions to facilities.
- Ensuring that recycled water for irrigation is used at agronomic rates and is not used during rainfall events.

In the event that a customer wishes to make personnel changes to the User Supervisor position, the City must be notified in writing, and the newly designated User Supervisor must attend the City's training program with 120 days of the personnel change.

3.2 Maintenance

In order to keep the recycled water system in compliance with the City's rules and regulations, the User Supervisor shall perform the following preventive maintenance activities:

- Perform no less than weekly inspections of the entire recycled water system including sprinkler heads, drip irrigation system emitters, spray patterns, piping and valves, pumps, storage facilities and controllers.
- Immediately repair all broken sprinkler heads, faulty spray patterns, leaking pipes or valves, or any other noted condition that violates the recycled water use requirements.
- Check all recycled water identification signs, tags, stickers, and above grade pipe markings for their proper placement and legibility. Replace damaged, unreadable, or missing signs, tags, stickers, and pipe markings.
- Check spray patterns to eliminate ponding, runoff and wind blown spray conditions. If evidence of ponding or runoff is noted, affected areas should be indicated on a sketch and sprinkler heads should be adjusted to prevent further ponding or runoff. Evidence of mosquitoes breeding within ponding should be noted and immediately eliminated.
- Establish and maintain an accurate record keeping system of all inspections, modifications and repair work in a bound inspection logbook. A summary of the observations made during water

recycling area inspections and a brief discussion of any corrective actions taken or planned shall be submitted to the City Recycled Water Supervisor at the end of the calendar year.

- Coordinate with the City to complete a cross-connection test annually.
- Coordinate with the City to complete annual testing of backflow assemblies.

3.3 Hours of Irrigation

Hours of irrigation with recycled water must be maintained to minimize potential public contact with recycled water. The following guidelines should be followed:

- For areas that are easily accessed by the general public, irrigation must take place during the periods of least use by the general public, generally between the hours of 9 pm and 7 am.
- For areas where public access is restricted or prohibited, such as highway medians, commercial nurseries, etc., irrigation may occur at all times.
- Irrigation of areas accessed by the general public may be performed at other times provided that the irrigation system is operated manually and is supervised (someone present at all times) to avoid inadvertently exposing any members of the general public.
- Irrigators should take into account that irrigated areas might need a reasonable dry-out period before the area is accessed by the general public.

3.4 System Modifications

The User Supervisor must obtain approval from the City before modifying an approved recycled water system. This includes converting any piping used for recycled water back to potable water, such as switching from a recycled water system to a backup potable water system.

Emergency modifications necessary to prevent contamination, damage or a public health hazard can be made without the prior approval of the City. However, immediately after the modification has been implemented and the hazard has been abated, the User Supervisor shall notify the City of the emergency modifications and file a written report.

3.5 Personnel Training

The User Supervisor is responsible for training all personnel involved with recycled water so they are familiar with the rules and regulations. At a minimum, the training program should convey the following information:

- The City's recycled water, although highly treated, is non-potable and must never be used for human consumption.
- Regulations prohibit ponding, windblown spray and runoff of recycled water.
- Working with non-potable recycled water is safe if common sense is used and appropriate regulations are followed.
- State law prohibits a connection between the recycled water and the potable water systems.

3.6 Annual Self Inspection Report

An annual self-inspection must be conducted at each use site while the recycled water system is in use. The User Supervisor must conduct this inspection and provide a report of the inspection to the City using the City's Annual Self Inspection Report form. The original report must be submitted to the City with a copy kept on-site by the User Supervisor.

The following questions must be answered as part of the annual inspection report:

- Is there evidence of recycled water runoff from the site? If yes, the User Supervisor must submit a sketch showing the affected area(s) and estimated volume of runoff.
- Is there an odor due to recycled water at the site? If yes, the User Supervisor must provide a description of the apparent source, characterization, direction of travel, and any public use areas or off-site facilities affected by the odors.
- Is there evidence of recycled water ponding, and/or evidence of mosquitoes breeding due to ponded water?
- Is there evidence of leaks or breaks in the system piping or tubing?
- Are advisory signs, tags and above-ground pipe markings in good condition and properly posted to inform public that water is recycled?
- Is there evidence of plugged, broken or otherwise faulty drip irrigation system emitters or spray irrigation sprinklers on the site?
- In the past year or since the last annual site inspection report, were there any modifications of the piping or the recycled water system as approved by DPH? If yes, the User Supervisor must submit a Facilities Modification Log.
- In the past year or since the last annual site inspection report, has the site owner, tenant or user supervisor changed?
- What corrective actions are being taken to correct any problems noted in the report?

3.7 Emergency Procedures

3.7.1 Disaster Response

Following a major earthquake, flood, fire or other incident which could likely damage a customer's recycled or potable water systems, the User Supervisor should inspect the potable and recycled water systems for damage as soon as it is safe to do so. If either system appears damaged, both the potable and recycled water systems should be shut off at their points of connection. If the User Supervisor cannot inspect the site and damage is expected, then both water systems should be shut off at their points of connection. The Supervisor should immediately contact the City for further instruction.

3.7.2 Unauthorized Discharge

An unauthorized discharge is defined as any amount of recycled water that leaves the designated use site. If a system failure results in an unauthorized discharge, every effort should be made to contain the recycled water and prevent it from entering the storm drain system. The User Supervisor is responsible for immediately notifying the City of the unauthorized discharge, and then preparing and submitting a report estimating the volume of discharge, identifying the cause of the system failure and discussing the corrective measures taken.

In the event that an unauthorized discharge occurs that may endanger public health or the environment, the City will verbally notify the RWQCB and DPH within 24 hours.

3.7.3 Cross-Connection

If a cross-connection occurs or is suspected of having occurred on the customer's premises, the User Supervisor must immediately implement the following Emergency Cross-Connection Response Plan:

- Immediately following discovery of the cross-connection, shut down the recycled water supply to the facility.
- Notify the City and DPH of the incident by telephone immediately. The City will notify the local health department. Within 24 hours the customer's verbal notifications must be followed by a written notice to the City that includes an explanation of the nature of the cross-connection, date

and time discovered, steps taken to mitigate the cross-connection and the contact information of the person reporting the cross-connection.

- Keep the potable system pressurized and post "Do Not Drink" signs at all potable water fixtures and outlets.
- Provide bottled water for employees until the potable water system is deemed safe to drink.
- Coordinate with the Cit to collect water samples from the potable water system and perform a 24-hour bacteriological analysis. If the bacteriological analysis is positive, chlorinate the potable water system maintaining a chlorine residual of at least 50 mg/l for 24 hours, flush the potable water system after the 24-hour chlorination period and perform standard bacteriological analysis. The City shall assist with the sampling and laboratory analyses at the customer's expense.
- Coordinate with the City to conduct a cross-connection test to verify that all cross-connections were eliminated.
- Follow additional procedures outlined by the City, DPH and local health department.

After final approval has been obtained from DPH and the local health department, the City will bring the recycled water system back into service and inform the customer to remove the "Do Not Drink" signs from all potable water fixtures and outlets.

3.8.1 Quarterly Reporting

G. RECYCLED WATER USE AREA MONITORING

1. The quantity of reclaimed water distributed to each reuse site shall be recorded on a weekly basis. Total flows shall be metered or estimated based on irrigation run times and distribution system design flow rates. Total as applied flows shall be compared to Supplier effluent flow rates.
2. All nutrient (fertilizer) additions during the quarter are to be reported. At a minimum, the application date, nutrient, application quantity, % content and any other appropriate information are to be included.
3. During periods of recycled water application the Distributor or Users, as applicable, shall inspect the irrigation use areas no less frequently than weekly to verify and document compliance with Order No R3-2008-0069. The visual inspections shall be noted in a bound inspection logbook(s) and at a minimum shall document proper sprinkler operation, runoff, erosion, saturated surface conditions, and odors. The logbook(s) shall be made available to the City of Hollister, Water Board and DPH upon request. A summary of observations made during water recycling area inspections and a brief discussion of any corrective actions taken or planned shall be included with each annual monitoring report.
4. Each individual User Reclaimed Water Site Supervisor shall provide quarterly updates to the Distributor regarding irrigation frequency and flow rates, proposed system modifications, system peculiarities, and to verify employee training. The Distributor shall keep a record of all system modifications and document that all work is conducted in accordance with the Cross Connection Control Plan and applicable regulations.

The Quarterly monitoring reports shall be submitted as follows:

Monitoring Period	Report Due Date
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January 1 – March 31	April 15 th
April 1 – June 30	July 15 th
July 1 – September 30	October 15 th
October 1 – December 31	January 15 th

Appendix A - Definitions

Whenever the following terms occur in this document, the intent and meaning shall be interpreted as follows:

AIR GAP	A physical separation between the free flowing discharge end of a water supply pipeline and an open or non-pressure receiving vessel. An approved air gap must be at least twice the diameter of the water supply pipe measured vertically above the overflow rim of the vessel, and in no case less than one inch.
APPROVED USE	An application of recycled water in a manner, and for a purpose, designated in a Recycled Water Use Permit issued by the City and in compliance with all applicable Regulatory Agency requirements.
APPROVED USE AREA	A site with well-defined boundaries designated on the approved Site Drawings, to receive recycled water for an approved use and acknowledged by all applicable regulatory Agencies.
CROSS-CONNECTION	Any physical connection between any part of a water system used or intended to supply water for drinking purposes and any source or system containing water or substance that is not or cannot be approved for human consumption. This includes direct piping between the two systems, regardless of the presence of valves, backflow prevention devices, or other appurtenances.
CUSTOMER	Any person, persons or firm including any public utility, municipality or other public body or institution issued a Recycled Water Use Permit by the City. They may be the owner, tenant, or property manager as appropriate.
INSPECTOR	Any person authorized by the City, the local authorities or the local or state health agencies to perform inspections on or off the customer's site before construction, during construction, after construction and during operation.
LANDSCAPE IMPOUNDMENT	A body of recycled water used for aesthetic enjoyment or which otherwise serves a function not intended to include public contact.
NONPOTABLE WATER	Water that has not been treated for human consumption in conformance with the latest edition of the United States Public Health Service Drinking Water Standards, the California Safe Drinking Water Act, or any other applicable standards.
ON-SITE	Designates or relates to all irrigation facilities downstream of the recycled water meter.
OVERSPRAY	The spray of recycled water outside of the approved irrigation area.
OWNER	Any holder of legal title, contract purchaser, or lessee under a lease with an unexpired term of more than one (1) year, for property for which recycled water service has been requested or established.
POINT OF CONNECTION	This is the point where the customer's system ties to the City's system. This is usually at the water meter.
PONDING	Unauthorized retention of recycled water on the surface of the ground or other natural or manmade surface for a period following the cessation of an

POTABLE WATER	approved recycled water use activity. Water that is authorized for human consumption according to the latest edition of the California Safe-Drinking Water Act, or other applicable standards.
PUBLIC	Any person or persons other than the site owner or employees who may come in contact with facilities and/or areas where recycled water is approved for use.
RECYCLED WATER (or NONPOTABLE RECYCLED WATER)	Water that meets California Administration Code Title 22, Division 4 of the Environmental Health Water Reclamation Criteria and is approved for purposes other than human consumption. For the purpose of these rules and regulations, “recycled water” refers to “Nonpotable recycled water.”
RECYCLED WATER USE PERMIT	A permit issued to the customer as required by DPH and the RWQCB that outlines monitoring, self-inspection, reporting, and site-specific requirements.
RUNOFF SERVICE	Recycled water that is allowed to drain outside the approved use area. The furnishing of recycled water to a customer through a metered connection to the onsite facilities.
STANDARD PIPE LENGTH	A section of pipe 18 to 20 feet in length that has no joints.
UNAUTHORIZED DISCHARGE	Any release of recycled water that violates the rules and regulations of City or applicable Federal, State or local statutes, regulations, ordinances, contracts or other requirements.
USER SUPERVISOR	The responsible person designated by the customer to provide liaison with the City. This person must have the authority to carry out any requirements of the City, must be responsible for the operation and maintenance of the recycled water system, and must prevent potential violations.
VIOLATION	Noncompliance with any condition of the Recycled Water Use Permit by any person, action or occurrence, intentional or unintentional.

Appendix B - Flow Chart

